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1 Linear Algebra

1.1 Column wise decomposition.

Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be decomposed into the sum of its columns:

$$\mathbf{A} = \sum_{i=1}^{n} \mathbf{A}_{:j} e_j^\top, \tag{1}$$

where e_j are standard basis vectors of $\mathbb{R}^n$. Notice that this is a rank 1 decomposition.

1.2 Row wise decomposition.

Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be decomposed into the sum of its rows:

$$\mathbf{A} = \sum_{i=1}^{m} e_i \mathbf{A}_{i:}^\top, \tag{2}$$

where e_i are standard basis vectors of $\mathbb{R}^m$. Notice that this is a rank 1 decomposition.

1.3 Trace-vec identities.

The notation $\otimes$, we wil be using for tensor (Kronecker) product. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$, then

$$\mathbf{A} \otimes \mathbf{B} \in \mathbb{R}^{(mp) \times (nq)},$$

is the block matrix defined by

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} \mathbf{A}_{11} \cdot \mathbf{B} & \dots & \mathbf{A}_{1n} \cdot \mathbf{B} \\ \vdots & \vdots & \vdots \\ \mathbf{A}_{m1} \cdot \mathbf{B} & \dots & \mathbf{A}_{mn} \cdot \mathbf{B}. \end{pmatrix}$$

Some properties involve:

- $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$.
- $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$.
- If $\mathbf{A}, \mathbf{B}$ are invertible, $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$.

The notation $\operatorname{vec}(\mathbf{A})$, means vectorize the matrix $\mathbf{A}$ by stacking the columns of $\mathbf{A}$ into a single long column vector.

- **Core identity 1:** $\operatorname{vec}(\mathbf{XYZ}) = (\mathbf{Z}^\top \otimes \mathbf{X}) \operatorname{vec}(\mathbf{Y})$.
- **Core identity 2:** $\operatorname{tr}\left(\mathbf{X}^\top \mathbf{Y}\right) = \operatorname{vec}(\mathbf{X})^\top \operatorname{vec}(\mathbf{Y})$.

2 Multithreading

Per-thread scratch buffers (aka thread-local workspaces) are a very common pattern to cut allocation overhead and lock contention in multi-threaded code. In this way we do not create lots of short-lived temporaries, and have better cache locality and to avoid false sharing on shared buffers.

3 LLM Training

3.1 Scaling the logits after LLM head.

We usually apply RMS norm to normalize (along the last dimension E, where it stands for model dimension, B means batch size and L means sequence length) the tensor $\mathbf{X} \in \mathbb{R}^{B\times L\times E}$ we feed into LLM head, and obtain the corresponding logits l. After RMS normalization, each tensor corresponding to the token $x_i \in \mathbb{R}^E$ will then have $\mathrm{RMS}(x_t) = 1$. Now notice that for each coordinate $x_{i,t}, t \in [E]$, treating as a random variable, its variance is given by

$$\mathrm{Var}\left(x_{i,t}\right) = \mathbb{E}\left[x_{i,t}^2\right] - \left(\mathbb{E}\left[x_{i,t}\right]\right)^2, \tag{3}$$

and if it is zero-mean (or small), then $\mathrm{Var}\left(x_{i,t}\right) \simeq \mathbb{E}\left[x_{i,t}^2\right]$, which is to say that second moment reflects the variance.

The next step is to use the empirical observation that for linear layers, hidden vectors tend to be approximatedly rotation-invariant (isotropic), i.e., each coordinate behaves like the others, so we can use the second moment over the coordinate in a token to replace the actual second moment. And the former, is given by

$$\mathrm{Var}\left(x_{i,t}\right) \simeq \frac{1}{E}\sum_{t=1}^{E} x_{i,t} = 1. \tag{4}$$

Now we start to consider the logits, which is generated by

$$l_{j,i} = w_j^\top x_i = \sum_{t=1}^{E} w_{j,t} x_{i,t}.$$

If we assume each weight entry $w_{j,t}$ are i.i.d. with variance σ^2 the logits variance is give by

$$\mathrm{Var}\left(l_{j,i}\right) = \sum_{t=1}^{E} \sigma^2 \,\mathrm{Var}\left(x_{t,i}\right) \simeq E\sigma^2.$$

So the standard deviation $\sim \sqrt{E}$. To ensure that logits do not scale with the model dimension, we scale it by $\sqrt{E}$.

4 Attention

4.1 Multihead Self Attention (MHA)

Consider an input tensor $\mathbf{X} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$ to an attention layer, where B is the batch size, L is the sequence length, and d_{model} is the model dimension.

1. The first step involves computing queries, keys, and values. We have three matrices, $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and simultaneously perform the following operations

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_q; \quad \mathbf{K} = \mathbf{X}\mathbf{W}_k; \quad \mathbf{V} = \mathbf{X}\mathbf{W}_v. \tag{5}$$

   These operations are vectorized, meaning that for each sequence b in the batch of size B, we do

$$\mathbf{Q}_b = \mathbf{X}_b \mathbf{W}_q \quad \forall b \in [B].$$

   $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are trainable parameters shared across the entire batch. The resulting $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ have the shape $\mathbb{R}^{B \times L \times d_{\text{model}}}$.

2. Next, for multihead attention, we reshape $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ from shape $\mathbb{R}^{B \times L \times d_{\text{model}}}$ into $\mathbb{R}^{B \times H \times L \times d_{\text{head}}}$, where H is the number of attention heads and d_{head} is the dimension of each head. To achieve this, we first divide d_{model} into H heads, resulting in shapes of $\mathbb{R}^{B \times L \times H \times d_{\text{head}}}$. Then we rearrange into $\mathbb{R}^{B \times H \times L \times d_{\text{head}}}$. Conceptually, each head uses a subset of dimensions from d_{model} to compute scores between queries and keys along the sequence dimension L. We will use the following notations $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h$ to denote the per head tensor in $R^{B \times 1 \times L \times d_{\text{head}}}$ for each head $h \in [H]$.

3. In the next step, we perform the attention calculation:

$$\begin{aligned} \mathbf{S}_h &:= \text{Scores}_h(\mathbf{Q}_h, \mathbf{K}_h) = \frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_{\text{head}}}} + \mathbf{M} \\ \mathbf{A}_h &:= \text{Attention}_h(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\mathbf{S}_h\right) \mathbf{V}_h, \quad \forall h \in [H]. \end{aligned}$$

   The scaled multiplication of $\mathbf{Q}_h$ and $\mathbf{K}_h^\top$ is vectorized, resulting in $\mathbf{S}_h \in \mathbb{R}^{B \times 1 \times L \times L}$ and $\mathbf{S} \in \mathbb{R}^{B \times H \times L \times L}$.[1] Optionally, we could use a mask matrix to mask out certain tokens, an example would be the causual self attention. To stabilize the gradients, we element-wise divide raw scores by $\sqrt{d_{\text{head}}}$. This scaling choice can be justified because each element of $\mathbf{Q}_h \mathbf{K}_h^\top$ represents a dot product between vectors of dimension d_{model}. The variance of this dot product scales as $\text{Var}\left(\langle q_h, k_h \rangle\right) \sim d_{\text{model}} \sigma_q^2 \sigma_k^2$. Since variance scales quadratically, we divide by $\sqrt{d_{\text{head}}}$. The softmax operation turns the scores after masking into probabilities, along the last dimension.[2] Imagine $z = [z_1, \ldots, z_L] \in \mathbb{R}^L$ is a row vector, then essentially, softmax defines the operation:

$$\sigma(z)_i := \frac{e^{z_i}}{\sum_{j=1}^{L} e^{z_j}}. \tag{6}$$

   Sometimes we use a numerically stable version to replace it

$$\tilde{\sigma}(z) := \frac{e^{z_i - \max(z)}}{\sum_{j=1}^{L} e^{z_j - \max(z)}}. \tag{7}$$

   It is worth mentioning that in single head attention (scaled dot product), the complexity of computation is $\mathcal{O}\left(BL^2 d_{\text{model}}\right)$, while for multihead attention, it is the same since we do $\mathcal{O}\left(H \times BL^2 d_{\text{head}}\right) = \mathcal{O}\left(BL^2 d_{\text{model}}\right)$.

[1] Here $\mathbf{S}$ is the stack of $\mathbf{S}_h$ along dimension H.

[2] This is to say that for each $L \times L$ matrix, we softmax every row.

4. Finally, we concatenate and mix attention outputs from all heads. Concatenation involves first transposing $\mathbf{A}_h$ to $\mathbb{R}^{B\times L\times H\times d_{\text{head}}}$ and then merging the last two dimensions into $\mathbf{A} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$. This concatenated result is projected using a matrix $\mathbf{W}_O$, as follows:

$$\text{MHA}(\mathbf{X}) = \mathbf{A}\mathbf{W}_O. \tag{8}$$

The final output retains the shape $\mathbb{R}^{B\times L\times d_{\text{model}}}$.

There a bunch of reasons why we are using multi heads instead of scaled dot product attention.

- **Diversity of learned attention patterns**: Each head learns different attention patterns in parallel. A single attention head computes only one set of attention scores.
- **Subspace specialization**: Instead of operating in d_{model}, each head projects to a lower dimension subspace d_{head}. This suggests that each head operates in a distinct feature subspace.
- **Improved gradient flow and representation mixing**: Independent paths improve gradient flow and richness of learned representations.

Notice that the computational cost are the **SAME**!

4.2 Multi Query Attention (MQA)

In MQA, different heads have its own query, but share the same key and value. Specifically, for a head h, we have

$$\begin{aligned}\mathbf{S}_h &:= \text{Scores}(\mathbf{Q}_h, \mathbf{K}) = \frac{\mathbf{Q}_h\mathbf{K}^\top}{\sqrt{d_{\text{head}}}} + \mathbf{M}\\ \mathbf{A}_h &:= \text{Attention}_h(\mathbf{Q}_h, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{S}_h)\mathbf{V}, \quad \forall h \in [H].\end{aligned}$$

This means that for each head h, we have a separate $\mathbf{Q}_h \in \mathbb{R}^{B\times 1\times L\times d_{\text{head}}}$ and shared $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{B\times 1\times L\times d_{\text{head}}}$.

Compared to standrad MHA, MQA has the following features:

- **Reduced parameter count**: Each head has its own query only, shared key and value.
- **Smaller activation size (memory usage)**: Now $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{B\times 1\times L\times d_{\text{head}}}$, so the activation size is smaller.
- **Reduced KV cache (fast inferencing)**: For transformer based models such as GPT, we generate text one token at a time. To avoid recomputing attention over all previous tokens on every step, we cache k, v (key and value vectors) for all previously seen tokens. Specifically, in standard MHA, for each layer and token generated, we neeed $2BHLd_{\text{head}}$ for cached $\mathbf{K}, \mathbf{V}$. In MQA, we share $\mathbf{K}$ and $\mathbf{V}$ so that the cost becomes $2BLd_{\text{head}}$.
- **Minimal accuracy loss.** Used in GPT-3.5, PaLM, LLaMA, etc.

4.3 KV cache

The motivation for KV caching is to enable efficient inference — both in terms of compute time and memory bandwidth. At inference time only, autoregressive models input a sequence of tokens $\{x_t, \ldots, x_{t+L-1}\}$ to generate the next token x_{t+L}. To avoid recomputing key and value vectors for all previous tokens every time, we cache the k, v pairs corresponding to the tokens $x_t \ldots, x_{t+L-1}$ in the forward pass. Then, when generating x_{t+L+1}, we can reuse the vectors for cached $x_{t+1}, \ldots, x_{t+L-1}$ and we only need to compute k, v for x_{t+L}. This mechanism is known as the **KV cache**.

4.4 Grouped Query Attention (GQA)

GQA is like an interpolation between MQA and MHA, where we ask groups of heads to share $\mathbf{K}, \mathbf{V}$. Specifically, let $g(h)$ be a function that maps a head h to its corrsponding group index, then we have

$$\mathbf{S}_h := \text{Scores}(\mathbf{Q}_h, \mathbf{K}_{g(h)}) = \frac{\mathbf{Q}_h \mathbf{K}_{g(h)}^\top}{\sqrt{d_{\text{head}}}} + \mathbf{M}$$
$$\mathbf{A}_h := \text{Attention}_h(\mathbf{Q}_h, \mathbf{K}_{g(h)}, \mathbf{V}_{g(h)}) = \text{softmax}(\mathbf{S}_h)\mathbf{V}_{g(h)}, \quad \forall h \in [H].$$

Benefits:

1. Less memory than MHA.
2. Flexible compute/memory tradeoff by controlling the number of k, v heads.

It is used in LLaMA 2 and Mistral.

4.5 Multihead Latent Attention (MLA)

Before we go into details, we need to first differentiate between self attention and cross attention.

- Self attention is the case when Q, K, V comes from the same input sequence. Its typically used in encoder blocks of BERT, GPT, LLaMA, etc, and decoder blocks in GPT, T5, etc.
- Cross attention refers to the case when Q comes from one sequence but K, V comes from another sequence. It is typically used in the case that decoder attends to encoder outputs (T5, BART), and the case of vision language models where text attends to image, and perceiver-style latent attention.

We can formulate cross attention in the following way: Let $\mathbf{Z} \in \mathbb{R}^{B \times M \times d_{\text{model}}}$ (expanded from $\mathbb{R}^{1 \times M \times d_{\text{model}}}$.) be a target sequence (queries) and $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ be a source sequence (keys and values),

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}_q; \quad \mathbf{K} = \mathbf{X}\mathbf{W}_k; \quad \mathbf{V} = \mathbf{X}\mathbf{W}_v. \tag{9}$$

Notice that M could be different than L. In the case of cross-attention with a latent array, we often have, $M << L$, which significantly reduces computational cost by avoiding full self-attention over the entire input sequence. We then compute the attention scores and softmax:

$$\mathbf{S}_h := \text{Scores}(\mathbf{Q}_h, \mathbf{K}_h) = \frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_{\text{head}}}} \tag{10}$$
$$\mathbf{A}_h := \text{Attention}_h(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}(\mathbf{S}_h)\mathbf{V}_h, \quad \forall h \in [H]. \tag{11}$$

Notice that in this case $\mathbf{S}_h \in \mathbb{R}^{B \times 1 \times M \times L}$ and $\mathbf{A}_h \in \mathbb{R}^{B \times 1 \times M \times d_{\text{head}}}$ for head $h \in [H]$. After concatenation, we result in $\mathbf{A} \in \mathbb{R}^{B, M \times d_{\text{model}}}$, which is like we are focusing on a smaller sequence. In MLA, latent vector it self is a learnable parameter and shared accross a batch. These latents act like information bottleneck that extract useful features from the long input $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$.

4.6 Latent Transformer Block

This is a key design of Perceiver (2021, DeepMind), Set Transformer and efficient transformers for long inputs (e.g., audio, video, documents). Essentially it can be viewed as cross attention followed by latent self-attention. Mathematically speaking, we are giving a vector $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$. First we are using the

latent vector $\mathbf{Z} \in \mathbb{R}^{1\times M\times d_{\text{model}}}$ (expanded accross batch dimension.) and the input $\mathbf{X} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$, we have

$$\mathbf{A}_1 = \text{Attention}(\mathbf{X}_q = \mathbf{Z}, \mathbf{X}_k = \mathbf{X}, \mathbf{X}_v = \mathbf{X}),$$

which essentially asks "What should I learn from all of you tokens?". After this step: each latent now contains information extracted from the input. Notice that now $\mathbf{A}_1 \in \mathbb{R}^{B\times M\times d_{\text{model}}}$ is a compressed representation of $\mathbf{X}$, extracted by the latent array. Then we do normal self attention on the latent variable:

$$\mathbf{A}_2 = \text{Attention}(\mathbf{X}_q = \mathbf{A}_1, \mathbf{X}_k = \mathbf{A}_1, \mathbf{X}_v = \mathbf{A}_1)$$

where each latent vector is allowed to look at other latents, share what they learned and refine itself.

```
for each block:
  z = z + CrossAttention(q ← z, k ← x, v ← x)
  z = z + SelfAttention(q ← z, k ← z, v ← z)
```

Before we actually feed the $\mathbf{Z}$ and $\mathbf{X}$ into the attention block and the final feed forward layer, we first do normlization (LayerNorm in the case of my code).

4.7 Pre- and Post- Normlization

In general, there are two ways of doing layer normlization, Post-LN and Pre-LN. In the original implementation of transformer, post-LN is used. However, pre-LN has become the modern default, which is used in GPT-2/3/4, T5, LLaMA, PaLM, Perceiver, etc. The benefits of using pre-LN includes the follows:

- **(Help gradient flow & increasing training stability)**: In a deep stack, residual paths carry the untouched signal forward. With Pre-LN, those residual paths also carry unit-variance, zero-mean activations (because they are already normalized). That keeps gradients well-scaled and prevents the exploding / vanishing issues that appeared when stacking 24 - 100+ layers with Post-LN. Empirically, Pre-LN lets you train hundreds (even thousands) of layers with a stable learning rate schedule, whereas Post-LN often needed warm-up tricks or gradient clipping.

- **(Easier optimization of very long sequences)**: Cross-entropy loss is applied after the final LayerNorm. With Post-LN every sub-layer's output is renormalized, the network must constantly "undo" those shifts. Pre-LN leaves the residual branch untouched, so the model can accumulate information across time steps or tokens without repeatedly rescaling it.

- **(Faster convergence)**: Many ablations show 1.3 - 1.5x faster convergence for GPT/T5 style models when switching from Post-LN $\rightarrow$ Pre-LN. This is because every tensor that flows straight down the stack (both forward activations and backward gradients through the residual skip) has mean 0 and variance 1, which helps stabilize second-moment estimate quickly for Adam.

- **(Safer with half-precision / mixed-precision)**: Normalizing before the high-variance matrix multiplications keeps activations in a narrower numeric range, reducing overflow/underflow risk in FP16/BF16 training.

LayerNorm: Mathematically speaking, consider an input $\mathbf{X}$ in the space $\mathbb{R}^{B\times L\times d_{\text{model}}}$, for the l-th token in the b-th batch

$$\mathbf{L}[b, l, :] = \text{LayerNorm}(\mathbf{X}[b, l, :]) = \gamma \cdot \frac{X[b, l, :] - \mu_{n,l}}{\sqrt{\sigma_{b,l}^2} + \epsilon} + \beta,$$

where

$$\mu_{b,l} = \frac{1}{d_{\text{model}}} \sum_{i=1}^{d_{\text{model}}} \mathbf{X}[b,l,i], \quad \sigma_{b,l}^2 = \frac{1}{d_{\text{model}}} \sum_{i=1}^{d_{\text{model}}} \left(\mathbf{X}[b,l,i] - \mu_{b,l}\right)^2,$$

$\gamma, \beta \in \mathbb{R}^{d_{\text{model}}}$ are learned shift and scale vectors. Notice that statistics are computed per sample, per position, no batch coupling, so the network behaves the same in training and inference and is robust to batch-size 1. Its benefits includes:

- **(Zero-mean, unit-var features).** Keeps dot-products in a predictable range, resulting in stable softmax gradients.
- **(Identical behaviour in training / inferencing).** Important for autoregressive generation where batch size changes.
- **(Works with any sequence length).** No running-average statistics needed.

In a simpler form, layernorm can be written as

$$\mathbf{L} = \gamma \odot \frac{\mathbf{X} - \mu}{\sqrt{\sigma + \epsilon}} + \beta,$$

where all operations are elementwise. Notice that $\gamma, \beta \in \mathbb{R}^{d_{\text{model}}}$ are learnable parameters.

4.8 Dropout

During Training: This refers to During training we randomly set a fraction of the sub-layer output activations to zero and scale the survivors. Specifically, for $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ and the drop out rate $0 < p < 1$, we sample a binary traning mask $\mathbf{M} \in \{0,1\}^{B \times L \times d_{\text{model}}}$, using the Bernoulli($q$) distribution where $q = 1 - p$, we then apply

$$\widehat{\mathbf{X}} = \frac{1}{q} \mathbf{X} \odot \mathbf{M}.$$

q is referred to as the keep probability, and we are trying to make $\mathbb{E}\left[\widehat{\mathbf{X}}\right] = \mathbf{X}$.

During Inferencing: At inference time the mask is removed and $\mathbf{X}$ passes through unchanged. Code:

```
torch.nn.Dropout(p=p)
```

4.9 DCA

This one seems to be less well known, on medical image tasks.

4.10 Linear Attention

Let us first look at the asymptotic time complexity and memory complexity of attention.

Operation	**Complexity**
Projections $\mathbf{X} \mapsto (\mathbf{Q},\mathbf{K},\mathbf{V})$	$\mathcal{O}\left(BLd_{\text{model}}^2\right)$
Score computation $\mathbf{QK}^\top$ (per head)	$\mathcal{O}\left(BHL^2 d_{\text{head}}\right)$
Softmax over scores	$\mathcal{O}\left(BHL^2\right)$
Weighted sum $\mathrm{softmax}(\mathbf{QK}^\top)\mathbf{V}$	$\mathcal{O}\left(BHL^2 d_{\text{head}}\right)$
Output projection (concat heads $\to d_{\text{model}}$)	$\mathcal{O}\left(BLd_{\text{model}}^2\right)$

Table 1: Computation complexity of the main steps in a standard self-attention block. Here B = batch size, L = sequence length, H = number of heads, d_{head} = head dimension, and $d_{\text{model}} = H \cdot d_{\text{head}}$.

For naive self-attention, we can see that as sequence length becomes longer, the dominating term would be $\mathcal{O}(BHL^2 d_{\text{head}})$, which becomes $\mathcal{O}(BHL_q L_k d_{\text{head}})$, which is quadratic in L.

Memory consumption:

- Store $\mathbf{Q}, \mathbf{K}$ and $\mathbf{V}$: $\mathcal{O}(BLd_{\text{model}})$
- Store attention scores/probs: $\mathcal{O}(BHL^2)$
- Output activations for backprob add similar terms.
- By FlashAttention / PyTorch SDPA (streaming): we keep compute the same $\mathcal{O}(BHL^2 d_{\text{head}})$ but reduce memory from $\mathcal{O}(BHL^2)$ to roughly $\mathcal{O}(BLd_{\text{model}})$ by not materializing the full $L \times L$ matrix.

Scenario	**Time (leading term)**	**Memory (leading term)**
Self-attention (train)	$O(BHL^2 d_{\text{head}})$	$O(BHL^2)$
Cross-attention	$O(BHL_q L_k d_{\text{head}})$	$O(BHL_q L_k)$
With FlashAttention / SDPA	same time	$O(BLd_{\text{model}})$
Autoregressive decoding (per token at step t)	$O(Htd_{\text{head}})$	cache $O(Ld_{\text{model}})$

Table 2: Asymptotic compute and memory complexity for different attention scenarios. Here B = batch size, L = sequence length (train), H = number of heads, d_{head} = per-head dimension, $d_{\text{model}} = H \cdot d_{\text{head}}$, L_q/L_k = query/key lengths for cross-attention, and t = current decoding step in autoregressive inference. Autoregressive decoding is with KV cache.

Linear attention: The linear attention is introduced to avoid the quadratic dependence using attention kernels which allow us to obtain linear dependence. Note that the attention of token i is given by

$$\text{attn}_i = \frac{\sum_{j=1}^{L} \exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_{\text{head}}}}\right) \mathbf{v}_j}{\sum_{j=1}^{L} \exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_{\text{head}}}}\right)}.$$

To compute it, we need to perform $\mathbf{QK}^\top$ and apply softmax rowwise. We introduce the kernel trick just as we did in SVM, imagine we can write

$$\exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_{\text{head}}}}\right) \approx \phi(\mathbf{q}_i)^\top \phi(\mathbf{k}_j),$$

for some feature map $\phi : \mathbb{R}^{d_{\text{head}}} \mapsto \mathbb{R}^{d_\phi}$ with non-negative outputs, then we have

$$\begin{aligned}
\text{Numerator:} \qquad & \sum_{j=1}^{L} \phi(\mathbf{q}_i)^\top \phi(\mathbf{k}_j)\mathbf{v}_j^\top = \left(\sum_{j=1}^{L} \mathbf{v}_j \phi(\mathbf{k}_j)^\top\right) \cdot \phi(\mathbf{q}_i), \\
\text{Denominator:} \qquad & \sum_{j=1}^{L} \phi(\mathbf{q}_i)^\top \phi(\mathbf{k}_j) = \left(\sum_{j=1}^{L} \phi(\mathbf{k}_j)^\top\right) \cdot \phi(\mathbf{q}_i).
\end{aligned}$$

Now denote $\mathbf{S_V} := \sum_{j=1}^{L} \mathbf{v}_j \phi(\mathbf{k}_j)^\top \in \mathbb{R}^{d_{\text{head}} \times d_\phi}$, and $\mathbf{S_K} := \sum_{j=1}^{L} \phi(\mathbf{k}_j)^\top \in \mathbb{R}^{1 \times d_\phi}$, we would have

$$\text{attn}_i = \frac{\mathbf{S_V}\phi(\mathbf{q}_i)}{\mathbf{S_K}\phi(q_i) + \epsilon}.$$

Notice that we may compute $\mathbf{S_V}$ and $\mathbf{S_K}$ in an linear fasion dependent on $\mathcal{O}(L)$ (since there are in total L terms). In this way we get a better dependence.

Causal (autoregressive) case: All we have to change is that we maintain running sum: at time (token) t:

$$\mathbf{S}_\mathbf{V}^{(t)} := \sum_{j \le t} \mathbf{v}_j \phi(\mathbf{k}_j)^\top \qquad \text{and} \qquad \mathbf{S}_\mathbf{K}^{(t)} := \sum_{j \le t} \phi(\mathbf{k}_j)^\top.$$

and

$$\text{attn}_t = \frac{\mathbf{S}_\mathbf{V}^{(t)}\phi(\mathbf{q}_i)}{\mathbf{S}_\mathbf{K}^{(t)}\phi(q_i) + \epsilon}.$$

There are a bunch of common choices for the feature maps,

1. **Positive feature maps (kernel trick for softmax)**: (1) ELU + 1 ($\phi(x) = \text{ELU}(x) + 1$), (2) ReLU / Square ReLU, (3) Exp with scaling $\phi(x) = \exp\left(\frac{x}{d_{\text{head}}}\right)$, (4) Random Fourier features ...
2. **Other maps**: Orthogonal / normalized maps; Exponential-normalized maps ...

4.11 Sparse Attention

5 Normalizations

Besides the layer norm mentioned in the previous chapter, there are other norms used quite often.

5.1 BatchNorm

Batch norm is rarely used in language modeling especially in modern architectures. It is not sequence or position aware, and it requires consistent statistics over a batch, with variable-length sequences in language modeling, the batch statistics can be unstable and unreliable. Furthermore, it mixes information from each sequences, but ideally we want to keep it separate. However, it is great for vision tasks

Mathematically, given an input $\mathbf{X} \in \mathbb{R}^{B\times d}$, for each feature j, we have

$$\mu_j = \frac{1}{B}\sum_{i=1}^{B}\mathbf{X}[i,j], \quad \sigma_j^2 = \frac{1}{B}\sum_{i=1}^{B}(\mathbf{X}[i,j]-\mu_j)^2.$$

Then each value is normlized in a way that

$$\hat{X}[i,j] = \gamma[j]\frac{X[i,j]-\mu_j}{\sqrt{\sigma_j^2+\epsilon}} + \beta[j],$$

where $\gamma, \beta \in \mathbb{R}^d$ are learnable parameters per feature / channel.

If we are given an input 2D/Convolutional data $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, we have

$$\mu_c = \frac{1}{BHW}\sum_{b,h,w}\mathbf{X}[b,c,h,w], \quad \sigma_c^2 = \frac{1}{BHW}\sum_{b,h,w}(\mathbf{X}[b,c,h,w]-\mu_c)^2,$$

basically, we are computing mean and variance per channel. Notice that **LayerNorm is still the default in most Vision Transformer (ViT-style) backbones**. In pratice, when we do the training, we can simply use the batch mean and batch variance, but in the inference stage, we have to use the running statistics (momentum apply to generate running mean and running variance). This is because that during inference, the batchsize might be 1 or different from that is used in the training phase.

In CNN, sometimes GroupNorm replaces BatchNorm because it's batch-size independent and mixes well with vision features.

5.2 RMSNorm

Its name is Root Mean Square Layer Normalization, which has become popular in some recent LLMs, especially as a lighter and sometimes more numerically stable alternative to LayerNorm. It normalizes only by root mean square of the features[3] instead of mean and variance. Given $X \in \mathbb{R}^{B\times L\times d_{\text{model}}}$

$$\text{RMS}_{b,l} = \sqrt{\frac{1}{d_{\text{model}}}\sum_{i=1}^{d_{\text{model}}}\mathbf{X}[b,l,i]^2+\epsilon}$$

Then, each element in $\mathbf{X}$ is normalized by

$$\hat{X}[b,l,i] = \frac{\mathbf{X}[b,l,i]}{\text{RMS}_{b,l}}\cdot\gamma_i, \qquad \forall i \in [d],$$

where γ_i is the i-th component of the learnable scaling factor $\gamma \in \mathbb{R}^{d_{\text{model}}}$.

LLaMA (and most of Meta-Llama 1/2/3 checkpoints) swap the original LayerNorm for pre-norm RMSNorm.

Below is a complete list of RMSNorm's features.

[3] Features in this context refer to the elements in dimension d_{model}

- **(Rescales by RMS only & scaling only)**.
- **(Compute & Memory efficiency)**. 30 % cheaper per norm op; overall 2-6 % faster end-to-end in large LLMs.
- **(Numerically stable)**. Works well with very deep pre-norm Transformers.

When does it shine: (1) Ultra-deep LLMs, (2) Inference-first or edge deployments, (3) Pre-norm architectures.

RMSNorm really does skip the "subtract-the-mean" step, so a single normalization needs one less reduction-operation and a bit less memory traffic (removing the mean saves one vector reduction, one broadcast, and one add.). This skip would not affect training stability, because pre-norm residuals absorb the offset: in modern transformers RMSNorm sits before each residual branch, any mean shift can be compensated by the next linear layer's bias. To see this: consider the following procedure, (assuming $x \in \mathbb{R}^d$)

$$\boxed{x} \xrightarrow{\text{RMSNorm}} \boxed{y = \gamma \cdot \frac{x}{\|x\|_{\text{RMS}}}} \xrightarrow{\mathbf{W},b} \boxed{u = \mathbf{W}y + b} \rightarrow \boxed{z = x + u}$$

y indeed does not have mean zero, denote it as μ_y, since the next operation is affine, we can rewrite it as

$$u = \mathbf{W}(y - \mu_y \mathbf{1}) + (b + \mathbf{W}\mu_y \mathbf{1}),$$

which means that it is equivalent to adjusting the bias. During training, back-prop will simply nudge b so the network learns whatever overall shift is optimal, it does not care where that shift originates. After that u is added to the original x, but the drift in μ_y will not accumulate unchecked because: (1) The residual path still carries the original, unshifted activations, (2) The next block starts with another RMSNorm, which rescales its input (including any offset) back to a controlled RMS before new computations begin.

6 Positional Embedding

6.1 Sinusoidal Positional Embedding

We just add the postion matrix $\mathbf{P} \in \mathbb{R}^{L \times d_{\text{model}}}$ to the input embedding $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, where L is the maximum sequence length. To be specific, we construct $\mathbf{P}$ as follows:

$$\mathbf{P}[l, 2i] = \sin\left(\frac{l}{10000^{\frac{2i}{d_{\text{model}}}}}\right), \quad \mathbf{P}[l, 2i+1] = \cos\left(\frac{l}{10000^{\frac{2i}{d_{\text{model}}}}}\right),$$

where $l \in [0, L-1]$ and $i \in [0, \frac{d_{\text{model}}}{2} - 1]$. This way, each dimension of the positional encoding corresponds to a sinusoid of different frequency. The wavelengths form a geometric progression. We just perform

$$\mathbf{X}' = \mathbf{X} + \mathbf{P},$$

after constructing $\mathbf{P}$.

6.2 RoPE (Rotary Positional Embedding)

RoPE, instead, directly encodes relative position information into the attention mechanism by **rotating the query and key vectors** based on their positions. Let us denote $\mathbf{Q} \in \mathbb{R}^{L \times d_{\text{head}}}$ and $\mathbf{K}^{L \times d_{\text{head}}}$ as the query matrix and the key matrix (We temporarily hide the B, H dimensions.). For each query vector

$$\mathbf{q}_l = \mathbf{Q}[l, :] = (\mathbf{q}_{l,1}, \ldots, \mathbf{q}_{l,d_{\text{head}}}) \in \mathbb{R}^{d_{\text{head}}}, \tag{12}$$

we group the dimensions into pairs

$$(\mathbf{q}_{l,1}, \mathbf{q}_{l,2}), (\mathbf{q}_{l,3}, \mathbf{q}_{l,4}), \ldots, (\mathbf{q}_{l,d_{\text{head}}-1}, \mathbf{q}_{l,d_{\text{head}}}), \tag{13}$$

assunimg d_{head} is even. This gives $\frac{d_{\text{head}}}{2}$ pairs and the same for $\mathbf{K}$. We now define angle frequencies

$$\theta_i = 10000^{-\frac{2(i-1)}{d_{\text{head}}}}, \qquad \text{for } i = 1,2,\ldots \frac{d_{\text{head}}}{2}.$$

We let each pair $\mathbf{q}_i^{(l)} = (\mathbf{q}_{l,2i-1}, \mathbf{q}_{l,2i})$ be rotated so that

$$\begin{pmatrix} \mathbf{q}'_{l,2i-1} \\ \mathbf{q}'_{l,2i} \end{pmatrix} = \begin{pmatrix} \cos(l\theta_i), & -\sin(l\theta_i) \\ \sin(l\theta_i), & \cos(l\theta_i) \end{pmatrix} \begin{pmatrix} \mathbf{q}_{l,2i-1} \\ \mathbf{q}_{l,2i} \end{pmatrix} = \mathbf{R}_i(l)\mathbf{q}_i^{(l)}.$$

Notice that

$$\begin{pmatrix} \cos(s\theta_i), & -\sin(s\theta_i) \\ \sin(s\theta_i), & \cos(s\theta_i) \end{pmatrix}^\top \cdot \begin{pmatrix} \cos(t\theta_i), & -\sin(t\theta_i) \\ \sin(t\theta_i), & \cos(t\theta_i) \end{pmatrix} = \begin{pmatrix} \cos((t-s)\theta_i), & -\sin((t-s)\theta_i) \\ \sin((t-s)\theta_i), & \cos((t-s)\theta_i) \end{pmatrix}$$

Now define the rotated vector:

$$\tilde{\mathbf{q}}_l = \tilde{\mathbf{Q}}[l,:] = \mathbf{R}(l)\mathbf{Q}[l,:], \qquad \tilde{\mathbf{k}}_l = \tilde{\mathbf{K}}[l,:] = \mathbf{R}(l)\mathbf{K}[l,:], \tag{14}$$

where $\mathbf{R}(l)$ is a block diagonal rotation matrix built from $d_{\text{head}}/2$ rotation blocks, this can be intepreted as if we rotate clockwisely for angle t and counter-clockwisely for angle s, the combination of the result is to rotate clockwisely for angle $t-s$.

Difference: The core difference between RoPE and Sinusoidal is where and how the position information enters the model. For sinusoidal, we create a position vector for each index p and we add it to the token embedding at the input.

$$\mathbf{x}'_p = \mathbf{x}_p + \text{PE}(p).$$

So the embedding is additive and usually treated as an absolute postion encoding. And it is usually applied before the transformer layers.

But for RoPE, we do not add a position vector to the token embedding, instead, we apply a position-dependent rotation to the query and key vectors inside the attention.

With sinusoidal embedding, position is given to the model as an input feature, while for RoPE, position is built directly into the attention score, which makes RoPE naturally encodes relative position differences.

Key result: When computing attention, the score between query at position s and key at position t is

$$\begin{aligned} \tilde{\mathbf{Q}}[s,:]^\top \tilde{\mathbf{K}}[t,:] = \sum_{i=1}^{d_{\text{head}}/2} \left(\tilde{\mathbf{q}}_i^{(s)}\right)^\top \cdot \tilde{\mathbf{k}}_i^{(t)} &= \sum_{i=1}^{d_{\text{head}}/2} \left(\mathbf{q}_i^{(s)}\right)^\top \mathbf{R}_i(s)^\top \mathbf{R}_i(t)\mathbf{k}_i^{(t)} \\ &= \sum_{i=1}^{d_{\text{head}}/2} \left(\mathbf{q}_i^{(s)}\right)^\top \mathbf{R}_i(t-s)\mathbf{k}_i^{(t)}. \end{aligned} \tag{15}$$

As one can see, the attention score only depends on the relative position $t-s$, this suggests that the attention mechanism in this case is dependent on relative position only, which is a desirable property.[4]

[4] If d_{head} is odd, then there is no one pairing with the last dimension, we can simply leave it unchanged.

Notice also that we are essentially spreading the frequencies by varying θ_i, lower index pairs have high-frequency oscillations which allows them to capture small relative displacements, while higher index pairs have low-frequency oscillations, which allows them to capture large relative displacements. The Fourier encoding of positions with exponentially decreasing frequencies means the set of basis functions spans a huge range, covering both short-term periodicity and long-range uniqueness, which is why the maximum context length of sinusoidal/RoPE embeddings is effectively determined by the lowest frequency (largest wavelength). This is similar to the sinusoidal positional embedding, but here we are directly applying it to the query and key vector.

Dynamic position range: One drawback of learned absolute positional embeddings is that they are fixed to the maximum sequence length used during training, and extrapolation to longer sequences is poor. RoPE, like sinusoidal embeddings, can generalize to longer sequences because the rotation matrices can be computed for any position index. However, the choice of base frequency (10000 in the original paper) determines the effective range of relative positions that can be represented before the rotations start to repeat.

To be clear, at position p, for the i-th pair, the angle is $p\theta_i = p \cdot 10000^{-\frac{2(i-1)}{d_{\text{head}}}} = p \cdot \omega_i$, we say ω_i is the frequency and $\lambda_i = \frac{2\pi}{\omega_i}$ is the wavelength, ω_i decreases exponentially with i, while λ_i increases exponentially with i. For those pairs with small i, they have high frequency (small wavelength), which means they can only represent small relative positions before the rotations start to repeat, this is typically a few thousands tokens with base 10000, which is the reason why naive extrapolation beyond the trained length often fails.

In practice, there are several strategies to adjust to the desired context length

- **Position interpolation (NTK(Neural Tangent Kernel)-aware scaling).** Scale positions before feeding them to RoPE, let p' be the new position, and p be the original index, we have $p' = \frac{p}{L_{\text{new}}} \cdot L_{\text{train}}$.
- **Change base frequency.** Instead of 10000, use a larger base to spread out the frequencies more. Larger bases = slower frequencies = longer usable context, but worse local resolution.
- **Exponential extrapolation (YaRN, etc.)** Instead of letting the angle grow linearly with p, modulate it smoothly so the growth is slower and more controlled at large positions. An example of YaRN is to rescale the base frequencies non-uniformly across dimensions.
- **Hybrid interpolation + fine-tuning. NOT FINISHED!!!**

7 Precisions

Here is a summary provided by Chat-GPT o3.

Format	Bits	Exp/Mant	Dynamic range†	ULP@1	Typical use
FP32	32	8/23	1.2×10^{-38}-3.4×10^{38}	2^{-23}	Master weights, optimiser
FP16 (IEEE)	16	5/10	6.1×10^{-5}-6.6×10^{4}	2^{-10}	Fwd/Bwd on Volta & T4
bfloat16	16	8/7	1.2×10^{-38}-3.4×10^{38}	2^{-7}	Default on A100/H100, TPU
TF32*	19	8/10	same as FP32	2^{-10}	GEMMs on Ampere

Table 3: Mid-/high-precision dtypes widely used during LLM training.†Smallest positive *normal* value to largest finite.*TF32 is a compute mode; tensors stored as FP32.

A floating point number in IEEE-style formats is stored as:

Format	Bits	Exp/Mant	Dynamic range†	ULP@1	Typical use
FP8-E4M3	8	4/3	9.2×10^{-8}-4.5×10^{2}	2^{-3}	Research train / fast inf. (H100)
FP8-E5M2	8	5/2	3.0×10^{-10}-5.7×10^{4}	2^{-2}	Wider range variant (H100)
INT8 / INT4	8/4	—	±127 / ±7	1 / $1/16$	Post-training inference quant.

Table 4: Very low-precision formats used for efficient training research (FP8) or deployment quantisation (INT).

```
Sign bit | Exponent bits | Mantissa (fraction) bits
```

1. **Sign bit**: (1 bit), 0 positive and 1 negative.
2. **Exponent bits**: control the scale (powers of 2). More exponent bits means wider dynamic range.
3. **Mantissa bits**: control the precision (how many distinct numbers you can represent between powers of 2). More mantissa bits means smaller ULP[5] (more precise).

A number is typically constructed in this way:

$$(-1)^{\text{sign}} \times (1.\text{mantissa bits}) \times 2^{\text{exponent}-\text{bias}}.$$

- **FP32** is used as the safe baseline in deep learning, it is almost always used for the master weigts[6] and optimizer state because it is numerically stable.
- **FP16** is used in forward/backward pass to save memory in Volta(Nvidia V100)/T4 mixed-precision training. Prone to overflow/underflow unless scaled (hence loss scaling).
- **Loss scaling** refers to the practice we multiply our loss by a large constant S before back propagation: $L_{\text{scaled}} = S \cdot L$, this scales up gradients which avoids gradient underflow. However, after computing the gradients, we divide them by S before updating the weights to restore correct magnitude. NVIDIA's AMP (Automatic Mixed Precision) does dynamic loss scaling, adjusting S automatically to prevent overflow.
- **bfloat16** is Brain floating point came from Google Brain, whose exp bits are same as **FP32**, but the mantissa bits are much fewer, this avoids the over/under-flow problem in **BF16**. This comes at a price of lower precision, but it is the default choice on A100 and H100.

  FP16 on Volta/T4 is fast but fragile, and requires loss scaling, which is why bfloat16 became popular. It enjoys the same range as FP32, but no scaling headaches.

  Its advantage includes: (i) no losss scaling needed to handle with over/under flow, (ii) half the memory of **FP32**, (iii) speed up in matmul on A100/H100, (iv) can used as a drop in for **FP32**, because the dynamic range is the same.

  For downsides: (a) precisiob loss, (b) optimizers often still keep master weights in **FP32** to avoid cumulative rounding error, (c) numerically sensitive operations such as softmax/ normalization, kernel still uses **FP32** internally.

[5] Unit in the Last Place: The smallest possible difference between two representable floating-point numbers around a given value

[6] Master weight refers to the full-precision copy of the model's parameter that we keep during mixed-precision training.

- **TF32**: NVIDIA introduced with Ampere GPUs (A100, RTX 3000 series), default in cuBLAS/cuDNN matmul on Ampere if you pass FP32 inputs. FP32's size and range, FP16's precision.
- **FP8-E4M3, FP8-E5M2**: We can now tell directly from their names that FP8-E4M3 has a narrower dynamic range at a higher precision. For FP8-E4M3, ULP@1 is $\frac{1}{2^3} = 0.125$ and for FP8-E5M2, the ULP@1 becomes $\frac{1}{2^2} = 0.25$ which is very coarse.

  E4M3: good for weights and activations that stay within a moderate range.

  E5M2: good for gradients or loss-related values that can swing wildly in magnitude.

How they (FP8) are used on H100: It can be applied both for training and inferencing.
During **training:**

1. Keep master weights in FP32 (like with FP16/bfloat16 training).
2. Cast activations/gradients to FP8 for GEMMs inside the forward/backward pass.
3. Apply per-tensor or per-channel **scaling** to map values into FP8's limited range.
4. Often mix E4M3 for forward activations, E5M2 for backward gradients.

During **Inferencing:** Post-training quantization to FP8 for ultra-fast inference with minimal memory footprint.

Scaling is not optional: FP8's numeric range is so tiny that, without actively scaling tensors before casting to FP8, we'll either hit overflow or underflow constantly. The fix is that we multiply the tensor by a scaling factor S before FP8 conversion,

$$x_{\text{scaled}} = x \times S.$$

We may choose S so that $\max x_{\text{scaled}}$ fits nicely into FP8's max normal value, and we store S as a separate FP32 number. We can later undo the the scaling after computation:

$$y = y_{\text{scaled}} \times S^{-1}.$$

This keeps numbers inside FP8's safe zone while still preserving the original magnitude relationship.

Per channel vs. per tensor: Often, there are **per channel** scaling and **per tensor** scaling. Let us image that we are doing a fully-connected layer, where we have this weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. Per channel scaling means that we have a scaling factor for each output feature (each row), so there will be d_{out} scalars, while for per tensor scaling, we only have 1 scaling factor. In this sense, per tensor scaling is simpler, but will be suboptimal is some channels are loud and the others are quiet, while per channel scaling has better precisions, especially in convolutional layers or transformers where variance differs accross each head / filter.

Automation on H100: Notice that there is automated scaling on NVIDIA H100, which tracks running max values for each tensor, and picks E4M3 (more precision) for forward activations, E5M2 (more range) for backward gradients. The engine applies scaling transparently, so we mostly see FP8 without manually tuning the scaling factor S.

As an example of scaling, let us consider the previous example of fully connected layer, where we are expected to do $\mathbf{O} = \mathbf{W}\mathbf{X}$, originally, both $\mathbf{W}$ and $\mathbf{X}$ are FP32 (master weights and full precision activations). Before GEMM[7], we independently scaled them to FP8 (to fit the range) by

$$\mathbf{X}_{\text{scaled}} = \mathbf{X} \cdot S_{\mathbf{X}}; \qquad \mathbf{W}_{\text{scaled}} = \mathbf{W} \cdot S_{\mathbf{W}},$$

then we performed the matmul in FP8,

$$\mathbf{O}_{\text{scaled}} = \text{FP8GEMM}(\mathbf{W}_{\text{scaled}}, \mathbf{X}_{\text{scaled}}).$$

To restore the original magnitude, we take advantage of the scaling factor $S_{\mathbf{X}}, S_{\mathbf{W}}$ who are FP32,

$$\mathbf{O} = \mathbf{O}_{\text{scaled}} \cdot \frac{1}{S_{\mathbf{X}} S_{\mathbf{W}}}.$$

8 Encoder and Decoder Structure

Encoder Typically, encodes looks like:

As we have discussed before, the Pre-LayerNorm fashion is the current state of the art as its training is more stable, as well as its convergence. It helps gradient flow, and deep model stability, does not requries learning rate warm up and works better with long context. It do has downside though, Pre-LN has a weaker normalizing effect so sometimes people add an extra final LayerNorm after the last layer for output stability ("final LN" in GPT models).

Decoder As a further explanation besides the figure: this $\mathbf{M}$ is the matrix of encoder hidden states produced by the last encoder layer. It is what the decoder attends to in cross attention. Let us image that an input $\mathbf{X} \in \mathbb{R}^{B \times L_{\text{enc}} \times d_{\text{model}}}$ is fed into the encoder blocks. After embedding and positional encoding, we have

$$\mathbf{H}^0 \in \mathbb{R}^{B \times L_{\text{enc}} \times d_{\text{model}}}.$$

Each encoder will do

$$\mathbf{H}^{k+1} = \text{Encoder}(\mathbf{H}^k) \in \mathbb{R}^{B \times L_{\text{enc}} \times d_{\text{model}}}, \quad \forall k \in [N_{\text{enc}}],$$

And $\mathbf{M}$ is exactly $\mathbf{M} = \mathbf{H}^{N_{\text{enc}}}$. As a result, for the cross attention layer: $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{B \times L_{\text{enc}} \times d_{\text{model}}}$ and $\mathbf{Q} \in \mathbb{R}^{B \times L_{\text{dec}} \times d_{\text{model}}}$, so the output of decoder block will always be $\mathbb{R}^{B \times L_{\text{dec}} \times d_{\text{model}}}$.

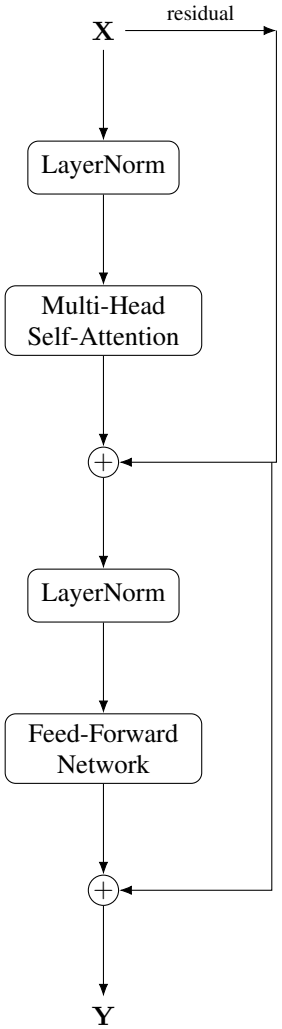
The two types of blocks are different:

- Encoder: Self attention only, no masking, aiming to build a contextual representation of the entire source sequence.
- Decoder: Masked self attention, which prevents seeing future tokens so that the generation is autoregressive. Cross attention, which lets each target position attend to the encoder's output $\mathbf{M}$ (source context) The purpose overall is to generate the target sequence one step at a time, with access to both past target tokens and the entire source sequence.

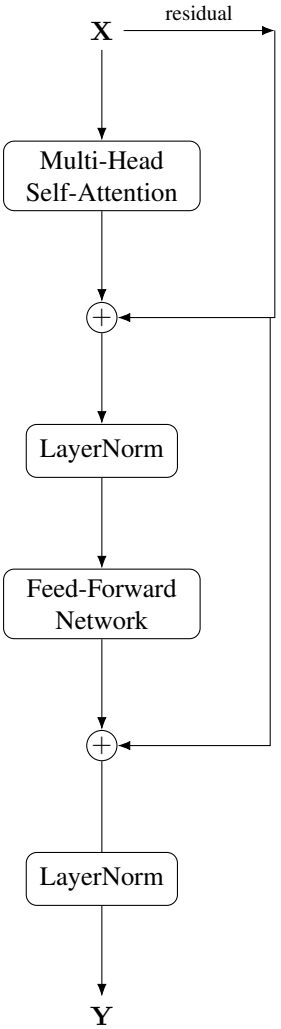
What happens if we are using one type?

- Encoder: leaking future target tokens during training (no causal mask), so the model wouldn't learn autoregressive generation.
- Decoder: useless masked self-attention in the encoder that blocks half the context for no reason, waste compute on a cross-attention sublayer when no encoder memory exists yet.

[7] GEneral Matrix-Matrix Multiplication, in the form of $\mathbf{D} = \alpha \cdot \mathbf{AB} + \beta\mathbf{C}$, where α, β are scalars. This is the workhorse of modern deep learning.



Pre-LayerNorm Transformer block



Post-LayerNorm Transformer block

The original seq2seq transformer: Things happens in two phases,

```
Source tokens ---> [Encoder stack] ----------------> M (memory)
                                                |
                                                v
Shifted target tokens ------> [Decoder stack] ---> Output logits
```

Encoder runs once on the full source, decoder runs once (training) or incrementally (inference), always starting with shifted target embeddings as its own input stream. The input of encoder is the source sequence $x = [x_1, x_2, ..., x_{L_{\text{enc}}}]$, after embedding we get $\mathbf{X}$, while the input of the decoder is the target sequence y so far, but shifted ($\hat{y}$) so that the model predicts the next token.

1. We take the original gold target sequence $y = [y_1, y_2, ..., y_{L_{\text{dec}}}]$.
2. Shift right by one and add the <BOS> special token, and obtain $\hat{y} = [\text{<BOS>}, y_1, ..., y_{L_{\text{dec}}-1}]$, we then embed to get $\hat{\mathbf{Y}}$ ans use it as an input to the decoder blocks.

8.1 Encoder only models

BERT, RoBERTa, DeBERTa, ELECTRA, Sentence-BERT (SBERT).

The architecture mostly are: • Stacks of encoder blocks only, • Full self attention, • Input sequence length stay fixed, • output contextualized embeddings for every token.

Purpose: Understand text: classification, regression, retrieval, token-level labeling (NER, POS tagging, QA span prediction). In those cases the model needs bidirectional context: token sees both left and right neighbors.

For many NLP tasks, we already have the full text and just need to analyze it, not generate it.

Advantages:

- **Better context capture:** every token attends to all others.
- **More efficient training** for non-generative tasks (no need to autoregress).
- **Easier fine-tuning** for classification tasks: just take the [CLS] embedding[8].

The goal of encoder-only models are to learn representations of the entire input sequence. Great for tasks like classification, NER, QA span extraction.

BERT pretraining in detail

1. **Input Preparation:**
   - **Tokenization:** Sentences are tokenized into WordPiece subwords. Special tokens are added such as [CLS] at the start (used for sequence level classification), [SEP] at the end to separate two sentences.
   - **Segment embeddings:** BERT was designed to handle pairs of sentences, this is to let the model know which tokens belong to which sentence, so we add a segment embedding on top of the token and positional embeddings. Sentence A tokens $\rightarrow$ segment embedding $\mathbf{0}$ ((vector for "sentence A").). Sentence B tokens $\rightarrow$ segment embedding $\mathbf{1}$ ((vector for "sentence B").).
   - **Position embeddings:** Added to encode order of tokens.
2. **Masked Language Modeling (MLM):**
   - **Random masking:** We select 15% of the input tokens for possible prediction. For each selected token, 80% of the time we replace it with [MASK], 10% of the time we replace it with a random word and 10% of the time we keep it unchanged. This trick prevents the model relies too heavily on [MASK] and improves robustness.
   - **Forward pass:** Feed the corrupted sentence into the encoder stack. For each masked position t, get hidden representation $\mathbf{h}_t$. Then we apply softmax over the vocabulary

     $$p_\theta(\mathbf{x}_t \mid \mathbf{x}_{\setminus t}) = \text{softmax}\left(\mathbf{W}\mathbf{h}_t\right),$$

     we then do cross-entropy only on masked positions

     $$\mathcal{L}_{\text{MLM}}(\theta) = -\sum_{t \in \text{Masked}} \log p_\theta(\mathbf{x}_t \mid \mathbf{x}_{\setminus t}),$$

     where $\mathbf{x}_{\setminus t}$ denotes the other positions than t.

[8] In BERT and similar models, we prepend a special token [CLS] ("classification") to the start of every input sequence before feeding it to the encoder. [CLS] has its own trainable embedding vector in the model's vocabulary, just like any word. It is treated as position 0 in the sequence and goes through all encoder layers along with the other tokens. After the final encoder layer, [CLS] has a contextualized vector $h \in \mathbb{R}^{d_{\text{model}}}$, which encodes information from the entire sequence (thanks to self-attention).

3. **Next Sentence Prediction (NSP):**

   - **Sentence pair construction:** 50% of the time, we set sentence B follows sentence A in corpus (label = isNext), and the other 50% of the time, sentence B is randomly sampled (label = notNext), we then construct them into the following format

     ```
     [CLS] Sentence A [SEP] Sentence B [SEP]
     ```

   - **Classification head:** we take the [CLS] embedding h_{CLS} from the final layers, and we pass through a classification layer

     $$y = \text{softmax}(\mathbf{W}_{\text{NSP}}\mathbf{h}_{\text{CLS}}),$$

     where y is the predicted probability output of the model, $y \in (0, 1)$, we will use $y^\star \in \{0, 1\}$ to denote the true label.

   - **Loss:** we use the binary classification loss,

     $$\mathcal{L}_{\text{NSP}}(\theta) = -y^\star \log y + (1 - y^\star) \log(1 - y),$$

     This is a classic binary cross entropy loss, which can be interpreted as follows, if the sentence is indeed in such a relation, then we maximize the log likelihood this is the case, otherwise if it is not in such a relation, then we maximize the log likelihood that this is not the case.

4. **Total Loss:** The two objectives are combined,

   $$\mathcal{L}(\theta) = \mathcal{L}_{\text{MLM}}(\theta) + \mathcal{L}_{\text{NSP}}(\theta),$$

   and we continue to optimize it.

5. **Extensions:** We may easily notice that BERT actually produces contextual embeddings for the entire sequence via [CLS] token or pooled output, so we can actually add a classification head to predict many kinds of relations. For examples: (i) Entailment / Contradiction (NLI) $\rightarrow$ [Premise] [SEP] [Hypothesis], (ii) Semantic similarity / Paraphrase $\rightarrow$ [Sentence1] [SEP] [Sentence2] (iii) Retrieval-style relevance $\rightarrow$ [Query] [SEP] [Document] (iv) Dialogue coherence $\rightarrow$ [Previous turn] [SEP] [Candidate response]

6. **Representations:** We note that using the [CLS] token, we can easily get a representation of the sequence (sentence) we are working with. A natural question arises, which is how we obtain such a representation for a document. In practice, if we are looking at short paragraphs (< 512 tokens: hard cap of vanilla BERT), the [CLS] token might be the right choice. For medium docs (2-3 pages), we may chunk it into 512 tokens and then using max/mean/attention pooling (learn weights to combine chunks). For larger docs, we may consider using a hierarchical BERT (imagine here sentences are tokens) or a long-document transformer.

8.2 Decoder only models

GPT family (GPT-2, GPT-3, GPT-4, LLaMA, Mistral, Falcon, etc.); BLOOM, OPT, Pythia; Code generation models (CodeLLaMA, StarCoder)

Architecture: • Stack of decoder blocks only • Causal self-attention (mask future positions) • **No cross-attention** • Input length = current sequence length during generation.

Purpose: Generate text: language modeling, code generation; model learns

$$P(\text{next token} \mid \text{prev tokens})$$

Advantages:

- **Simpler**: do not need an encoder, since we aim to predict the next token given the past. Same architecture works for both pretraining (predict next token) and inference (sample next token)
- **Massive scalability**: can ingest any text corpus, no need for aligned parallel data.
- **Flexible prompts**: can condition on arbitrary text in-context.

The goal of decoder-only models is to autoregressively generate text, good for open-ended generation.

8.3 Encoder-decoder models

Often on seq2seq task, requires paired data[9]. Take an input sequence, encode it, then decode into an output (good for seq2seq tasks like translation, summarization).

8.4 Prefix-decoder models

Slightly less main stream variant in the tranformer family, similar to an interpolation between decoder-only and encoder-decoder models.

Key difference: In prefix decoder blocks, we only have one prefix self attention layer (causal self attention layer with prefix mask), and we do not have two attentions (cross attention + self attention) pattern as in the normal decoder block. We are essentially changing the externel information from encoder to prefix region in the mask.

Imagine now we take a sequence of length $L = m + n$

$$\mathbf{x} = [\underbrace{p_1, \ldots, p_m}_{\mathbf{p}}, \underbrace{g_1, \ldots, g_n}_{\mathbf{g}}],$$

where $\mathbf{p}$ is the prefix tokens which are given in full before text generation starts. This could be tokens from a source sentence (like the "encoder output" flattened into tokens), learnable virtual tokens (prefix tuning), encoded representations from another modality (e.g., image embeddings mapped to token space),or just a long text prompt. $\mathbf{g}$ here is the generation tokens that we will generate causually one by one.

A standard decoder (decoder-only transformers) uses a strict causal mask, which is a lower triangular matrix,

$$\text{Mask}[i, j] = \begin{cases} 1 & \text{if} \quad j \leq i \\ 0 & \text{otherwise} \end{cases}$$

for all $i, j \in [L]$, this says that token i can only attends to positions $j \leq i$. While for a prefix decoder, we relaxes the prefix region

$$\text{Mask}[i, j] = \begin{cases} 1 & \text{if} \quad i \leq m \text{ and } j \leq m & \text{prefix attends to all prefix} \\ 1 & \text{if} \quad i \geq m \text{ and } j \leq m & \text{generation attends to all prefix} \\ 1 & \text{if} \quad i > m, m < j \leq i & \text{generation attends to its own past} \\ 0 & \text{otherwise} \end{cases}$$

An example is that when we want to generate a translation for a sentence $\mathbf{t}$ (of tokens), we will use it as a prefix, and we know that $\mathbf{t}_0$ is translated into $\mathbf{s}_0$ already, in this case, our prefix would be $[\mathbf{t}_0, \text{<sep>}, \mathbf{s}_0, \text{<BOS>}]$ and our generation tokens would be $\mathbf{t}$. We include <BOS> in the prefix because it is a known token which is used to anchor.

We prefer prefix decoder sometimes, because

[9] An example is that machine translation needs paired data: Source: "I like apples", Target: "J'aime les pommes".

- **Fewer Parameters:** Parameter in the attention part drop roughly half.
- **Lower compute cost:** When prefix length is modest compared to generation length, prefix decoder can be fast overall, especially on hardware where fusing the attention into one pass matters.
- **Simpler architecture:** Do not need a separate encoder-decoder split
- **Easier caching for autoregressive generation**: In standard decoder with cross attention, we need to cache (i) decoder past key/values for masked self attention (ii) the encoder outputs for cross attention. In a prefix decoder, the prefix tokens are just part of the same KV-cache.

Down sides: (1) less flexibility in terms of shared layer stack and parameters. (2) for long prefixes, attention is more expansive.

8.5 Summary

As a summary:

Model type	Pros	Cons
Encoder-only	• Strong understanding with bidirectional context • Great for classification, retrieval, and embedding learning	• Not suited for generative tasks
Decoder-only	• Simpler architecture • Easy to train generatively • Highly flexible for any prompt-based generation	• Lacks bidirectional context • Weaker for pure understanding tasks without adaptation
Encoder–decoder	• Best for sequence-to-sequence tasks (translation, summarization, speech-to-text)	• Requires paired data • Heavier compute at inference (two stacks)

Table 5: Comparison of encoder-only, decoder-only, and encoder–decoder Transformer architectures.

8.6 The Linear Layers

FFNs (Feed Forward Networks) are a part of the encoder, decoder block, there are several design choices here. They are also known as MLPs (Multi-Layer Perceptrons), and they are used to transform the input features into a higher-dimensional space, apply non-linear activation, and then project back to the original dimension. For the 1-d case, imagine we have a vector $x \in \mathbb{R}^{d_{\text{model}}}$, then the FFN does the following:

$$\text{FFN}(x) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 x + b_1) + b_2,$$

where $mW_1 \in \mathbb{R}^{d_{\text{ffn}} \times d_{\text{model}}}$ is the matrix that projects the input to a higher dimension, σ is a non-linear activation function (e.g., ReLU, GELU), and $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ffn}}}$ projects it back to the original dimension. d_{ffn} is often 2 $4\times$ d_{model}. Now if we consider the true input $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, then basically, for each sequence in the batch, we apply the same FFN to each token, and there will be no interactions between different tokens.

Why do we put FFNs in the encoder/decoder blocks? The FFN is used to introduce non-linearity into the model, allowing it to learn more complex representations. Self-attention is linear in the feature dimension for a fixed set of attention weights. It mixes tokens but does not increase the per-token expressivity much. To see this, consider the following attention formula:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \underbrace{\text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_{\text{model}}}}\right)}_{:=\mathbf{A}} \mathbf{V},$$

if the attention weight matrix $\mathbf{A}$ is fixed, then the output is a linear combination of the input $\mathbf{V}$ which is linear in $\mathbf{X}$, i.e, linear in the feature dimension.

If we **remove the FFN**, the model would reduce to mostly linear mixing layers accross tokens, which leads to collapse of expressivity (model underfits complex transformations), and much worse performance.

Design choices for FFNs: There are several design choices for FFNs, which can affect the model's performance and efficiency.

- Expansion ratio: $d_{\text{ffn}}/d_{\text{model}}$.

This is typically around 2 $4\times$. In the original Transformer paper, it is $4\times$, while in LLaMA, it is $2\times$. In many large models, FFNs account for $50\% - 60\%$ of total parameters.

- Activation function: ReLU, GELU, SiLU, etc.

The original Transformer paper uses ReLU, but GELU is more popular in modern models (BERT, GPT-2/3).

ReLU: piecewise linear, fast, but can lead to dead neurons (zero gradients for negative inputs).

$$\text{ReLU}(x) = \max(0, x).$$

GELU: a smoother, probabilistic activation function that approximates the Gaussian distribution.

$$\text{GELU}(x) = x \cdot \Phi(x) = x \cdot \frac{1}{2}\left(1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right)\right),$$

where $\Phi(x)$ is the cumulative distribution function of the standard normal distribution, and erf is the error function.

- Pros: **Smooth, i.e., differentiable everywhere**[10], which enables better gradient flows. It often yields better performance in transformers, understanding and generation tasks.
- Cons: **Computationally more expensive**, sligltly more than ReLU.

SiLU: another smooth activation function, models such as EfficientNet used it.

$$\text{SiLU}(x) = x \cdot \sigma(x),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function. It is sometimes called the Swish-1 (Swish with $\beta = 1$.)

[10]Smoothness in this context refers to infinitely differentiable functions C^∞, not the smoothness in the optimization theory sense.

8.7 Gated Linear Units

GLU (Gated Linear Unit) takes an input, splits it into two parts (value & gate), and uses one to gate the other via an elementwise product. Mathematically, for $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, we first apply two independent linear projections using $\mathbf{W}_g, \mathbf{W}_v \in \mathbb{R}^{d_{\text{ffn}} \times d_{\text{model}}}$,

$$\mathbf{G} = \mathbf{W}_g \mathbf{X} + \mathbf{B}_g, \qquad \mathbf{V} = \mathbf{W}_v \mathbf{X} + \mathbf{B}_v;$$

We then apply a nonlinear activation to the gate part, after which we do elementwise multiplication.

$$\mathbf{G}' = \sigma(\mathbf{G}), \qquad \text{GLU}(\mathbf{X}) = \mathbf{V} \odot \mathbf{G}'.$$

Note that we are applying activation to $\mathbf{G}$ (the gate) instead of $\mathbf{V}$, because we are applying the nonlinearity here to shape its gating behaviour. This separation between gate and value allows us to learn specialized gating pattern, which acts like an learned feature selector.

SwiGLU: we are essentially replacing the original activation function $\sigma(\cdot)$ with $\text{SiLU}(\cdot)$, which is smoother. This is used in LLaMA, PaLM, etc.

GEGLU: we are essentially replacing the original activation with $\text{GELU}(\cdot)$ or its approximations with tanh.

8.8 Mixture-of-Experts FFNs

In a standard Transformer block, the FFN is a single feed-forward network (sometimes gated like SwiGLU) applied to every token. In MoE, instead of one FFN, we have E separate FFN "experts". For each token, a router chooses a subset (often 1 or 2) of experts to run. The outputs are combined according to the router's weights. Its advantages include

- **Scaling parameters without scaling compute:** We can have lots of experts trained but we only invoke one of them.
- **Specialization:** The model specializes experts for different types of inputs (topics, syntactic structures, etc.).

Mathematically, let $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, E be the number of experts in this case.
The forward pass:

(i) Router logits:

$$\mathbf{R} = \mathbf{W}_r \mathbf{X} + \mathbf{B}_r \quad \in \mathbb{R}^{B \times L \times E},$$

where $\mathbf{B}_r = \mathbf{1} b_r^\top$ is the extended bias matrix.

(ii) Gating probabilities:

$$\mathbf{P} = \text{softmax}(\mathbf{R}) \quad \in \mathbb{R}^{B \times L \times E}.$$

(iii) Top-k routing: Keep the best-k experts per token according to the probabilities, and renormalize. Specifically, for each token l in each batch b, denote $\mathcal{T}_k(b,l)$ as the indices of top-k experts according to $\mathbf{P}[b,l,:]$, we do

$$\tilde{\mathbf{P}}[b,l,d] = \frac{\mathbf{P}[b,l,d]}{\sum_{j=1}^{\mathcal{T}_k(b,l)} \mathbf{P}[b,l,j]}, \quad \forall d \in \mathcal{T}_k(b,l), \quad |\mathcal{T}_k(b,l)| = k,$$

and sets the other coordinates to be zero, which allows us to obtain $\tilde{\mathbf{P}}$. The renormalization here is to avoid suppressing magnitude purely because we dropped low-prob experts.

(iv) Dispatch: For each expert e, define the packed input by selecting tokens routed to e as $\mathbf{X}_e$. Let N_e be the number of tokens whose top-k includes expert e, $\pi_e(\cdot)$ be the corresponding map of expert e (from $\{1,\dots,N_e\} \mapsto \{[b,l]\}$) to the original indices pair (b,l) and let $\eta_e(\cdot)$ be the map from $\{[b,l]\}$ to the index in $[N_e]$. Then the input for an expert is

$$\mathbf{X}_e \in \mathbb{R}^{N_e \times d_{\text{model}}}, \quad \text{where} \quad \mathbf{X}_e[j,:] = \mathbf{X}[\pi_e(j),:], \qquad \forall j \in [N_e]$$

with the renormalized gate weight

$$g_e \in \mathbb{R}^{N_e}, \quad \text{where} \quad g_e[j] = \tilde{\mathbf{P}}[\pi_e(j), e].$$

(v) Expert maps: Depending on the expert type (dense FFN / Gated FFN), we create

$$\mathbf{Y}_e = \text{FFN}_e(\mathbf{X}_e) \quad \in \mathbb{R}^{N_e \times d_{\text{model}}}.$$

(vi) Gate scaling and combine: Apply per token gate (weight) to the output

$$\tilde{\mathbf{Y}}_e = \text{diag}(g_e) \cdot \mathbf{Y}_e, \qquad (\text{in ML literature} \qquad g_e \odot \mathbf{Y}_e \text{ using broadcast.})$$

Then the output $\mathbf{Z} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ would be filled such that

$$\mathbf{Z}[b,l,:] = \sum_{e \in \mathcal{T}_k(b,l)} \tilde{\mathbf{Y}}_e[\eta_e(b,l),:]$$

before outputing.

Capacity constraint: In practical implementation, we often set **capacity constraint** in its formalization, which is the limit on how many tokens each expert is allowed to process in a single forward pass. The reason to have it is

- **Load imbalance is common:** one "hot" expert could get far more tokens than the average, causing GPU memory overflow for that expert, and slower steps since we need to wait for it.
- **Hardware needs fixed allocation:** In distributed training, each expert lives on one or more GPUs, memory buffers for expert input/output must be pre-allocated. If the number of tokens per expert varies wildly, we cannot pre-allocate efficiently without wasting huge amounts of memory.

We set a capacity per expert:

$$C_e = \left\lfloor \alpha \cdot \frac{BL}{E} \right\rfloor,$$

where E is the number of experts and $\alpha \geq 1$ is the capacity factor to allow slackness. If an expert get more than C_e tokens:

1. **Dropping:** (common in Switch Transformers) Excess tokens are simply dropped for that expert: they don't get processed there, and their contribution from that expert is zeroed.

2. **Rerouting** (less common) Excess tokens are sent to a backup expert.

If an expert get fewer than C_e tokens: one may use **padding** we pad its buffer to C_e entries for vectorization but those padded entris are ignored.

Load balancing auxiliary loss: We need an extra loss term during training so that the router doesn't collapse onto just a few experts. The reasion is that, without any extra incentive, the model may prefer some experts a lot more than others, leaving many experts underused or unused. We have

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathcal{L}_{\text{aux}},$$

where $\mathcal{L}_{\text{task}}$ is the task loss, and in the common GShard / Switch Transfomer[11], we have

$$\mathcal{L}_{\text{aux}} = E \cdot \sum_{e=1}^{E} f_e \cdot m_e.$$

f_e is defined as the fraction of tokens assigned to expert e,

$$f_e = \frac{N_e}{BL} = \frac{1}{BL} \sum_{b=1}^{B} \sum_{l=1}^{L} \mathbf{1} \left\{ e \in \mathcal{T}_k(b,l) \right\},$$

where $\{e \in \mathcal{T}_k(b,l)\}$ is the indicator vector (sum of corresponding one hot vector) in $\mathbb{R}^{B \times L}$ suggesting if expert e is selected by this token, while m_e is defined as

$$m_e = \frac{1}{BL} \sum_{b=1}^{B} \sum_{l=1}^{L} P[b,l,e],$$

which is the average probability of assigning to expert e. One can show that the loss is smaller when f_e and m_e are more evenly distributed across experts. $\mathcal{L}_{\text{aux}}$ can be though of as an load balancing regularizer.

The all to all trick: In distributed training, experts are sharded across devices (e.g., 1 expert per GPU or multiple experts per GPU.) Naively, we must

1. Send tokens to the device hosting their assigned expert.

2. Process them locally in the expert's feed-forward network.

3. Send back the processed outputs to the original device to continue the model pipeline.

This is a typical many-to-many communication patter, where every GPU may need to send tokens to every other GPU. The most straight forward way is to let all tokens sent to all devices before filtering using a series of all_gather operations, but this can waste bandwitdh.

We can do it with the **all to all** trick, which use a collective communication primitive where each GPU directly sends only the tokens that the other GPUs need, in one coordinated call. In NCCL and similar libraries

[11] Both of these are specific MoE implementations from Google that popularized large-scale sparse FFNs. Switch is basically a simplified, more efficient GShard.

```
torch.distributed.all_to_all_single(output, input, ...)
```

In practice, MoE frameworks do two all-to-all per MoE layer:

1. **Dispatch:** tokens → owning expert GPU.
2. **Combine:** expert outputs → original token order.

In this way, redundant communication is reduced. Real world examples includes GShard / Switch Transformer / DeepSpeed-MoE / Megatron-MoE.

Processes and threads: The legacy nn.DataParallel (**DP**) is based on one process multi-threading, which is generally slower and less scalable (single optimizer state, host-side bottlenecks, GIL contention, extra device = host hops). This is why **DDP** (multiprocess, one rank per GPU) is recommended.

Reasons:

1. **CUDA context is per-process:** Each process gets a clean, single-GPU context, which leads to simpler, safer memory management and fewer heisenbugs.
2. **Communication stacks (NCCL)** are designed around ranks = processes.
3. **Python GIL:** Python threads can't execute Python bytecode truly in parallel. Many CUDA ops release the GIL(Global Interpreter Lock), but coordination/launch logic still contends. One process per GPU avoids that bottleneck.
4. **Failure isolation:** If one GPU OOMs or a kernel errors, you don't take down every replica in the same process.
5. **Deterministic performance:** Independent schedulers per process avoid thread scheduling contention and reduce cross-device interference within a single interpreter.

8.9 Parallel Attention + FFN

The parallel design changes the flow so that attention and FFN happen at the same time on the same input (after normalization), instead of one after the other. Their outputs are summed together with the residual connection in a single step.

```
        ↑       → FFns       →     ↓
x → LayerNorm → Attention → Summation
↓             → Residual  →     ↑
```

Why do we do this:

- **Parallelism:** Attention and FFN can be computed concurrently.
- **Fewer LayerNorms:** Saves computation.
- **Information Flow:** In sequential design, the FFN only sees the post-attention representation. In parallel, FFN sees the original representation directly, possibly preserving more raw token information.
- **Better gradient flow:** Shorter path between input and output in the computation graph.

This architecture is used in some newer LLMs, e.g. PaLM and GPT-JT, to improve speed without hurting performance.

Some practical notes:

- **Scaling:** Some designs scale the attention and FFN outputs to balance their contributions.
- **DropPath / Stochastic depth:** In residual networks, instead of always computing the residual branch, we may randomly drop it during training according to some probabilities, so that the model learns to rely on multiple paths, not just one. In parallel attention, we may consider dropping attention branch and FFN branch independently. In this way we reduces overfitting by introducing randomness, and encourages robustness.
- **Variance stability:** When we sum two residual branches instead of one, we risk blowing up the variance of activations early in training. This is another reason why we do scaling:

$$x' = x + \frac{1}{\sqrt{2}}\left(\text{attn_out} + \text{ffn_out}\right)$$

We can also control by careful weight initializations, or considering set a learnable scaling parameters like α_{attn} and α_{ffn} which is small intially and train up.

9 Computing the Number of Parameters

10 Fintuning

Alignment: making the behavior of a large language model match human goals, values, or expectations.

10.1 Pretraining Loss (self-supervised)

As we know, in pretraining, we are basically doing a self-supervised learning task on predicting the next token. Given an sequence $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_L)$, we are trying to model

$$\mathbb{P}_\theta\left(\mathbf{x}_1, \ldots, \mathbf{x}_L\right) = \prod_{t=1}^{L} \mathbb{P}_\theta\left(\mathbf{x}_t \mid \mathbf{x}_{<t}\right),$$

where θ represnets all the trainable parameters of the language model. The standard choice for it is the causal language modeling (CLM) loss:

$$\mathcal{L}_{\text{pretrain}}(\theta) = -\frac{1}{L}\sum_{t=1}^{L} \log\left(\mathbb{P}_\theta\left(\mathbf{x}_t \mid \mathbf{x}_{<t}\right)\right), \tag{CLM}$$

which is the negative log-likelihood over the sequence. This can be written equivalently as the cross entropy between the model's predicted distribution $\mathbb{P}_\theta\left(\cdot \mid x_{<t}\right) \in \mathbb{R}^{|\mathcal{V}|}$ and the one-hot true token $\mathbf{y}_t \in \{0,1\}^{|\mathcal{V}|}$, where $\mathcal{V}$ is the vocabulary,

$$\mathcal{L}_{\text{pretrain}}(\theta) = -\frac{1}{L}\sum_{t=1}^{L}\sum_{v\in\mathcal{V}} \left(\mathbf{y}_t\right)_v \log \mathbb{P}_\theta\left(v \mid \mathbf{x}_{<t}\right), \tag{Cross Entropy}$$

where $\left(\mathbf{y}_t\right)_v = \mathbf{1}\left\{v = \mathbf{x}_t\right\} \in \mathbb{R}^{|\mathcal{V}|}$. Basically, we are summing over all possible vocabularies. The equivalence is clear once we see that, in (Cross Entropy) $\mathbf{y}_t$ itself is a one hot vector, and it is 1 nowhere but when $v = \mathbf{x}_t$, with everywhere else 0. The training itself reduces to a **maximum likelihood** problem given a trainable set of parameter θ. A quick implementation is that

```
loss = F.cross_entropy(
    logits[:, :-1, :].reshape(-1,  vocab_size), # (B, L, V) -> (BL, V)
    tokens[:, 1:].reshape(-1) # (BL, V)
  )
```

10.2 Perplexity

Perplexity is defined as the defined as the exponential of the average loss. For each token position, we have a prediction of the true next token $\mathbf{y}_t$, our model outputs a probability distribution and the loss is the corresponding negative log likelihood given by

$$-\log p_\theta(\mathbf{y}_t \mid \mathbf{y}_{<t}).$$

The perplexity is defined as

$$\exp(\mathbb{E}\left[-\log p_\theta(\mathbf{y}_t \mid \mathbf{y}_{<t})\right]),$$

which is the exponential of the mean loss. The exponential transforms the original loss which is in the log space and brings it back to the original probability space.

Intuitively, imagine reading a sentence word by word. At each step, the model has to choose from many possible next tokens. If perplexity $= k$, it means the model is as "confused" as if it had to choose uniformly among k equally likely options at each step. That's why ppl is often described as the model's average uncertainty.

Some examples include,

1. **Perfect model** (always assigns probibility 1 to the correct token): in this case loss $= 0$, which leads to ppl $= 1$.

2. **Random guessing model** loss $= \log(\text{vocab size})$, ppl $=$ vocab size.

From this point it is easy to understand the lower the perplexity is, the better predictive power the model has.

10.3 Finetuning Loss

When adapting to downstream tasks or human preferences, loss functions differ. Typical approaches include Supervised Fine-Tuning (SFT), Parameter-Efficient Fine-Tuning (PEFT) (such as LoRA and QLoRA), Reinforcement Learning from Human Feedback (RLHF)[12], DPO / IPO, continual pretraining, domain adaptation.

The reason for perform such finetuning is that without such adaptations, LLMs only predict next tokens, if we ask a question, they might continue the text in a way that doesn't look like an answer.

10.3.1 Gradient Checkpointing

Normally, during backpropagation, PyTorch needs to store all intermediate activations so it can compute gradients later. That eats a lot of GPU memory. Gradient checkpointing is the technique that

- Instead of storing every intermediate activation, the model discards most of them during the forward pass.

- Later, during backward pass, it recomputes those activations on the fly.

[12] Reinforcement learning with a reward model.

- Saves GPU memory, but uses extra compute time.

This technique is useful when fine-tuning very large models on limited GPU memory, often lets you increase batch size or sequence length at the cost of some speed.

10.3.2 Gradient Accumulation

Sometimes, when the GPU does not fit the full effective batch in memory. In this case, we split the large batch into N micro-batches, and for each micro batch, we run (i) forward-backward and accumulate the gradient, but **do not** step (ii) scale the loss down by $\frac{1}{N}$ so that the accumulated gradient is correct. After N steps, we do optimizer.step() once, in this way, we can simulate a large batch size while staying within memory limits.

10.4 Supervised Fine-Tuning (SFT)

Our goal is to make the LLM output exactly match human-provided target completions given prompts. Our loss is still cross-entropy loss, but computed **only** over the target part of the sequence.

Given a prompt $\mathbf{p}$ of length $L_{\mathbf{p}}$ and a target $\mathbf{y}$ of length $L_{\mathbf{y}}$, we compute

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{L_{\mathbf{y}}} \sum_{t=1}^{L_{\mathbf{y}}} \log \left(\mathbb{P}_\theta \left(\mathbf{y}_t \mid \mathbf{y}_{<t}, \mathbf{p} \right) \right) = \frac{1}{L_{\mathbf{y}}} \sum_{t=1}^{L_{\mathbf{y}}} \sum_{v \in \mathcal{V}} \left(\mathbf{y}'_t \right)_v \log \left(\mathbb{P}_\theta \left(v \mid \mathbf{p}, \mathbf{y}_{<t} \right) \right), \tag{SFT}$$

where $\left(\mathbf{y}'_t\right)_v = \mathbf{1}\left\{v = \mathbf{y}_t\right\} \in \mathbb{R}^{|\mathcal{V}|}$. Notice that only completetion tokens are included in the sum (masking the prompt tokens in loss calculation). In practice we will just mask in the following way.

$$\underbrace{\text{<bos> prompt_tokens}}_{\text{Mask}} \text{ target_tokens <eos>.}$$

10.5 Instruction Tuning

As we have said, pretrained LLMs (like GPT, LLaMA, Falcon, etc.) are trained on massive text corpora with the next-token prediction objective. They learn a lot of knowledge and general linguistic ability, but they are not naturally aligned with human intent, if we ask for a list or code snippet, they may not format it correctly or if we ask a question, they might continue the text in a way that doesn't look like an answer. Instruction tuning is a way to bridge the gap: we explicitly train the model to respond to natural-language commands (prompts) in the way humans expect. Formally, instruction tuning $\iff$ SFT of a pretrained LLM on a dataset of **(instruction, response)** pairs. The dataset is some times called an instruction set $\mathcal{D}_{\text{instr}}$. Let $\mathbf{x}$ be instructions and $\mathbf{y}$ be a desired response, then

$$\mathcal{L}^{\text{IT}} = -\sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}_{\text{instr}}} \log \pi_\theta \left(\mathbf{y} \mid \mathbf{x} \right),$$

which is exactly the same as the standard language modelling loss used in finetuning (SFT).

Common instruction datasets FLAN (Google): mixture of 60 NLP tasks rephrased as natural language instructions; Alpaca: 52k instruction-response pairs generated from GPT-3; OpenAssistant, Dolly, UltraChat, ShareGPT, etc.

10.6 Continued Pretraining

Continuted pretraining is the process of taking a large pretrained language model (LM) and further training it with the same pretraining objective (causal LM, masked LM, span corruption, etc.) but on a new corpus that represents the target domain, task data, or language. This is an unsupervised adaptation method: no labels or preference data are needed. The goal is to shift the model's distribution closer to the new data distribution without discarding general knowledge.

For a formal definition, let us image that we have θ_0, which is the pretrained parameters from the base model, denote P_{base} as the distribution of the original pretraining data, and P_{target} as the distribution of the new data corpus (domain/task/language). The continued pretraining then solves:

$$\theta^\star = \arg\min_\theta \mathbb{E}_{\mathbf{x}\sim P_{\text{target}}} \left[\mathcal{L}\left(\theta; x\right)\right].$$

which is identical on the original pretraining objective but with a shifted data distribution.

10.7 Parameter-Efficient Fine-Tuning (PEFT)

Updating all parameters of a model is costly. To resolve this, one way is to use the so called "adaptors". Other approaches include LoRA, QLoRA.

Adaptors In parameter-efficient fine-tuning (PEFT), adapters are small trainable modules inserted into a frozen pre-trained model's layers, typically after the feed-forward (FFN) or attention sublayer. The core idea is that,

- We freeze the original model weights.
- Insert a lightweight bottleneck layer (down-projection $\rightarrow$ non-linearity $\rightarrow$ up-projection) inside each transformer block.
- Train only these new parameters,

Mathematically, for an ouput $\mathbf{h} \in \mathbb{R}^{d_{\text{model}}}$ (we temporarily skip the batch and sequence length dimension as we are doing this to each token's represenetation.)

$$\text{Adapter}(\mathbf{h}) = \mathbf{h} + \mathbf{W}_{\text{up}} \cdot \sigma\left(\mathbf{W}_{\text{down}}\mathbf{h}\right),$$

We often have $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d_{\text{down}} \times d_{\text{model}}}$ where $d_{\text{down}} << d_{\text{model}}$ so that the total number of trainable parameters are drastically reduced ($< 5\%$ of the full model).

Low-Rank Adaptation of LLMs Our goal here remains the same, we are trying to fine-tune large models with far fewer trainable parameters by only learning a low-rank update to certain weight matrices.

- **LoRA**

Formally, consider a weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ in the pre-trained model, in stead of update its pretrained parameter $\mathbf{W}_0$ directly, we keep it frozen, and learn:

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W}, \qquad \Delta\mathbf{W} = \frac{\alpha}{r}\mathbf{B}\mathbf{A},$$

where $\mathbf{A} \in \mathbb{R}^{r \times d_{\text{in}}}$, $\mathbf{B} \in \mathbb{R}^{d_{\text{out}} \times r}$, r is the low rank parameter, α is a scaling factor.

1. Usually, LoRA is applied to attention projection matrices ($\mathbf{W}_q$, $\mathbf{W}_v$ and sometimes $\mathbf{W}_k$, $\mathbf{W}_o$).
2. Rank r and scaling α are hyperparams.

Its advantage includes:

1. Drastic parameters savings.

2. No added inference latency if merged into $\mathbf{W}_0$ after training.

Notice that empirically, touching $\mathbf{W}_q$ and $\mathbf{W}_v$ is basically enough since it recovers almost all the performance of full fine-tuning, while touching fewer parameters. Intuitively, $\mathbf{W}_q$ and $\mathbf{W}_k$ affactes the matching space of attention. Since $\mathbf{QK}^\top$ is bilinear, it often suffices to change $\mathbf{W}_q$. $\mathbf{W}_v$ controls the attented values and changing it also affects the post projection controlled by $\mathbf{W}_o$.

There is a reason why do we apply a scaling factor here:

1. **Training stability:** If $\mathbf{A}$, $\mathbf{B}$ are initialized with small random values, the magnitude of their product varies with r, and without scaling, high r can produce much larger updates, destabilizing training.

2. **rank normalization** This keeps the expected variance of the update constant across different r values. α then allows manual controls (dependent on sensitivity).

Empirically $\alpha = r$ or $\alpha = 1$.

The parameter r controls the capacity of the low rank update. Small r means fewer trainable parameters and lower expressive power, works fine for minor domain shifts. Large r means more trainable parameters and expressive power, but if it is too large, then there is a risk of overfitting especially if training data is small. Empirically, $r = 4$ to 16 common for LLaMA, GPT-J, BLOOM, etc. Higher r used in complex multi-task fine-tuning. Futher improving r often gives negligible benefits.

Usually, we initialize the matrix $\mathbf{A}$ with normal initialization like kaiming_uniform_, but we initialize $\mathbf{B}$ as all zeros. The reason is that we need to prevent the activations we get from $\mathbf{A}$ from collapsing and we also need to make sure that initially the LoRA adaptor contributed nothing to the pretrained language model, by initializing $\mathbf{B}$ as all zero.

```
self.lora_A = nn.Parameter(torch.zeros(r, d_in))
self.lora_B = nn.Parameter(torch.zeros(d_out, r))
nn.init.kaiming_uniform_(self.lora_A, a=math.sqrt(5))
nn.init.zeros_(self.lora_B)
```

- **QLoRA** This actually refers to Quantized LoRA, which allows fine-tuning very large language models (e.g., 65B params) on a single GPU without running out of memory. It combines 4-bit quantization for the frozen base model and LoRA.

1. Quantize $\mathbf{W}_0$ to 4-bit NormalFloat (NF4)[13], which cuts memory usage by 4.

2. Keep $\mathbf{W}_0$ frozen in quantized form.

3. Train LoRA modules $\mathbf{A}$, $\mathbf{B}$ in FP16 / bfloat16.

4. During forward pass:

   - Dequantize $\mathbf{W}_0$ on the fly.
   - Add $\Delta\mathbf{W}$ from LoRA, and proceed as usual.

Note that there are double quantization effective in QLoRA,

[13] It is a 4-bit quantization scheme that preserves more precision than plain int4 by using a learned normal distribution mapping.

1. **First quantization:** The original pretrained weight tensor $\mathbf{W}_0$ is quantized from FP16/BF16 to 4-bit NF4 values plus per-block scaling factors (distribution-aware mapping, so it preserves more accuracy than uniform int4).

$$\mathbf{W}_0 = s \cdot \mathbf{W}_0^{\text{quantized}}.$$

2. **Second quantization:** The scaling factors s themselves are quantized into a lower-precision format (8-bit) plus a meta-scaling factor.

10.8 Reinforcement Learning from Human Feedback (RLHF)

The goal is to train (finetune) a language model so its outputs align with human preferences by using reinforcement learning where the "reward" comes from human judgments.

Mathematically, let us first define the concept of the policy. Given $\mathbf{x}$ as an input sequence (prompt, context) and $\mathbf{y}$ of length T as an output sequence (continuation, response), let $\pi_0(\mathbf{y} \mid \mathbf{x})$ be defined as the probability that he model generates $\mathbf{y}$ given $\mathbf{x}$, defined token-by-token:

$$\pi_0(\mathbf{y} \mid \mathbf{x}) = \prod_{t=1}^{T} \pi_0(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}).$$

We are using π as our notation for it is standard in reinforcement learning[14].

Stage 1: Supervised Finetuning We prepare a supervised dataset $\mathcal{D}_{\text{SFT}} = \{(\mathbf{x}_i, \mathbf{y}_i^\star)\}_{i=1}^N$ where $\mathbf{x}_i$ is prompt tokens, $\mathbf{y}_i$ is gold-standard human-written completion (target tokens). Typically, the dataset size is $10^3 - 10^6$, in this stage we do standard SFT trying to minimize the cross-entropy loss on the target tokens, which can be equivalently formulated as

$$\theta_{\text{SFT}} = \arg\min_\theta \mathbb{E}_{(\mathbf{x},\mathbf{y}^\star \sim \mathcal{D}_{\text{SFT}})} \left[-\log \pi_\theta(\mathbf{y}^\star \mid \mathbf{x}) \right],$$

where

$$\log \pi_\theta(\mathbf{y}^\star \mid \mathbf{x}) = \sum_{t=1}^{L_{\mathbf{y}^\star}} \log \pi_\theta \left((\mathbf{y}^\star)_t \mid \mathbf{x}, (\mathbf{y}^\star)_{<t} \right).$$

Basically[15], we are only computing the cross entropy loss on the human labeled completions (target sequences) without touching the prompt.

Stage 2: Reward Modelling We prepare a preference dataset $\mathcal{D}_{\text{pref}} = \left\{ (\mathbf{x}_j, \mathbf{y}_j^+, \mathbf{y}_j^-) \right\}_{j=1}^M$ where $\mathbf{x}$ is the prompt, $\mathbf{y}_j^+$ is the preferred output and $\mathbf{y}_j^-$ is the less preferred output, M is the total number of pairs of preference pairs (10^4 to 10^6) coming from human labelers ranking multiple model completions for the same prompt.

Padding: We define the reward model $R_\phi(\mathbf{x}, \mathbf{y})$ whose input is the (input prompt tokens, completion), i.e. $(\mathbf{x}, \mathbf{y})$ and outputs a scalar reward score. Typically, we start with a base language model architecture (often the same type as the policy, e.g., Transformer decoder), and we replace the usual LM head (which

[14] In RL, statues s_t is current context (prompt + generated tokens so far), action a_t is next token to generate, policy $\pi(a_t \mid s_t)$ gives the probability distribution. The pretrained LM is thus an autoregressive policy that maps states (text histories) to action distributions (token probabilities).

[15] Here we use $(\cdot)_t$ to denote the t-the component of the vector.

produces a distribution over next tokens) with a scalar regression head. Without the language modeling head, the output is a hidden state for each $(\mathbf{x}, \mathbf{y})$

$$\mathbf{h} = (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{B \times (L_{\mathbf{x}} + L_{\mathbf{y}}) \times d_{\text{model}}}$$

One may notice that for each pair of $\mathbf{x}_i, \mathbf{y}_i$, the length $L_{\mathbf{x}_i} + L_{\mathbf{y}_i}$ is not the same, so we must do padding in order to make it a batch **before** inputting them to the model. We may pad all sequences on the left/right to ensure that they are the same length $L_{\max}$ to form a batch (not the global maximum, otherwise too wasteful)[16], and record a corresponding attention mask $\mathbf{m}_i \in \{0, 1\}^{B \times L_{\max}}$ (1 for real token). Often the paddings are treated as normal tokens and its embedding is either a learned vector or all zeros.

Batch Forming: Now for notational convenience, let us define

$$\mathbf{s}^+ = (\mathbf{x}, \mathbf{y}^+) \in \mathbb{N}^{L_{\mathbf{x}} + L_{\mathbf{y}^+}} \quad \text{and} \quad \mathbf{s}^- = (\mathbf{x}, \mathbf{y}^-) \in \mathbb{N}^{L_{\mathbf{x}} + L_{\mathbf{y}^-}},$$

we padded them locally to form a batch $\mathbf{S} = \left\{\hat{\mathbf{s}}_1^+, \mathbf{s}_1^-, \ldots \mathbf{s}_B^+, \mathbf{s}_B^-\right\}$ where each element is now $L_{\max}$ long, with its attention mask $\mathbf{M} \in \{0, 1\}^{2B \times L_{\max}}$. We feed them to the headless model and obtain

$$\mathbf{H} \in \mathbb{R}^{2B \times L_{\max} \times d_{\text{model}}}.$$

Pooling: We now need to do pooling to choose a single vector for the entire sequence to be used for pairwise comparison. We can choose

- (A) Last-token pooling (most common), which chooses the representation of the last non-PAD tokens (since it contains context from all preceding tokens because of casual attention).
- (B) Mean pooling over completions (ignore prompt tokens if we only want to summarize the generated output).
- (C) Special token pooling (Insert a <EOS> token and take its hidden state), works as [CLS] pooling in BERT.

After those steps, $\mathbf{H}$ becomes $\mathbf{P} \in \mathbb{R}^{2B \times d_{\text{model}}}$.

Scalar Reward: After we obtain $\mathbf{P}$, we apply a reward head to it, which forms the reward

$$R_\phi(\mathbf{x}, \mathbf{y}) = \mathbf{W}_r \mathbf{P} + \mathbf{b}_r \in \mathbb{R}^{2B},$$

where we get a scalar value for each sequence.

Computing Loss: We appear the pairwise Bradley-Terry loss:

$$\mathcal{L}_{\text{BT}}(\phi) = -\frac{1}{B} \sum_{i=1}^{B} \log \sigma \left(R_\phi \left(\mathbf{x}_i, \mathbf{y}_i^+ \right) - R_\phi \left(\mathbf{x}_i, \mathbf{y}_i^- \right) \right),$$

The above loss encourages $R_\phi \left(\mathbf{x}_i, \mathbf{y}_i^+ \right)$ to be much larger than $R_\phi \left(\mathbf{x}_i, \mathbf{y}_i^- \right)$.

This is not the only way we can choose this loss, we can also pick hinge loss:

$$\mathcal{L}_{\text{Hinge}}(\phi) = \frac{1}{B} \sum_{i=1}^{B} \max \left\{ 0, m - \left(R_\phi \left(\mathbf{x}_i, \mathbf{y}_i^+ \right) - \mathbb{R}_\phi \left(\mathbf{x}_i, \mathbf{y}_i^- \right) \right) \right\},$$

where we tries to enforce a margin $m > 0$ and if that is satified, no gradient will be applied. Compared to BT-loss, it is non-smooth.

[16] This is called dynamic padding and is standard in Hugging Face DataCollatorWithPadding

Mean-squared error loss can also be used:

$$\mathcal{L}_{\text{MSE}}(\phi) = \frac{1}{B}\sum_{i=1}^{B}\left(R_\phi\left(\mathbf{x}_i, \mathbf{y}_i^+\right) - \mathbb{R}_\phi\left(\mathbf{x}_i, \mathbf{y}_i^-\right) - t\right)^2,$$

where t is a target gap which assumed to be known (which is also the reason why it is used).

If we have more than 2 candidates ranked, we can normalize the exponentiated rewards into a softmax, say we have $\mathbf{y}^1, \mathbf{y}^2, \ldots, \mathbf{y}^K$

$$P_\phi(\mathbf{y}^j \mid \mathbf{x}) = \frac{\exp\left(R_\phi\left(\mathbf{x}, \mathbf{y}^j\right)\right)}{\sum_{j=1}^{K}\exp\left(R_\phi\left(\mathbf{x}, \mathbf{y}^j\right)\right)},$$

then minimize the cross entropy loss based on this,

$$\mathcal{L}(\phi) = -\sum_{i=1}^{M}\sum_{j=1}^{K}\text{prob_of_rank}_j \cdot \log\left(P_\phi(\mathbf{y}^j \mid \mathbf{x})\right),$$

where this prob_of_rank can be 1 for the top choice and 0 for others, or fractional.

Note that the starting point of ϕ is the pretrained headless model. We can choose to use the version with/without SFT, most pipelines use the one with SFT.

1. Pros 1: RM is specialized to the distribution of completions the RL stage will actually produce
2. Pros 2: KL penalty in PPO is between two models (policy and reference) that are both in-domain with respect to RM's training data
3. Cons 1: RM might overfit to the stylistic quirks of the SFT model and undergeneralize to very different policies

Stage 3: RL fine-tuning using PPO (Proximal Policy Optimization) with KL regularization. We have the reference policy after SFT, which we denote as $\pi_{\text{SFT}} = \pi_{\text{ref}}$ and the reward model $\mathbb{R}_\phi$. Assume that now we have a prompt set $\mathcal{D}_{\text{RL}}$ which are unlabeled instructions / prompt drawn from a prompt distribution (can be the same pool used in SFT, plus extra domains).

RL objective: We intialize our policy as $\pi_\theta = \pi_{\text{ref}}$, we want to now update π_θ so that it scores high under R_ϕ and stay close to π_{ref} (to avoid drifting into low-quality or unsafe behavior), which can be described by the following KL-regularized objective:

$$\max_\theta \mathbb{E}_{\substack{\mathbf{x}\sim\mathcal{D}_{\text{RL}} \\ \mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}}\left[R_\phi(\mathbf{x},\, \mathbf{y}) - \beta \cdot \mathrm{D}_{\text{KL}}\left(\pi_\theta(\cdot \mid \mathbf{x}) \parallel \pi_{\text{ref}}(\cdot \mid x)\right)\right], \tag{16}$$

where $\beta > 0$ is the KL-coeffficient[17]. Notice that to compute the sequence level KL divergence as given above, we may decompose it into token level contribution. For one sequence $\mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_T)$ generated from prompt $\mathbf{x}$,

$$\mathrm{D}_{\text{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{x}\right) \parallel \pi_{\text{ref}}\left(\cdot \mid \mathbf{x}\right)\right) = \mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\left[\sum_{t=1}^{T}\log\frac{\pi_\theta\left(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}\right)}{\pi_{\text{ref}}\left(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}\right)}\right],$$

[17]Note that this is two nested expectation, first we can sample $\mathbf{x}$ from the dataset, then because that the policy it self is stochastic, we have multiple possible y.

we can see that the token level contribution is given by

$$\mathrm{KL}_t\left(\mathbf{x}, \mathbf{y}_{<t}, \mathbf{y}\right) = \log \frac{\pi_\theta\left(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}\right)}{\pi_{\text{ref}}\left(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}\right)}.$$

If we think about it, this discourages the case that the probability of policy $\pi_\theta\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right)$ generates $\mathbf{y}_t$ is smaller than that of policy $\pi_{\text{ref}}\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right)$. In practiccal training, rollouts (states, actions) are sampled from the old policy in the current outer iteration θ_{old}, which is the reason why the advantage-weighted likelihood ratio ($o_t(\theta; \theta_k)$ defined later) appears.

About KL divergence: Token level definition of KL divergence : at timestep t, the KL divergence between current policy π_θ and the reference π_{ref} is

$$\mathrm{D}_{\mathrm{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right) \| \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right)\right) = \sum_{\mathbf{a}} \pi_\theta\left(\mathbf{a} \mid \mathbf{x}, \mathbf{y}_{<t}\right) \cdot \log\left(\frac{\pi_\theta\left(\mathbf{a} \mid \mathbf{x}, \mathbf{y}_{<t}\right)}{\pi_{\theta_{\text{ref}}}\left(\mathbf{a} \mid \mathbf{x}, \mathbf{y}_{<t}\right)}\right),$$

which is an expectation over all possible next tokens $\mathbf{a}$. The previously defined log ratio $\mathrm{KL}_t\left(\mathbf{x}, \mathbf{y}_{<t}, \mathbf{y}\right)$ is an estimate of this true $\mathrm{D}_{\mathrm{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right) \| \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right)\right)$ based on the one token $\mathbf{y}_t$, which is a random variable. We have

$$\mathbb{E}_{\mathbf{y}_t \sim \pi_\theta}\left[\mathrm{KL}_t\left(\mathbf{x}, \mathbf{y}_{<t}, \mathbf{y}\right)\right] = \mathrm{D}_{\mathrm{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right) \| \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}, \mathbf{y}_{<t}\right)\right).$$

Even if we view it at a squence level, notice that

$$\sum_{t=1}^{T} \mathrm{KL}_t\left(\mathbf{x}, \mathbf{y}_{<t}, \mathbf{y}\right) = \sum_{t=1}^{T} \log\left(\frac{\pi_\theta\left(\mathbf{a} \mid \mathbf{x}, \mathbf{y}_{<t}\right)}{\pi_{\theta_{\text{ref}}}\left(\mathbf{a} \mid \mathbf{x}, \mathbf{y}_{<t}\right)}\right) \overset{(a)}{=} \log\left(\frac{\pi_\theta\left(\mathbf{y} \mid \mathbf{x}\right)}{\pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right)}\right),$$

where (a) is due to the chain rule. This still is an estimate in the sense that

$$\mathbb{E}_{\mathbf{y} \in \pi_\theta(\cdot \mid \mathbf{x})}\left[\sum_{t=1}^{T} \mathrm{KL}_t\right] = \mathrm{D}_{\mathrm{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{x}\right) \| \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}\right)\right).$$

In practice, during rollouts we only compute the log-prob ratio for the tokens we sampled, because that gives us a Monte Carlo estimate of the KL penalty. We don't compute the full expectation KL (that would require summing over the whole vocab).

Sampling: We can apply certain **sampling strategies** for turning the model's per-token probability distribution into actual token choices during rollout.

- **Temperature sampling:** assume the size of the vocabulary is V, for logits $\mathbf{z}_t \in \mathbb{R}^V$ generated by the model at the current step t, we have the softmax probability

$$\mathbf{p}_i = \frac{\exp(\mathbf{z}_i)}{\sum_{j=1}^{V} \exp(\mathbf{z}_j)},$$

however, we can add another parameter which is the so called temperature $T > 0$ and rescales the logits before softmax

$$\mathbf{p}_i^{(T)} = \frac{\exp(\mathbf{z}_i / T)}{\sum_{j=1}^{V} \exp(\mathbf{z}_j / T)}.$$

Observation:

1. $T = 1$: normal sampling from model's predicted distribution.
2. $T > 1$: flatter, more uniform, more randomness.
3. $T < 1$: sharper, more peaked, less randomness, more deterministic.

In a word, is T is high, we move to regions of smaller values which is more random, model may produce incoherent text (off-policy noise), and if T is low, model keeps generating the same high-probability responses (no diversity).

- **Top-p sampling (nucleus sampling):** We sort tokens by probability $\mathbf{p}_i$ in descending order, and take the smallest set $\mathcal{S}$ so that

$$\sum_{i \in \mathcal{S}}^{\mathbf{p}_i} \mathbf{p}_i \geq \mathbf{p}_{\text{threshold}},$$

where $\mathbf{p}_{\text{threshold}}$ (e.g. 0.9) is the top-p parameter. Its effect is to keep most probable tokens, while cut off the long tailed improbable tokens.

- **Top-k sampling:** similar to other top-K.
- We may use them in combination, such as temperature in $[0.7, 1]$, top $p \in [0.9, 0.95]$. Note that they influence not only the KL divergence but also the reward model score and the learning signal diversity.

Computation: Let us consider the k-th iteration for PPO and see what exactly we are going to do following the standard reinforcement learning road map. We denote the policy parameters at the start of iteration as θ_k and the critic parameter as ψ_k.

1. **Rollout**: We freeze policy $\theta_{\text{old}} = \theta_k$, and we sample a minibatch of prompts $\left\{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(B)}\right\} \sim \mathcal{D}_{\text{RL}}$, for each prompt $\mathbf{x}^{(j)}$ we generate a completion $\mathbf{y}^{(j)} = \left\{\mathbf{y}_1^{(j)}, \ldots, \mathbf{y}_{T_j}^{(j)}\right\}$ with θ_{old} (with potential temperature sampling and others). For each token t in each sequence, we form state $\mathbf{s}_t^{(b)} = \left(\mathbf{x}^{(b)}, \mathbf{y}_{<t}^{(b)}\right)$, and we let action $\mathbf{a}_t^{(b)} = \mathbf{y}_t^{(b)}$. We record the old log probability under $\pi_{\theta_{\text{old}}}$,

$$\log \pi_{\theta_{\text{old}}}\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right),$$

and the log probability under the reference policy

$$\log \pi_{\theta_{\text{ref}}}\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right),$$

we also evaluate the critic value $V_{\psi_k}(\mathbf{s}_t^{(b)})$ at roll-out time.

2. **Learning Signals (Inner):** We may take a minibatch $\mathcal{M}$ (token-step minibatch) of token steps $\{b, t\}$ (optionally with a padding mask $m_t^{(b)} \in \{0, 1\}$ if sequences have different lengths.) , we initialize $\theta = \theta_{k,0} = \theta_k$ and $\psi = \psi_{k,0} = \psi_k$ where we plan to run E steps of update (via SGD). Since we have the current inner iteration policy as θ, we can compute the log probability,

$$\pi_\theta\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right),$$

which allows us to compute token level KL to the reference

$$\text{KL}_t^{(b)} = \log \pi_\theta\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right) - \log \pi_{\theta_{\text{ref}}}\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right),$$

and further the KL-shaping reward:

$$r_t^{\mathrm{KL},(b)} = -\beta \cdot \mathrm{KL}_t^{(b)}.$$

Then we get the per token reward

$$r_t^{(b)}(\theta) \;=\; \begin{cases} r_t^{\mathrm{KL},(b)}(\theta), & t < T_b, \\ r_t^{\mathrm{KL},(b)}(\theta) + R_\phi\Big(\mathbf{x}^{(b)}, \mathbf{y}^{(b)}\Big), & t = T_b. \end{cases}$$

We further obtain the discounted return as

$$G_t^{(b)}(\theta) = \sum_{l=0}^{T_b - t} \gamma^l r_{t+l}^{(b)}(\theta),$$

and the TD residual

$$\delta_t^{(b)} = r_t^{(b)}(\theta) + \gamma(1 - \mathrm{done}_t^{(b)}) V_\psi(\mathbf{s}_{t+1}^{(b)}) - V_\psi(\mathbf{s}_t^{(b)}),$$

as well as the GAE through backward recursion

$$\hat{A}_t^{(b)}(\theta) = \delta_t^{(b)}(\theta) + \gamma\lambda(1 - \mathrm{done}_t^{(b)}) \hat{A}_{t+1}^{(b)}(\theta), \qquad \hat{A}_{T_b}^{(b)}(\theta) = 0.$$

Then we construct the value target

$$\hat{G}_t^{(b)}(\theta) = \hat{A}_t^{(b)}(\theta) + V_\psi(\mathbf{s}_t^{(b)}).$$

3. **PPO losses (Inner):** We first compute the importance ratio

$$o_t^{(b)} = \exp\left(\log \pi_\theta\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right) - \log \pi_{\theta_{\text{old}}}\left(\mathbf{a}_t^{(b)} \mid \mathbf{s}_t^{(b)}\right)\right),$$

and we compute the actor loss

$$\mathcal{L}^{\text{actor}}(\theta) = \frac{1}{|\mathcal{M}|} \sum_{(b,t)\in\mathcal{M}} \min\Big(o_t^{(b)}(\theta)\, \hat{A}_t^{(b)}(\theta),\, \mathrm{Clip}\big(o_t^{(b)}(\theta), 1-\epsilon, 1+\epsilon\big)\, \hat{A}_t^{(b)}(\theta)\Big),$$

and critic loss

$$\mathcal{L}^{\text{critic}}(\psi) = \frac{1}{|\mathcal{M}|} \sum_{(b,t)\in\mathcal{M}} \big(V_\psi(\mathbf{s}_t^{(b)}) - \hat{G}_t^{(b)}(\theta)\big)^2,$$

as well as entropy

$$\mathcal{L}^{\text{entropy}}(\theta) = \frac{1}{|\mathcal{M}|} \sum_{(b,t)\in\mathcal{M}} \left[-\sum_{\mathbf{a}} \pi_\theta(\mathbf{a} \mid \mathbf{s}_t^{(b)}) \cdot \log \pi_\theta(\mathbf{a} \mid \mathbf{s}_t^{(b)}) \right].$$

To this end, we may form the PPO loss

$$L^{\text{PPO}}(\theta, \psi) = L^{\text{actor}}(\theta) - c_v \cdot L^{\text{critic}}(\psi) + c_e \cdot L^{\text{entropy}}(\theta),$$

where we can do gradient descent on $-\mathcal{L}^{\text{PPO}}$. Note that we can also do an optional mask for padded tokens, if sequence are padded, we just multiplly each summand by a binary mask $m_t^{(b)}$ and divide by $\sum_{(b,t)\in\mathcal{M}} m_b^{(t)}$ instead of $|\mathcal{M}|$ (Token-step minibatch convention does not need it.).

10.9 The Bigger Picture in Reinforcement Learning (PPO)

In RL theory, the formal setting is a Markov Decision Process (MDP):

$$(\mathcal{S}, \mathcal{A}, P, r, \gamma)\,,$$

where

1. State $\mathcal{S}$: all possible situations the agent could be in. In RLHF, a state $\mathbf{s}_t = (\mathbf{x}, \mathbf{y}_{<t})$.
2. Action $\mathcal{A}$: choices the agent can make. In RLHF, it stands for picking the next token from vocabulary.
3. Transition function $P(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$: how actions change the state. In RLHF: appending the chosen token to the sequence.
4. Reward function $r(\mathbf{a}_t, \mathbf{s}_t)$: numeric feedback,in RLHF it is to construct KL penalty in most steps and plus the terminal reward at last.
5. Discount factor γ: how much future rewards are worth compared to immediate ones. (In RLHF we set this to 1 since episodes[18] are short.)

Objective: Assuming we have a policy $\pi_\theta(\mathbf{a} \mid \mathbf{s})$, which is a probability distribution over actions, parameterized by θ (the weights of LLM). The objective in RL is

$$\theta^\star = \arg\max_\theta J(\theta),$$

where $J(\theta)$ is defined as

$$J(\theta) := \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r(s_t, a_t)\right]. \tag{17}$$

That is maximize expected total reward. Note that the expectation actually means that we are sampling episodes (trajectories) using π_θ and for each trajectory, we compute the total discounted reward. We average those totals over all possible trajectories, weighted by their probability of occurring under π_θ. For example, assume τ is a trajectory:

$$\tau = (\mathbf{s}_0, \mathbf{a}_0, \mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_T, \mathbf{a}_T)\,,$$

with the probability $p_\theta(\tau)$

$$p_\theta(\tau) = p(\mathbf{s}_0) \prod_{t=0}^{T} \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) \cdot p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t).$$

we can rewrite $J(\theta)$ as

$$J(\theta) = \sum_\tau p_\theta(\tau) \cdot \left[\sum_{t=0}^{T} \gamma^t r(s_t, a_t)\right].$$

Notice that we can of course think of using gradient ascent to solve this. However, if we define $R(\tau) := \sum_{t=0}^{T} \gamma^t r(s_t, a_t)$, notice that R is the return (sum of rewards) by the environment, and it is only implicitly dependent on θ in terms of the randomness of the trajectory and if we fix one trajectory, the dependence on θ is gone. We cannot backpropagate through this function properly.

[18] An episode is one rollout: starting from a prompt, generating tokens until we hit an end condition (EOS token or max length).

Policy Gradient: The idea here is that

$$J(\theta) = \mathbb{E}_{\mathbf{y}\sim\pi_\theta}\left[r(\mathbf{y})\right] = \sum_{\mathbf{y}} \pi_\theta(\mathbf{y}) r(\mathbf{y}),$$

itself is not differentiable because of the sampling process, but we can manipilate it mathematically by differentiate the expectation itself,

$$\nabla_\theta J(\theta) = \nabla_\theta \sum_{\mathbf{y}} \pi_\theta(\mathbf{y}) r(\mathbf{y}) = \sum_{\mathbf{y}} r(\mathbf{y}) \nabla_\theta \pi_\theta(\mathbf{y}).$$

At this stage, everything is differentiable but we cannot estimate by sampling, because it contains $\nabla_\theta \pi_\theta(\mathbf{y})$, which we does not know, so we use the log-derivative trick to handle it mathematically again

$$\nabla_\theta \pi_\theta(\mathbf{y}) = \pi_\theta(\mathbf{y}) \nabla_\theta \log \pi_\theta(\mathbf{y}),$$

which allows us to obtain

$$\nabla_\theta J(\theta) = \sum_{\mathbf{y}} r(\mathbf{y}) \pi_\theta(\mathbf{y}) \nabla_\theta \log \pi_\theta(\mathbf{y}),$$

which is

$$\nabla_\theta J(\theta) = \mathbb{E}_{\mathbf{y}\sim\pi_\theta}\left[r(\mathbf{y}) \nabla_\theta \log \pi_\theta(\mathbf{y})\right]$$

Now the gradient is expressed as an expectation over samples from π_θ, so we can approximate it with Monte Carlo

$$J(\theta) \approx \frac{1}{N} \sum_i r(\mathbf{y}_i) \nabla_\theta \log \pi_\theta(\mathbf{y}_i).$$

Details: As details, differentiate $J(\theta)$, we get

$$\begin{aligned}
\nabla_\theta J(\theta) &= \int \nabla_\theta p_\theta(\tau) R(\tau) \quad d\tau \\
&= \int p_\theta(\tau) \nabla_\theta \log p_\theta(\tau) \cdot R(\tau) \quad d\tau \\
&= \mathbb{E}_{\tau\sim p_\theta}\left[\nabla_\theta \log p_\theta(\tau) \cdot R(\tau)\right].
\end{aligned}$$

The environment dynamics $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$ and $p(\mathbf{s}_0)$ does not depend on θ which is just a const scaling on the original objective (17), we drop them and result in

$$\nabla_\theta J(\theta) = \mathbb{E}_\tau \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right) \cdot R(\tau)\right]. \tag{A}$$

This is called the **Reinforce**. Now define the return starting at time t: G_t, which is

$$G_t := \sum_{k=t}^{T} \gamma^{k-t} r(\mathbf{s}_k, \mathbf{a}_k).$$

It is a random variable depends on the future state after t, note that

$$R(\tau) = \underbrace{\sum_{k=0}^{t-1} \gamma^k r(\mathbf{s}_k, \mathbf{a}_k)}_{\text{past reward, const given } \mathbf{s}_t.} + \gamma^t \underbrace{\sum_{k=t}^{T} \gamma^{k-t} r(\mathbf{s}_k, \mathbf{a}_k)}_{G_t}.$$

For the t-th term in the summation in (A) and (17), notice that the expectation is already conditioned on $\mathbf{a}_t$ and $\mathbf{s}_t$, if we remove the past reward, the gradient will not change since it is equivalent to remove a constant on $J(\theta)$, this allows us to obtain

$$\nabla_\theta J(\theta) = \mathbb{E}_\tau \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot G_t \right]. \tag{18}$$

The vanilla Policy Gradient method is based on this and Monte Carlo, which is of high variance, no constraints on updates. Now defining the **action-value** function $Q^\pi(\mathbf{s}, \mathbf{a})$,

$$Q^\pi(\mathbf{s}, \mathbf{a}) := \mathbb{E}_\pi \left[G_t \mid \mathbf{s}_t = \mathbf{s}, \mathbf{a}_t = \mathbf{a} \right],$$

which is the expected return if we are at $\mathbf{s}$ state and take $\mathbf{a}$ as action now, following policy π thereafter. The expectation itself is over the random future trajectory. Using the tower property of expectation, we can further write

$$\begin{aligned}
\nabla_\theta J(\theta) &= \mathbb{E}_\tau \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot G_t \right] \\
&= \sum_{t=0}^{T} \mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot \mathbb{E} \left[G_t \mid \mathbf{s}_t, \mathbf{a}_t \right] \right] = \sum_{t=0}^{T} \mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot Q^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right]
\end{aligned}$$

Now, for variance reduction, we introduce the state-value function:

$$V^\pi(\mathbf{s}) := \mathbb{E}_{\mathbf{a} \sim \pi(\cdot \mid \mathbf{s})} \left[Q^\pi(\mathbf{s}, \mathbf{a}) \right] = \mathbb{E}_\pi \left[G_t \mid \mathbf{s}_t = \mathbf{s} \right],$$

which is the expected return starting in state $\mathbf{s}$ following π. The advantage function is then

$$A^\pi(\mathbf{s}, \mathbf{a}) := Q^\pi(\mathbf{s}, \mathbf{a}) - V^\pi(s),$$

which measure how much better the action $\mathbf{a}$ is compare to the average action in state $\mathbf{s}$. Notice that for any baseline function $b(\mathbf{s}_t)$ that depends only on the state $\mathbf{s}_t$ but not on $\mathbf{a}_t$, we observe that

$$\begin{aligned}
\mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot b(\mathbf{s}_t) \right] &= \mathbb{E}_{\mathbf{s}_t} \left[b(\mathbf{s}_t) \cdot \mathbb{E}_{\mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \right] \right] \\
&= \mathbb{E}_{\mathbf{s}_t} \left[b(\mathbf{s}_t) \cdot \sum_{\mathbf{a}_t} \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \right] \\
&\overset{\spadesuit}{=} \mathbb{E}_{\mathbf{s}_t} \left[b(\mathbf{s}_t) \cdot \sum_{\mathbf{a}_t} \nabla_\theta \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \right] \\
&= \mathbb{E}_{\mathbf{s}_t} \left[b(\mathbf{s}_t) \cdot \nabla_\theta \sum_{\mathbf{a}_t} \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \right] = \mathbb{E}_{\mathbf{s}_t} \left[b(\mathbf{s}_t) \cdot \nabla_\theta 1 \right] = 0
\end{aligned}$$

where $\spadesuit$ is due to chain rule. As, a result, we have

$$\nabla_\theta J(\theta) = \sum_{t=0}^{T} \mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot \left(Q^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) - b(\mathbf{s}_t) \right) (\mathbf{s}_t, \mathbf{a}_t) \right].$$

Now take $b(\mathbf{s}_t) = V^{\pi_\theta}(\mathbf{s}_t)$, we result in

$$\nabla_\theta J(\theta) = \sum_{t=0}^{T} \mathbb{E}_{\mathbf{s}_t, \mathbf{a}_t} \left[\nabla_\theta \log \pi_\theta \left(\mathbf{a}_t \mid \mathbf{s}_t \right) \cdot A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right].$$

This is called the **policy gradient theorem**. In practice, we still **cannot** compute Q^{π_θ} or V^{π_θ} exactly. So we train an auxiliary **value network** (Critic) to approximate V^{π_θ}, and we estimate A^{π_θ} from the rollout data using methods like **Generalized Advantage Estimation** (GAE), which trades bias for variance reduction, which gives us an estimate $\hat{A}_t$. To do this, we relies on three basic building blocks:

1. **Monte Carlo (MC) return**: $G_t = \sum_{u=t}^{T} \gamma^{u-t} r_u$. This is the unbiased estimate of the true return, which is very noisy and of high variance, especially if the horizon T is long.

2. **Critic (value function)**: $V_\psi \approx V^{\pi_\theta}(\mathbf{s}_t) = \mathbb{E}\left[G_t \mid \mathbf{s}_t\right]$, as θ is consistently changing, we need to train critic per iteration.

3. **TD residual**: $\delta_t := r_t + \gamma V_\psi(\mathbf{s}_{t+1}) - V_\psi(\mathbf{s}_t)$, which is the one step error signal telling us how much better/worse things turned out compared to the critic's expectation[19]. Notice that δ_t can be viewed as 1-step estimate of the advantage function, since by definition (we get immediate reward + discounted value of next state)

$$Q^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \simeq r_t + \gamma V_\psi(\mathbf{s}_{t+1})$$

From reward to policy update: We are solving a standard episodic RL problem.

Step 1: Rollout with current policy. Assume we are in iteration k, and we collect our trajectories by following π_{θ_k}, this gives us data $(\mathbf{s}_t, \mathbf{a}_t, r_t)$[20], while we are collecting we also store the log probability of each action $\log \pi_{\theta_k}(\mathbf{a}_t \mid \mathbf{s}_t)$.

Step 2: Advantage estimation. We would like to know whether each action was better or worse than average.

Step 2.1: Train a critic. So we train a critic V_{ψ_k} to approximate the function value V^{π_θ}. Usually, this V_ψ can be a MLP parametrized by ψ.[21] This result in an extra loss term based on MSE loss in the final PPO loss, which we name critic.

Step 2.2: Temporal Difference (TD) residuals. We use the trained critic to compute 1-step errors

$$\delta_t = r_t + \gamma(1 - \text{done}_t)V_\psi(\mathbf{s}_{t+1}) - V_\psi(\mathbf{s}_t) = \big(\text{obs reward} + \text{pred future value}\big) - \big(\text{pred value at } s_t\big),$$

where done_t is a termination flag, $\text{done}_t = 1$ means the episode has terminated at step t either because the agent reached a terminal state or a time limit was reached; $\text{done}_t = 0$ means the episode continues. When we reach the first termination iteration T, $1 - \text{done}_T = 0$, which means that $\delta_T = r_T - V_\psi(s_T)$[22]. One may wonder why we are setting the done flag, the reason is precisely that we are handling batched data, so episodes have variable length. As a result, we need to pad them so that the batch is aligned, and the done flag lets us distinguish real continuation from padding or termination. In this way, advantage/return computation stops at the right place, even inside a large batch.

If $\delta_t > 0$, the return is better than expected and the action was better than average, if $\delta_t < 0$, vice versa. To understand this intuitively, we only have to remember that V_{π_k} is the expected total discounted reward you will get if you start from state $\mathbf{s}$ then following policy π_{θ_k}.

Step 2.3: Generalized Advantage Estimation (GAE). We have two choices to estimate advantage, one is the 1-step estimation generated by TD-residual, which is very noisy because it only looks one step ahead,

[19] If the critic is perfect, we have $V_\psi(\mathbf{s}_t) = \mathbb{E}\left[r_t + \gamma V^{\pi_\theta}(\mathbf{s}_{t+1})\right]$, this is known as Bellman expectation equation which tells us what a critic would satisfy.

[20] During training, we sample many such $\mathbf{x}$ from $\mathcal{D}_{\text{RL}}$ in parallel, and collect a batch of rollouts at once.

[21] In complex environments (e.g. Atari, robotics, LLMs), it can be a convolutional net, a transformer, or any architecture suited for the state representation.

[22] In RL, we use bootstrapping to denote estimating a return by combining the observed reward with the critic's estimate of the future, (instead of waiting until the very end of the episode to know the true return). Here $\gamma(1 - \text{done}_t)V_\psi(\mathbf{s}_{t+1})$ is the bootstrap part.

especially when we are in a long episode and there is a large reward at the end. From an opposite point of view, if we use $G_t = r_t + \gamma r_{t+1} + \dots$ and define

$$\hat{A}_t = G_t - V_{\psi_k}(\mathbf{s}_t).$$

MC returns are unbiased but high variance. Using a learned critic baseline reduces variance but can introduce bias if the critic is imperfect.

So we just combine them which results in GAE (instead of picking one horizon, we take a weighted average of all horizons). Formally,

$$\hat{A}_t = \delta_t + \gamma\lambda\delta_{t+1} + (\gamma\lambda)^2 \delta_{t+2} + \dots,$$

where $\gamma = 0.99$ and $\lambda = 0.95$ are common. Notice that γ is the discounting factor and λ controls how far we can look ahead. To compute is efficiently, we use backward recursion,

$$\hat{A}_t = \delta_t + \gamma\lambda\hat{A}_{t+1}.$$

In practice, due to the padding used in batch, we set

$$\hat{A}_t = \delta_t + \gamma\lambda(1 - \text{done}_t)\hat{A}_{t+1}, \qquad \hat{A}_T = 0 \tag{19}$$

where we use a termination flag to control bootstrapping, T is the index one past the last real step in the trajectory. From the estimate $\hat{A}_t$, we can reconstruct

$$\hat{G}_t = \hat{A}_t + V_\psi(\mathbf{s}_t), \tag{20}$$

which we will later use in loss.

Step 2.4: Gradient. After running GAE, we now have per-time-step advantage estimates $\hat{A}_t$, which allows us to scale the log-probability gradient

$$\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) \cdot \hat{A}_t, \tag{Policy Gradient}$$

where $\hat{A}_t$ increases the probability and vice versa.

Step 3: Policy update with PPO. Now we want to adjust π_{θ_k}, so that actions with positive advantage become more likely, and those with negative advantage less likely. If we use the naive policy gradient then the policy might change too much in one step. It could also collapse (e.g., probability mass shifts to a single action, causing instability), so we need to regularize the update. Notice that the naive objective after policy gradient transform and approximations can be written as

$$J^{\text{PG}}(\theta) = \mathbb{E}_{t,\mathbf{s}_t,\mathbf{a}_t}\left[\log \pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right) \cdot \hat{A}_t\right].$$

Historically, we use **TRPO** (Trust Region Policy Optimization), where we do

$$\max_\theta \mathbb{E}_{t,\mathbf{s}_t,\mathbf{a}_t}\left[r_t(\theta;\theta_k)\hat{A}_t\right] \qquad \text{s.t.} \qquad \mathbb{E}_{\substack{\mathbf{s}\sim\rho_{\pi_{\theta_k}} \\ \mathbf{a}\sim\pi_{\theta_k}(\cdot|\mathbf{s})}}\left[\log \pi_{\theta_k}\left(\mathbf{a} \mid \mathbf{s}\right) - \log \pi_\theta\left(\mathbf{a} \mid \mathbf{s}\right)\right] \le \delta,$$

where $\rho_{\pi_{\theta_k}}$ denotes the (on-policy) state distribution under π_{θ_k} and

$$o_t(\theta;\theta_k) := \frac{\pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right)}{\pi_{\theta_k}\left(\mathbf{a}_t \mid \mathbf{s}_t\right)},$$

is the ratio. The constraint can also be written equivalently as

$$\mathbb{E}_{\mathbf{s}\sim\rho_{\pi_{\theta_k}}}\left[\mathrm{D}_{\text{KL}}\left(\pi_{\theta_k}\left(\cdot \mid \mathbf{s}\right) \parallel \pi_\theta\left(\cdot \mid \mathbf{s}\right)\right)\right] \le \delta.$$

There are two layers of expectations: (1) over states $\mathbf{s}$ that the old policy visits, (2) over actions $\mathbf{a}$ drawn from the old policy at each state (this is built into the KL). However, **TRPO** is too time consuming in the sense that it often requires second order optimization methods (conjugate gradient with Fisher-vector products to enforce the KL trust region).

Importance sampling based on old policy data. In practice, at iteration k we first collect trajectories using the current policy π_{θ_k}. The most direct way to update would be to compute the gradient of the policy gradient objective at $\theta = \theta_k$, which is exactly on-policy with respect to the data. However, modern methods like PPO typically perform many minibatch gradient updates on the same rollout data. After the first update, the parameters have moved from θ_k to some $\theta_{k,t}$ (where t indexes local iterations on this minibatch), while the data is still drawn from π_{θ_k}. This creates a distribution mismatch: the true policy gradient is an expectation under the new policy $\pi_{\theta_{k,t}}$, but we only have samples from the old one. To address this, PPO introduces an importance-sampling correction using the ratio function

$$o_t(\theta;\theta_k) \;=\; \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_k}(\mathbf{a}_t \mid \mathbf{s}_t)}.$$

The surrogate objective is then written as

$$\widehat{J}^{\mathrm{PG}}(\theta;\theta_k) \;=\; \mathbb{E}_{\substack{t \\ \mathbf{s}_t\sim\rho_{\pi_{\theta_k}} \\ \mathbf{a}_t\sim\pi_{\theta_k}}}\left[o_t(\theta;\theta_k)\,\hat{A}^t\right].$$

Differentiating gives

$$\begin{aligned}\nabla_\theta \widehat{J}^{\mathrm{PG}}(\theta;\theta_k) &= \mathbb{E}_{t,\mathbf{s}_t,\mathbf{a}_t}\left[\nabla_\theta o_t(\theta;\theta_k)\,\hat{A}^t\right] \\ &= \mathbb{E}_{t,\mathbf{s}_t,\mathbf{a}_t}\left[\frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_k}(\mathbf{a}_t \mid \mathbf{s}_t)}\,\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)\,\hat{A}^t\right] \\ &= \mathbb{E}_{t,\mathbf{s}_t,\mathbf{a}_t}\left[o_t(\theta;\theta_k)\,\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)\,\hat{A}^t\right].\end{aligned}$$

Thus the final expression is exactly the *importance-sampled version of the original policy gradient*, allowing us to reuse old-policy data π_{θ_k} while updating toward θ_{k+1}. This is easy to understand, we just first divide it by the probability it appears in the old disrtribution and then assigned the correct weight (the current probability) to it. For a formal proof, we only need to use the change of measure trick in (21).

Clipping. If we maximize this directly, the policy can change too aggressively, leading to instability. As a result, **PPO** in turn introduces a clipped surrogate,

$$J^{\mathrm{Clip}}(\theta;\theta_k) = \mathbb{E}_{\substack{t \\ \mathbf{s}_t\sim\rho_{\pi_{\theta_k}} \\ \mathbf{a}_t\sim\pi_{\theta_k}}}\left[\min\left\{o_t(\theta;\theta_k)\hat{A}^t, \mathrm{clip}\left(o_t(\theta;\theta_k), 1-\varepsilon, 1+\varepsilon\right)\cdot\hat{A}^t\right\}\right],$$

where the clipping operator does the following:

1. When $o_t \in [1-\varepsilon, 1+\varepsilon]$, we do nothing.
2. When o_t goes beyond, we replace o_t by the nearest boundary value.

Entropy bonus. If we only maximize rewards, the policy may collapse to a near-deterministic one (always picking the "best" action it has found so far). To counter this, we add an entropy bonus to the objective.

$$\mathcal{H}_{\pi_\theta(\cdot|\mathbf{s})} = -\sum_{\mathbf{a}} \pi_\theta\left(\mathbf{a} \mid \mathbf{s}\right)\log \pi_\theta\left(\mathbf{a} \mid \mathbf{s}\right),$$

if the policy is uniform then it is high.

Full PPO loss. The clipped surrogate is the policy gradient, but PPO also trains the critic and uses an entropy bonus.

$$\mathcal{L}^{\text{PPO}}(\theta, \psi; \theta_k) := -\underbrace{\mathbb{E}\left[\mathcal{J}^{\text{Clip}}(\theta; \theta_k)\right]}_{\text{actor}} + c_v \cdot \underbrace{\mathbb{E}\left[\left(V_\psi(s_t) - \hat{G}_t\right)^2\right]}_{\text{critic}} - c_e \cdot \underbrace{\mathbb{E}\left[\mathcal{H}\left(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right)\right)\right]}_{\text{exploration}},$$

where

$$\mathbb{E}\left[\mathcal{H}\left(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right)\right)\right] = \frac{1}{T}\sum_{t=0}^{T} \mathcal{H}(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right)).$$

Practical hint. In practice, the training process goes very similarly like local training where there is a nested loop:

In the k-th iteration, we do

1. **Rollout Phase**: We fix $\theta_{\text{old}} = \theta_k$ where θ_{old} denotes the frozen snapshot of the current policy.
   - We use $\pi_{\theta_{\text{old}}}$ to interact with environment.
   - We collect a batch of trajectories: states $\mathbf{s}_t$, actions $\mathbf{a}_t$ and reward r_t, we then compute the log probabilities $\log \pi_{\theta_{\text{old}}}\left(\mathbf{a}_t \mid \mathbf{s}_t\right)$, and the old values $V_{\psi_k}(\mathbf{s}_t)$, and the done flags.

   Imagine that there are T steps across N parallel environments, eventually, we got

   $$\mathcal{B} = \left\{(\mathbf{s}_t, \mathbf{a}_t, r_t, \log \pi_{\text{old}(\mathbf{a}_t|\mathbf{s}_t)}, V_{\psi_k}(\mathbf{s}_t), \text{done}_t)_i\right\}_{t\in[T], i\in[N]}$$

2. **Advantage + Target Computation:** We compute learning signals:

   $$\delta_t = r_t + \gamma\left(1 - \text{done}_t\right) V_{\psi_k}(\mathbf{s}_{t+1}) - V_{\psi_k}(\mathbf{s}_t),$$

   $$\hat{A}_t = \delta_t + \gamma\lambda\left(1 - \text{done}_t\right)\hat{A}_{t+1}, \qquad \hat{A}_T = 0,$$

   where T is the index one past the last real step in the trajectory, and the value target

   $$\hat{G}_t = \hat{A}_t + V_{\psi_k}(\mathbf{s}_t).$$

3. **Update Actor & Critic:** We initialize trainable parameter $\theta = \theta_{k,0} = \theta_{\text{old}}$ and $\psi = \psi_{k,0} = \psi_k$, we loop over minibatches $\mathcal{M}$ of $\mathcal{B}$ for several epochs E. There are actually two types of minibatching, one is **episode level minibatching** and we are dividing based on episodes, but this can waste parallelism if episodes lengths vary a lot. Fortunately, we can also choose to use the **token-step minibatching**, constructing $\mathcal{M}$ using token steps from different episodes. We use the later minibatching, and in this case $|\mathcal{M}|$ is just the total pairs of token-steps.

   (I) Policy distribution under current θ,

   $$\pi_\theta\left(\cdot \mid \mathbf{s}_t\right), \quad \log \pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right).$$

   (II) Importance ratio,

   $$o_t(\theta; \theta_{\text{old}}) = \exp\left(\log \pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right) - \log \pi_{\theta_{\text{old}}}\left(\mathbf{a}_t \mid \mathbf{s}_t\right)\right)$$

(III) Actor loss (clipped surrogate)

$$\begin{aligned} L^{\text{actor}} &= \mathbb{E}_t \left[\min \left\{ o_t(\theta;\theta_k)\hat{A}^t, \text{clip}\left(o_t(\theta;\theta_k), 1-\varepsilon, 1+\varepsilon\right) \cdot \hat{A}^t \right\} \right] \\ &= \frac{1}{|\mathcal{M}|} \sum_{(\mathbf{s}_t,\mathbf{a}_t)\in\mathcal{M}} \min \left\{ o_t(\theta;\theta_k)\hat{A}^t, \text{clip}\left(o_t(\theta;\theta_k), 1-\varepsilon, 1+\varepsilon\right) \cdot \hat{A}^t \right\}. \end{aligned}$$

Notice that we already do summation over time since $|\mathcal{M}|$ contains it.

(IV) Critic loss

$$\begin{aligned} L^{\text{critic}} &= \mathbb{E}_t \left[\left(V_\psi(\mathbf{s}_t) - \hat{G}_t \right)^2 \right] \\ &= \frac{1}{|\mathcal{M}|} \sum_{\mathbf{s}_t\in\mathcal{M}} \left(V_\psi(\mathbf{s}_t) - \hat{G}_t \right)^2. \end{aligned}$$

(V) Entropy bonus (exploration):

$$\begin{aligned} L^{\text{entropy}} &= \mathbb{E}_t \left[-\sum_{\mathbf{a}} \pi_\theta(\mathbf{a} \mid \mathbf{s}_t) \cdot \log \pi_\theta(\mathbf{a} \mid \mathbf{s}_t) \right] \\ &= \frac{1}{|\mathcal{M}|} \sum_{\mathbf{s}_t\in\mathcal{M}} \left(-\sum_{\mathbf{a}} \pi_\theta(\mathbf{a} \mid \mathbf{s}_t) \cdot \log \pi_\theta(\mathbf{a} \mid \mathbf{s}_t) \right). \end{aligned}$$

(VI) Total PPO objective

$$L^{\text{PPO}} = -L^{\text{actor}} + c_v \cdot L^{\text{critic}} - c_e \cdot L^{\text{entropy}}.$$

(VII) Take a gradient step, so that we obtain $\theta_{k,e+1}, \psi_{k,e+1}$ from $\theta_{k,e}$ and $\psi_{k,e}$.

4. We let $\theta_{k+1} = \theta_{k,E}$ and $\psi_{k+1} = \psi_{k,E}$ using gradient ascent (or gradient descent to minus PPO loss), and prepare to enter the next iteration.

Regarding the meaning of concepts

- Reward r_t: decided by the environment, independent of the policy, indicates how much can one get at moment t. Normally, it depends on $\mathbf{a}_t$ and $\mathbf{s}_t$, i.e., $r_t = r_t(\mathbf{a}_t, \mathbf{s}_t)$. The policy do affect the realized policy since it decides what $\mathbf{a}_t$ and $\mathbf{s}_t$ could be.
- Return G_t: discounted cumulated future rewards, which is the total one to get from now on.
- Discounted factor γ, which is a number in $[0, 1]$ controlling how much future rewards matter. $\gamma \approx 1$ means that we think in a long term fashion, and γ close to 0 suggests to live in the moment.
- Value function $V^\pi(\mathbf{s}) = \mathbb{E}_\pi \left[G_t \mid \mathbf{s}_t = \mathbf{s} \right]$, which is the expected return in state $\mathbf{s}$ following policy π.
- Action-value function $Q^\pi(\mathbf{s}, \mathbf{a}) = \mathbb{E}_\pi \left[G_t \mid \mathbf{s}_t = \mathbf{s}, \mathbf{a}_t = \mathbf{a} \right]$, which is the expected return starting in state $\mathbf{s}$, taking action $\mathbf{a}$.
- Advantage $A^\pi(\mathbf{s}, \mathbf{a}) = Q^\pi(\mathbf{s}, \mathbf{a}) - V^\pi(\mathbf{s})$, which has the meaning that is this move better than my usual move? This term is the key for variance reduction.

- Temporal difference $\delta_t = r_t + \gamma V(\mathbf{s}_{t+1}) - V(\mathbf{s}_t)$, where V is the baseline prediction of cumulative reward. δ_t is the prediction error. The true value satisfies

$$V^\pi(\mathbf{s}_t) = \mathbb{E}_\pi \left[r_t + \gamma V^\pi(\mathbf{s}_{t+1}) \mid \mathbf{s}_t \right],$$

  which is the Bellman equation.

- Optimal policy $\pi^\star$, policy that maximizes expected return from every state.

10.10 PPO-KL

If we are using raw PPO as an alignment method, we are just optimizing against the reward model R_ϕ, which may render the policy dropping into unsafe, verbose, or unnatural completions. The solution to this is to add a KL-penalty that discourages moving too far from the reference policy (usually the SFT model). Essentially, PPO-KL just adds an extra KL penalty term

$$J^{\text{PPO-KL}}(\theta) = J^{\text{PPO}}(\theta) - \beta \cdot \mathbb{E}_t \left[\mathrm{D}_{\text{KL}} \left(\pi_\theta \left(\cdot \mid \mathbf{s}_t \right) \parallel \pi_{\theta_{\text{ref}}} \left(\cdot \mid \mathbf{s}_t \right) \right) \right].$$

Here we follow the convention and use J as the objective to maximize while $\mathcal{L}$ is reserved for objective to minimize.

10.11 DPO (Direct Preference Optimization)

Instead of using PPO with a reward model, DPO directly optimizes the model's policy to match human preferences using a pairwise comparison dataset. We no longer need to train or sample from a reward model at each step and we do not need complex RL machinery (advantage estimation, rollout, etc.). Training reduces to a standard supervised loss on pairs of responses.

We will train (finetune) on a preference dataset $\mathcal{D}_{\text{pref}} = \left\{ (\mathbf{x}_j, \mathbf{y}_j^+, \mathbf{y}_j^-) \right\}_{j=1}^M$ where $\mathbf{x}$ is the prompt, $\mathbf{y}_j^+$ is the preferred output and $\mathbf{y}_j^-$ is the less preferred output, M is the total number of pairs of preference pairs (just like RLHF PPO stage 2). The core idea is that we assume that there is an underlying reward function $r(\mathbf{x}, \mathbf{y})$ such that

$$\mathbb{P}\left(\mathbf{y}^+ \succ \mathbf{y}^- \mid x \right) = \sigma \left(r(\mathbf{x}, \mathbf{y}^+) - r(\mathbf{x}, \mathbf{y}^-) \right), \tag{Bradley-Terry}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. DPO avoids learning an explicit reward model by connecting rewards to log-probabilities under the learned policy vs. a reference model (typically the SFT model):

$$r(\mathbf{x}, \mathbf{y}) \propto \beta \left(\log \pi_\theta \left(\mathbf{y} \mid \mathbf{x} \right) - \log \pi_{\theta_{\text{ref}}} \left(\mathbf{y} \mid \mathbf{x} \right) \right),$$

where $\beta > 0$ is the temperature that controls the regularization strength.

If we plug in this into the Bradley-Terry model, we get

$$\mathbb{P}\left(\mathbf{y}^+ \succ \mathbf{y}^- \mid x \right) = \sigma \left(\beta \left(\log \frac{\pi_\theta \left(\mathbf{y}^+ \mid \mathbf{x} \right)}{\pi_{\theta_{\text{ref}}} \left(\mathbf{y}^+ \mid \mathbf{x} \right)} - \log \frac{\pi_\theta \left(\mathbf{y}^- \mid \mathbf{x} \right)}{\pi_{\theta_{\text{ref}}} \left(\mathbf{y}^- \mid \mathbf{x} \right)} \right) \right).$$

Thus we choose to maximize the likelyhood of observed preference,

$$J^{\text{DPO}}(\theta) = \mathbb{E}_{(x, y^+, y^-) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(\frac{1}{\beta} \left(\underbrace{\log \frac{\pi_\theta(y^+|x)}{\pi_{\text{ref}}(y^+|x)}}_{\text{advantage of } y^+} - \underbrace{\log \frac{\pi_\theta(y^-|x)}{\pi_{\text{ref}}(y^-|x)}}_{\text{advantage of } y^-} \right) \right) \right],$$

which is just a binary logistic regression (here we maximize this likelihood) objective over pairs. The entire DPO algorithm can then be described as

1. We start with a supervised model both as π_θ and π_{ref}.
2. We take a minibatch of paris $\left\{\mathbf{x}_b, \mathbf{y}_b^+, \mathbf{y}_b^-\right\}_{b=1}^B$, and we compute the log probabilities from π_θ and π_{ref}.
3. We then compute the loss $\mathcal{L}^{\text{DPO}}$ and backpropagate and update θ.

Notice that the $\pi_{\theta_{\text{ref}}}(\cdot \mid \mathbf{x})$ term is there to ensure that the model doesn't drift too far from the SFT model (similar to the KL penalty in PPO-RLHF). Its advantages include

1. It directly aligns the policy with preference data.
2. Simple, stable, and efficient: training is just supervised cross-entropy with a custom loss.

Why do we assume r follows the specific form? Consider the per prompt optimization with reward and KL-regularization,

$$\max_{\pi_\theta} \left\{ \mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\left[r(\mathbf{x},\mathbf{y})\right] - \beta \cdot \mathrm{D}_{\text{KL}}\left(\pi_\theta(\cdot \mid \mathbf{x}) \parallel \pi_{\theta_{\text{ref}}}(\cdot \mid \mathbf{x})\right)\right\}.$$

The objective itself is strictly concave in $\pi_\theta(\cdot \mid \mathbf{x})$ which is a probability distribution (a vector of probabilities) over all possible $\mathbf{y}$. Clearly for the first term, it is linear in $\pi_\theta(\cdot \mid y)$, while for the second term, we have

$$-\beta \cdot \sum_{\mathbf{y}} \pi_\theta(\mathbf{y} \mid \mathbf{x}) \log \frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})},$$

which is also strictly concave in $\pi_\theta(\cdot \mid \mathbf{x})$ (KL-divergence itself is convex in $\pi_\theta(\cdot \mid \mathbf{x})$). If we write the write the Lagrangian with normalization muiltipler $\lambda(\mathbf{x})$:

$$\mathcal{L}(\pi_\theta, \lambda) = \sum_{\mathbf{y}} \pi_\theta(\mathbf{y} \mid \mathbf{x}) \cdot r(\mathbf{x},\mathbf{y}) - \beta \sum_{\mathbf{y}} \pi_\theta(\mathbf{y} \mid \mathbf{x}) \cdot \log \frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})} + \lambda(\mathbf{x}) \left(\sum_{\mathbf{y}} \pi_\theta(\mathbf{y} \mid \mathbf{x}) - 1 \right),$$

whose optimal condition implies

$$r(\mathbf{x},\mathbf{y}) - \beta\left(\log \pi_\theta(\mathbf{y} \mid \mathbf{x}) - \pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x}) + 1\right) + \lambda(\mathbf{x}) = 0.$$

Rearrange we obtain:

$$r(\mathbf{x},\mathbf{y}) = \beta\left(\log \pi^\star(\mathbf{y} \mid \mathbf{x}) - \log \pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})\right) + C(\mathbf{x}).$$

where $C(\mathbf{x}) = \lambda(\mathbf{x}) + \beta$. Since in the pariwise preference model the bias $C(\mathbf{x})$ cancels out, this justifies our choice. In a word, the DPO choice of r comes from the PPO + KL setup, but with simplifications such as implicit reward model and sequence level loss instead of per token.

10.12 GRPO (Group Relative Preference Optimization)

GRPO generalizes DPO further, in the sense that it instead of relying only on binary preferences (preferred vs dispreferred), it allows training with arbitrary reward functions (scalar signals, multiple preferences, or mixtures).

Consider the dataset $\mathcal{D}_{\text{pref}} = \{(\mathbf{x}_j, \mathbf{y}_j, r(\mathbf{x}_j, \mathbf{y}_j))\}_{j=1}^M$ (Different x_j could be the same.). We now define the log-ratio between policy and preference

$$z_\theta(\mathbf{x},\mathbf{y}) = \log \frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})}.$$

The GRPO loss is given by

$$J^{\text{GRPO}}(\theta) = \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\mathcal{D}_{\text{pref}}}\left[\Omega^\star\left(\beta z_\theta(\mathbf{x},\mathbf{y})\right) - \beta r(\mathbf{x},\mathbf{y}) z_\theta(\mathbf{x},\mathbf{y})\right],$$

where $\Omega^\star$ is the convex conjugate of a link function Ω,$\beta > 0$. The training follows maximizing this dual objective. In practice we use an equivalent form:

$$\widehat{L}(\theta) = \sum_{x\in\mathcal{D}_{\text{pref}}} \mathcal{L}_{\mathbf{x}}(\theta) = -\beta \sum_{\mathbf{x}} \sum_{i=1}^{K_{\mathbf{x}}} \omega_i(\mathbf{x}) \log \pi_\theta\left(\mathbf{y}_i \mid \mathbf{x}\right), \qquad \omega_i(\mathbf{x}) = \frac{e^{r_i(\mathbf{y}_i,\mathbf{x})/\beta}}{\sum_{j=1}^{K_{\mathbf{x}}} e^{r_i(\mathbf{y}_j,\mathbf{x})/\beta}}$$

The intuition: Fix a prompt $\mathbf{x}$, consider the standard KL-regularized objective

$$\max_\pi \left\{\mathbb{E}_{\mathbf{y}\sim\pi(\cdot|\mathbf{x})}\left[r(\mathbf{x},\mathbf{y})\right] - \beta \cdot \mathrm{D}_{\text{KL}}\left(\pi\left(\cdot \mid \mathbf{x}\right) \parallel \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}\right)\right)\right\}.$$

We will reparameterize the policy via a density ratio. Let

$$z(\mathbf{y}) = \log \frac{\pi\left(\mathbf{y} \mid \mathbf{x}\right)}{\pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right)}, \qquad u(\mathbf{y}) = \exp\left(z(\mathbf{y})\right) = \frac{\pi\left(\mathbf{y} \mid \mathbf{x}\right)}{\pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right)}.$$

Then $\pi\left(\mathbf{y} \mid \mathbf{x}\right) = \pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right) u(\mathbf{y})$ must normalize, i.e.,

$$\sum_{\mathbf{y}} \pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right) u(\mathbf{y}) = 1 \Leftrightarrow \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u(\mathbf{y})\right] = 1.$$

Moreover,

$$\mathrm{D}_{\text{KL}}\left(\pi\left(\cdot \mid \mathbf{x}\right) \parallel \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{x}\right)\right) = \mathbb{E}_{\mathbf{y}\sim\pi(\cdot|\mathbf{x})}\left[z(\mathbf{y})\right]$$

and

$$\mathbb{E}_{\mathbf{y}\sim\pi(\cdot|\mathbf{x})}\left[f(\mathbf{y})\right] = \sum_{\mathbf{y}} \pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right) u(\mathbf{y}) f(\mathbf{y}) = \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u(\mathbf{y}) f(\mathbf{y})\right]. \tag{21}$$

As a result, taking $f(\mathbf{y}) = r(\mathbf{x},\mathbf{y})$, we can reform the original objective as

$$\max_{u\geq 0, \mathbb{E}_{\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}[u(\mathbf{y})]=1} \left\{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u(\mathbf{y}) r(\mathbf{x},\mathbf{y})\right] - \beta \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u(\mathbf{y}) z(\mathbf{y})\right]\right\}, \qquad z(\mathbf{y}) = \log u(\mathbf{y}).$$

The same conclusion also holds in the continuous case. We form the Lagrangian for the fixed $\mathbf{x}$,

$$\begin{aligned}\mathcal{L}\left(u,\lambda\right) &= \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[ur\right] - \beta \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[uz\right] - \lambda\left(\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u\right] - 1\right) \\ &= \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[-\left(\beta uz - ur + \lambda u\right)\right] + \lambda.\end{aligned}$$

Differentiate pointwisely w.r.t. u, we get

$$r - \beta(\log u + 1) - \lambda = 0,$$

which suggests

$$\log u(\mathbf{y}) = \frac{r(\mathbf{x},\mathbf{y}) - \lambda - \beta}{\beta} \Rightarrow u^\star(\mathbf{y}) = C \cdot \exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right), \quad \text{where} \quad C := \exp\left(-\frac{\lambda+\beta}{\beta}\right).$$

Enforcing the constraint $\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}[u] = 1$, we get

$$C \cdot \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r}{\beta}\right)\right] = 1 \Rightarrow C = \frac{1}{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r}{\beta}\right)\right]}.$$

Therefore

$$u^\star(\mathbf{y}) = \frac{\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)}{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r}{\beta}\right)\right]} \quad \text{and} \quad \pi^\star(\mathbf{y} \mid \mathbf{x}) = \pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})\, u^\star(y) = \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x}) \exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)}{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)\right]}.$$

The optimal policy is obtained by reweighting the reference distribution, so that outcomes with higher reward get exponentially more weight, this reweighting is what people called **exponential tilt**. The optimal log ratio is given by

$$z^\star(\mathbf{y}) = \frac{r(\mathbf{x},\mathbf{y})}{\beta} - \log \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)\right].$$

Since the elementwise objective is strictly concave for $u > 0$ and the constraint is affine, KKT conditions apply; complementary slackness gives an interior solution $u^\star(\mathbf{y}) > 0$, which makes the stationary point above a unique maximizer. Plug it back into the primal gives the optimal value

$$\beta \log \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r}{\beta}\right)\right],$$

which is of the classic Donsker-Varadhan form, consistent with the exponential-tilt solution. In principle, if we have explicit $r(\mathbf{x},\mathbf{y})$ for all candidate responses $\mathbf{y}$ we could just set our policy equal to $\pi^\star$, but in practice, the reward is only given on the dataset, and even if the value is available, we need to compute

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r}{\beta}\right)\right],$$

which is log-sum-exp (log-partition function) all possible completions $\mathbf{y}$, which is not computable.

The insight of GRPO: Instead of trying to match π_θ to the global closed form $\pi^\star$, we can design a tractable convex surrogate that drives the model's log-ratio[23]

$$z_\theta(\mathbf{y}) = \log\left(\frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})}\right),$$

to behave as if it were the solution of that optimization. Now introducing the convex function (entropy link)

$$\Omega(u) = \beta\left(u \log u - u + 1\right), \qquad \text{for} \qquad u > 0,$$

whose convex conjugate is

$$\Omega^\star(t) = \sup_{u\geq 0}\{ut - \Omega(u)\} = \beta\left(e^{t/\beta} - 1\right).$$

Now we use the Fenchel-Young inequality,

$$ut - \Omega(u) \leq \Omega^\star(t)$$

[23] The question is simply: how do we train a parameterized π_θ so that its log-ratio z_θ behaves like $z_\star$ without that partition function.

with equality achieved if and only if $u = e^{\frac{t}{\beta}}$, which means that for any $\mathbf{y}$, we have

$$u^\star(\mathbf{y}) \cdot \beta z_\theta(\mathbf{y}) - \Omega(u^\star(\mathbf{y})) \leq \Omega^\star(\beta z_\theta(\mathbf{y}))$$

Taking expectation under $\pi_{\theta_{\text{ref}}}(\cdot \mid \mathbf{x})$, we have

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u^\star(\mathbf{y}) \cdot \beta z_\theta(\mathbf{y})\right] - \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega(u^\star(\mathbf{y}))\right] \leq \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right]. \tag{22}$$

For the first term, using the change of measure in (21), we have

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u^\star(\mathbf{y}) \cdot \beta z_\theta(\mathbf{y})\right] = \beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right],$$

while for the second term, we have

$$\begin{aligned}
\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega(u^\star(\mathbf{y}))\right] &= \beta\left(\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u^\star(\mathbf{y})\log u^\star(\mathbf{y})\right] - \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[u^\star(\mathbf{y})\right] + 1\right) \\
&= \beta \cdot \mathbb{E}_{\pi^\star(\cdot|\mathbf{x})}\left[\log u^\star(\mathbf{y})\right] \\
&= \beta \cdot \mathbb{E}_{\pi^\star(\cdot|\mathbf{x})}\left[\frac{r(\mathbf{x},\mathbf{y})}{\beta} + C\right] \\
&= \beta \cdot \mathbb{E}_{\pi^\star(\cdot|\mathbf{x})}\left[\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right] + \beta C.
\end{aligned}$$

Plus this into the original inequality (22), we obtain

$$\beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right] - \left(\beta \cdot \mathbb{E}_{\pi^\star(\cdot|\mathbf{x})}\left[\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right] + \beta C\right) \leq \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right],$$

which rearranges to

$$\begin{aligned}
\beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right] - \mathbb{E}_{\pi^\star(\cdot|\mathbf{x})}\left[r(\mathbf{x},\mathbf{y})\right] &\leq \beta C + \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right], \\
&= \beta\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[e^{z_\theta(\mathbf{y})} - 1\right]
\end{aligned}$$

where the inequality is tight when $\log z_\theta(\mathbf{y}) = u^\star(\mathbf{y}) = \log z^\star(\mathbf{y})$. This is essentially an convex upper bound that is tight when $z_\theta(\mathbf{y}) = z^\star(\mathbf{y})$. Notice that we are trying to optimize towards z, so we can write the above inequality as

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right] - \beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right] \geq \text{constant}.$$

We can define the following objective from Fenchel-Young,

$$\mathcal{S}_\mathbf{x}(z_\theta) := \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right] - \beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right] - \text{constant},$$

where $\mathcal{S}_\mathbf{x}(z_\theta) \geq 0$, $\mathcal{S}_\mathbf{x}(z_\theta) = 0$ if and only if $z = z^\star$ and clearly it is convex. Notice that this surrogates resembles the original GRPO loss. This surrogate serves as a way to drive z_θ to what we want without ever computing the log partition. Notice that for the first term, we have

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\Omega^\star(\beta z_\theta(\mathbf{y}))\right] = \beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[e^{z_\theta}\right] - \beta,$$

while

$$\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[e^{z_\theta}\right] = \sum_{y\sim\pi_\theta(\cdot|\mathbf{x})} \pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x}) \cdot \frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y} \mid \mathbf{x})} = 1.$$

Thus we can further write

$$\mathcal{S}_{\mathbf{x}}(z_\theta) = -\beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[z_\theta(\mathbf{y})\right],$$

leaving out the constant. Differentiate it w.r.t θ, and notice that

$$\nabla_\theta z_\theta(\mathbf{y}) = \nabla_\theta \log \pi_\theta\left(\mathbf{y} \mid \mathbf{x}\right).$$

Hence

$$\nabla_\theta \mathcal{S}_{\mathbf{x}}(z_\theta) = -\beta \cdot \mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[\nabla_\theta \log \pi_\theta\left(\mathbf{y} \mid \mathbf{x}\right)\right],$$

which is the population gradient we want to estimate from data. Notice that we can not compute the term $\mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[f(\mathbf{y})\right]$ directly because we do not know the distribution, however, we can estimate it, with rewards. We have

$$\begin{aligned}\mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[f(\mathbf{y})\right] &= \sum_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})} \pi^\star(\mathbf{y} \mid \mathbf{x}) \cdot f(\mathbf{y}) \\ &= \frac{\pi_{\theta_{\text{ref}}}\left(\mathbf{y} \mid \mathbf{x}\right) \exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right) \cdot f(\mathbf{y})}{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)\right]} = \frac{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right) \cdot f(\mathbf{y})\right]}{\mathbb{E}_{\mathbf{y}\sim\pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})}\left[\exp\left(\frac{r(\mathbf{x},\mathbf{y})}{\beta}\right)\right]}.\end{aligned}$$

Assume for prompt $\mathbf{x}$, we have K responses with rewards $\{(\mathbf{y}_i, r_i)\}_{i=1}^{K_{\mathbf{x}}}$, then we can estimate the expectation using[24]

$$\mathbb{E}_{\mathbf{y}\sim\pi^\star(\cdot|\mathbf{x})}\left[f(\mathbf{y})\right] \approx \frac{\frac{1}{K}\sum_{i=1}^{K_{\mathbf{x}}} e^{r_i/\beta} f(\mathbf{y}_i)}{\frac{1}{K_{\mathbf{x}}}\sum_{i=1}^{K_{\mathbf{x}}} e^{r_i/\beta}} = \sum_{i=1}^{K_{\mathbf{x}}} \omega_i(\mathbf{x}) f(\mathbf{y}_i), \qquad \omega_i(\mathbf{x}) := \frac{e^{r_i/\beta}}{\sum_{i=1}^{K_{\mathbf{x}}} e^{r_i/\beta}},$$

where the problematic As a result, the estimated gradient is given by

$$\nabla_\theta \mathcal{S}_{\mathbf{x}}(z_\theta) \approx -\beta \cdot \sum_{i=1}^{K_{\mathbf{x}}} \omega_i(\mathbf{x}) \nabla_\theta \log \pi_{\pi_\theta}\left(\mathbf{y}_i \mid \mathbf{x}\right),$$

which is a **weighted negative log-likelihood**. corresponding to

$$\widehat{\mathcal{L}}_{\mathbf{x}}(\theta) := -\beta \cdot \sum_{i=1}^{K_{\mathbf{x}}} \omega_i(\mathbf{x}) \log \pi_\theta\left(\mathbf{y}_i \mid \mathbf{x}\right),$$

if we further aggregate over prompts

$$\widehat{L}(\theta) = \sum_{x\in\mathcal{D}_{\text{pref}}} \mathcal{L}_{\mathbf{x}}(\theta) = -\beta \sum_{\mathbf{x}} \sum_{i=1}^{K_{\mathbf{x}}} \omega_i(\mathbf{x}) \log \pi_\theta\left(\mathbf{y}_i \mid \mathbf{x}\right),$$

which is clearly equivalent to the original GRPO dual loss, since it is our starting point (denoted as $\mathcal{S}_{\mathbf{x}}(z_\theta)$ here). Note that we are trying to minimize this loss as we are trying to minimizer $\mathcal{S}_{\mathbf{x}}$.

[24] This is the key, the expectation itself is easier to estimate than the distribution.

Relation to DPO DPO is related to GRPO, let a training item be a pair $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$ with the label $\mathbf{y}^+ \succ \mathbf{y}^-$. From the GRPO analysis, we have

$$z^\star(\mathbf{x}, \mathbf{y}) = \frac{r(\mathbf{x}, \mathbf{y})}{\beta} - \log \mathbb{E}_{\mathbf{y} \sim \pi_{\theta_{\text{ref}}}(\cdot|\mathbf{x})} \left[\exp\left(\frac{r(\mathbf{x}, \mathbf{y})}{\beta} \right) \right],$$

the second partition term is eliminated if we take

$$\Delta z^\star(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) = z^\star(\mathbf{x}, \mathbf{y}^+) - z^\star(\mathbf{x}, \mathbf{y}^-) = \frac{r(\mathbf{x}, \mathbf{y}^+) - r(\mathbf{x}, \mathbf{y}^-)}{\beta}.$$

If preferences follow a Bradley-Terry (logistic) model with temperature, then, we have

$$\mathbb{P}(\mathbf{y}^+ \succ \mathbf{y}^- \mid \mathbf{x}) = \sigma\left(\frac{r(\mathbf{x}, \mathbf{y}^+) - r(\mathbf{x}, \mathbf{y}^-)}{\beta} \right) = \sigma\left(\Delta z^\star(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \right).$$

To make z_θ mimics $z_\star$, we apply a pairwise logistic surrogate to the margin,

$$\begin{aligned} m_\theta(\mathbf{x}; \mathbf{y}^+, \mathbf{y}^-) &:= \frac{1}{\beta} \left(z_\theta(\mathbf{x}, \mathbf{y}^+) - z_\theta(\mathbf{x}, \mathbf{y}^-) \right) \\ &= \frac{1}{\beta} \left(\log \pi_\theta\left(\mathbf{y}^+ \mid \mathbf{x}\right) - \log \pi_\theta\left(\mathbf{y}^- \mid \mathbf{x}\right) - \log \pi_{\theta_{\text{ref}}}\left(\mathbf{y}^+ \mid \mathbf{x}\right) + \log \pi_{\theta_{\text{ref}}}\left(\mathbf{y}^- \mid \mathbf{x}\right) \right). \end{aligned}$$

The DPO loss is exactly the logistic loss that pushes m^θ to $\Delta z^\star$, i.e.,

$$L^{\text{DPO}}(\theta) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)} \left[\log \sigma\left(m_\theta(\mathbf{x}; \mathbf{y}^+, \mathbf{y}^-) \right) \right]$$

Note that this is something to minimize, and in the original introduction to DPO, we are writing the goal in terms of maximization. If we differentiate with respect to θ, we get

$$\begin{aligned} \nabla_\theta \left(-\log \sigma(m_\theta) \right) &= -\sigma(-m_\theta) \cdot \nabla_\theta m_\theta \\ &= -\frac{\sigma(-m_\theta)}{\beta} \left(\nabla_\theta \log \pi_\theta\left(\mathbf{y}^+ \mid \mathbf{x}\right) - \nabla_\theta \log \pi_\theta\left(\mathbf{y}^- \mid \mathbf{x}\right) \right) \\ &= -\frac{1}{\beta} \left(\sigma(-m_\theta) \nabla_\theta \log \pi_\theta\left(\mathbf{y}^+ \mid \mathbf{x}\right) - \sigma(-m_\theta) \nabla_\theta \log \pi_\theta\left(\mathbf{y}^- \mid \mathbf{x}\right) \right). \end{aligned}$$

As we can see, this is a weighted sum of the log-likelihood gradients, but the coefficient of the less preferred sample is negative, which means that one "weight" is negative. It is not representable as a weighted Negative Log Likelihood (NLL) form, GRPO generalizes DPO in spirits. To be specific, DPO uses a pairwise logistic surrogate focuses only on to relative differences, which leads to negative coefficients for the dispreferred sample.

Relation to PPO and its implementation Let us recall the entire PPO objective:

$$\begin{aligned} J^{\text{PPO}}(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \pi_{\text{old}}} \big[\min \Big\{ o_t(\theta; \theta_k) \hat{A}^t, \text{clip}\left(o_t(\theta; \theta_k), 1 - \varepsilon, 1 + \varepsilon \right) \cdot \hat{A}^t \Big\} \\ - \beta \cdot \mathrm{D}_{\text{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right) \parallel \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{s}_t\right) + \lambda_{\text{ent}} H(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right)) \right) \big]. \end{aligned}$$

There is one implicit constraint here, that is, we do clipping to ensure that the ration is in a certain range, which enforces we are not deviating too much from the old policy θ_{old}, this increases stability. There is also

an explicit constraint via KL divergence, which prevents the policy deviating too much from the reference policy θ_{ref}. Here the o_t is defined as:

$$o_t(\theta;\theta_k) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\text{old}}(\mathbf{a}_t \mid \mathbf{s}_t)} = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_k}(\mathbf{a}_t \mid \mathbf{s}_t)}.$$

In GRPO, we simpliy removes the arbitrary clipping term, and anchor everything to the referrence model $\pi_{\theta_{\text{ref}}}$. We can write the GRPO objective as:

$$J^{\text{GRPO}}(\theta) = \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\pi_{\theta_k}}\left[o_t(\theta;\theta_{\text{ref}})\hat{A}^t - \beta_k \cdot \mathrm{D}_{\text{KL}}\left(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right) \parallel \pi_{\theta_{\text{ref}}}\left(\cdot \mid \mathbf{s}_t\right) + \lambda_{\text{ent}} H(\pi_\theta\left(\cdot \mid \mathbf{s}_t\right))\right)\right].$$

which is the on policy version of the GRPO objective. Here the $o_t(\theta;\theta_{\text{ref}})$ is defined as:

$$o_t(\theta;\theta_{\text{ref}}) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{ref}}}(\mathbf{a}_t \mid \mathbf{s}_t)}.$$

Notice that we will use β_k as a dynamic controller to control the KL divergence, and it will be updated in each iteration. If the KL exceeds the target, we will increase β_k, otherwise, we will decrease β_k. Notice that sometimes we will see $o_t(\theta;\theta_{\text{ref}})$ which is called the ratio format, while sometimes we will see $\log \pi(\mathbf{a}_t \mid \mathbf{s}_t)$, which is the log format, and they are equivalent from the derivation of the policy gradient method. This is the on policy GRPO, sometimes we will see the off policy version, where we use $\pi_{\theta_{\text{ref}}} = \mu$, i.e., the behavior policy, the rollouts are generated from this behavoir policy and we need the importance sampling trick to correct the distribution mismatch. In practice, if we use $\mu = \theta_k$ which is the current policy, then it is the same as the on policy version, we can also let it be some old policy, which allows us to reuse the rollouts generated from the old policy, or we can let it be the reference policy, which allows us to reuse the rollouts generated from the reference policy. Note that PPO fixes $\mu = \pi_{\text{old}}$ which is the old policy.

Group In GRPO, we do not treat each response-prompt pair independently. Instead, we group multiple responses (and their rewards) that share the same prompt into a group, so we can compare them and apply relative regularization or preference-based objectives.

- **Preference-based objective:** For a pair of responses $\mathbf{y}^+$ and $\mathbf{y}^-$, a preference model (human or reward model) tells us $r^+ > r^-$. Then the preference-based loss (as in DPO or GRPO) tries to enforces

$$\log \pi_\theta(\mathbf{y}^+ \mid \mathbf{x}) - \log \pi_\theta(\mathbf{y}^- \mid \mathbf{x})$$

should be large. This is equivalent to minimize

$$-\log \sigma\left(\beta\left(\log \frac{\pi_\theta(\mathbf{y}^+ \mid \mathbf{x})}{\pi_\theta(\mathbf{y}^- \mid \mathbf{x})} - \log \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}^+ \mid \mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}^- \mid \mathbf{x})}\right)\right),$$

which is a relative formulation.

- **Relative regularization:** In GRPO, we extend this to groups of size ≥ 2 (say 4). For each group of responses $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4\}$ for the same prompt $\mathbf{x}$, we compute relative preferences using a reward model or normalized scores

$$\omega_i = \frac{\exp(\beta r_i)}{\sum_j \exp(\beta r_j)},$$

then optimize policy probabilities to match this distribution,

$$\min_\theta \mathrm{D}_{\text{KL}}\left(\omega_i \parallel \frac{\pi_\theta(\mathbf{y}_i \mid \mathbf{x})}{Z_\theta(x)}\right),$$

where $Z_\theta(x) = \sum_j \pi_\theta(\mathbf{y}_i \mid \mathbf{x})$. One thing to notice is that we do not actually compute the term explicitly, as it has already been captured by the advantage estimate.

Practical Hint: In practice, for each prompt $\mathbf{x}_i$, we sample a group of K resposnes,

$$\{\mathbf{y}_{i1}, \mathbf{y}_{i2}, \ldots, \mathbf{y}_{iK}\} \sim \pi_\theta(\cdot \mid \mathbf{x}_i),$$

and use the corresponding reward r_{ik} to compute the **group standardrized advantage**

$$\hat{A}_{ik} = \frac{r_{ik} - \bar{r}_i}{\sigma_i + \varepsilon}, \qquad \text{where } \bar{r}_i = \frac{1}{K}\sum_{k=1}^{K} r_{ik}, \sigma_i = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(r_{ik} - \bar{r}_i)^2}.$$

The GRPO loss can be written as

$$J^{\text{GRPO}} = \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\pi_\theta}\left[\hat{A}(\mathbf{x},\mathbf{y}) \cdot \log \pi_\theta(\mathbf{y} \mid \mathbf{x})\right].$$

Optionally, we can add a KL penalty term

$$J^{\text{GRPO-reg}} = \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\pi_\theta}\left[\hat{A}(\mathbf{x},\mathbf{y}) \cdot \log \pi_\theta(\mathbf{y} \mid \mathbf{x}) - \beta \cdot \mathrm{D}_{\text{KL}}\left(\pi_\theta \parallel \pi_{\theta_{\text{ref}}}\right)\right]$$

10.13 On Policy and Off Policy

This is one of the most fundamental distinctions in reinforcement learning

- **On policy** refers to the case that the agent is learning about the same policy that it is currently following. It gathers experience and then update that exact strategy.
- **Off policy** refers to the case that the agent wants to learn about a different policy than the one it is using to act. It might behave randomly or follow an old policy, but still update another (better) policy.

The key difference is, in on policy, we learn from what we actually do and in off policy, we learn from what we (or the others) did, but update toward a different target strategy.

Some examples include:

1. **PPO:** This is on policy, because PPO collects rollouts (trajectories) using the current policy π_θ, and immediately uses them to update the same policy.

2. **DPO:** This is not really RL in the classical sense, DPO is usually described as offline preference-based fine-tuning. So in strict RL terms: it's neither on-policy nor off-policy, because it doesn't collect trajectories with a policy. (It behaves like an offline / off-policy method, since it learns from pre-collected data.)

3. **GRPO:** Same as DPO — not classical on-policy RL. It's trained on a static dataset of human preferences, so it's closer to off-policy / offline RLHF-style training.

This is actually related to the broader range of reinforcement learning problem. In reinfocrement learning, we usually have three types of methods:

1. Value based methods: learn the value of states or actions (e.g. Q function), and then derive a policy by picking the best action. Representive algorithms include Q-learning on small problems, and DQN (Deep-Q Networks) on large state spaces. They are usually off-policy.

2. Policy based methods: directly learn the policy $\pi_\theta(\mathbf{a} \mid \mathbf{s})$. Representive algorithms include: REINFORCE (vanilla policy gradient), Actor-Critic (has both a policy and a value function helper), PPO (Proximal Policy Optimization), TRPO (Trust Region Policy Optimization). They are usually on policy. While they need fresh data each update, they work well with continuous or large action spaces.

3. Model based methods: learn a model of the dynamics of the environment (state transitions and rewards), then plan ahead inside that model. Representative algorithms include: Dyna-Q (classic), MuZero (modern, combines deep learning + planning). Although they can be very sample-efficient if the model is accurate, it is hard to learn good models for complex environments.

Value based methods: Let us recall the utimate goal of reinforcement learning:

$$\pi^\star = \arg\max_\pi \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t\right],$$

and the definition of a Q function

$$Q^\pi(\mathbf{s}, \mathbf{a}) = \mathbb{E}\left[\text{future reward} \mid \text{start in } s\text{, take } a\text{, then follow } \pi\right] = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \mathbf{s}_t = \mathbf{s}, \mathbf{a}_t = \mathbf{a}\right].$$

The idea is that if we know

$$Q^\star(\mathbf{s}, \mathbf{a}) = \max_\pi Q^\pi(\mathbf{s}, \mathbf{a}),$$

then we can find the corresponding best policy

$$\pi^\star(\mathbf{s}) = \arg\max_{\mathbf{a}} Q^\star(\mathbf{s}, \mathbf{a}).$$

SARSA: State-Action-Reward-State-Action (On policy) Update rule:

$$Q(\mathbf{s}_t, \mathbf{a}_t) \leftarrow Q(\mathbf{s}_t, \mathbf{a}_t) + \alpha\left[r_t + \gamma \cdot Q(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) - Q(\mathbf{s}_t, \mathbf{a}_t)\right],$$

which means that when we are in state $\mathbf{s}_t$ taking action $\mathbf{a}_t$ from our policy, we observe the reward r_t and land in the next state $\mathbf{s}_{t+1}$, our policy tells us to take some next action $\mathbf{a}_{t+1}$ in that state. In this case, the update rule can be intepreted as update our $Q(\mathbf{s}_t, \mathbf{a}_t)$ so that it is closer to reward now + discounted reward of the (next state, next action), α is like a step size controlling how fast we proceed.

SARSA, as a result, is on policy and if our policy is ϵ-greedy (sometimes exploring randomly instead of picking the action with the highest Q-value so far.), SARSA will learn the value of that ϵ-greedy policy (SARSA converges to the value function of the behavior policy we actually follow), not necessarily the optimal greedy one.

Q-learning (Off-policy) Update rule:

$$Q(\mathbf{s}_t, \mathbf{a}_t) \leftarrow Q(\mathbf{s}_t, \mathbf{a}_t) + \alpha\left[r_t + \gamma \max_{\mathbf{a}'} Q(\mathbf{s}_{t+1}, \mathbf{a}') - Q(\mathbf{s}_t, \mathbf{a}_t)\right].$$

Instead of looking at the action we actually took next, we imagine what if we take the best possible action in $\mathbf{s}_{t+1}$, which results in the term $\max_{\mathbf{a}'} Q(\mathbf{s}_{t+1})$, and we are updating toward reward now + discounted value of the best possible next step.

As a result, Q-learning is off-policy: our behavior policy might explore randomly, but our updates always push you toward the optimal greedy policy. That's why Q-learning converges (under conditions) to $Q^\star$.

10.14 Domain Adaption

In machine learning, we usually assume that training and test data come from the same distribution. But in practice, this often fails (for example: training a sentiment classifier on Amazon reviews and testing it on Twitter posts). This motivates domain adaptation (DA), a subfield of transfer learning. Formally, a domain is formally defined as a pair

$$\mathcal{D} = (\mathcal{X}, P(X)),$$

where $\mathcal{X}$ is an input space (images, sentences), and $P(X)$ is a mariginal distribution over $\mathcal{X}$ (we denote the original distribution as $P(\mathcal{X}, \mathcal{Y})$), and a task is defined as

$$\mathcal{T} = (\mathcal{Y}, f)\,,$$

where $\mathcal{Y}$ is the output (label) space, $f : \mathcal{X} \mapsto \mathcal{Y}$ is the true labeling function. We consider source domain $\mathcal{D}_S = (\mathcal{X}_S, P_S(X))$ and target domain $\mathcal{D}_T = (\mathcal{X}_T, P_T(X))$, we assume

1. **Different distributions:** $P_S(X) \neq P_T(X)$, but typically, $\mathcal{X}_S = \mathcal{X}_T$ and $\mathcal{Y}_S = \mathcal{Y}_T$.
2. **Same task (most classical setting):** $f_S = f_T = f$, variants exist where tasks differ, but then we are in broader transfer learning.
3. **Label scarcity:** Often, the source domain has many labeled samples: $\left\{\left(x_i^S, y_i^S\right)\right\}_{i=1}^{n_S} \sim P_S(\mathcal{X}, \mathcal{Y})$ while target domain may have few or zero labeled samples: $\left\{x_j^T\right\}_{j=1}^{n_T} \sim P_T(X)$.

We want to learn a hypothesis (model) $h : \mathcal{X} \mapsto \mathcal{Y}$ that minimizes target risk:

$$\epsilon_T(h) = \mathbb{E}_{(x,y)\sim P_T(x,y)}\left[\mathcal{L}(h(x), y)\right],$$

given most supervision comes from $\mathcal{D}_S$.

Typical approaches to handle this problem includes:

1. **Instance based methods**: reweight the the source samples so that they look like they were drawn from the target distribution.

$$\epsilon_T(h) = \mathbb{E}_{(x,y)\sim P_T(x,y)}\left[\frac{P_T(x)}{P_S(x)}\mathcal{L}(h(x), y)\right],$$

   there is no need to change the struture of the model, but the ratios are hard to estimate especially in high dimensions.

2. **Feature based methods**: Learn a transformation Φ, such that the source and target distributions are aligned in feature space $P_S(\Phi(X)) \approx P_T(\Phi(X))$, Φ can be viewed as a feature extractor (a mapping from the raw input space $\mathcal{X}$ to some latent space $\mathcal{Z}$), in practice, Φ is usually the encoder part of the model (e.g. the neural net before the final classifier head). For a neural model h_θ, we can think of

$$h_\theta(x) = g_\theta(\Phi(x)).$$

   The introduction of the feature map $\Phi(\cdot)$ allows us to transform soucre and target into a feature space $\mathcal{Z}$, where they will look very similar. For an exemplary pipeline: (i) an encoder Φ is trained so that features of source and target overlap. (ii) Classifier g_θ then works equally well across domains.

3. **Parameter based methods**: Assume the parameters of source and target models should share some structure, examples include shared encoder across domains, domain-specific classification heads. Useful when domains are similar but not identical. For a example, we assume some shared parameters across source and target, and some domain-specific parameters, we finetune a subset of layers on target domain data.

11 Retrieval-Augmented Generation

Retrieval-Augmented Generation (**RAG**) is a hybrid neural-information-retrieval + text-generation framework that augments a language model with access to an externel knowledge store during inference (and sometimes training). This is needed because an LLM's internal weight can store parametric knowledge that could be outdated or incomplete, and we need RAG to augments it with externel knowledge retrieved at run time.

Mechanism:

- **Encode the user query into a vector.**
- **Retrieve top-k relevant documents** from a pre-built knowledge index using vector similarity.
- **Fuse retrieval results** into the prompt.
- **Generate** the answer using the LLM conditioned on this augmented prompt[25].

In a word, it is feeding a richer prompt built from relevant retrieved context, and then it performs normal next-token generation. RAG can reduce hallucinations, increase factuality and/or ground the generation model on a specific dataset.

End to end training Normally, RAG has two separate components, the retriever η and the generator θ, and in standard practice, the retriever is frozen and only the generator is finetuned. However, it is possible to treat the retrieval step as a latent variable in the full log likelihood

$$\mathcal{L}(\theta, \eta) = -\log \sum_{\mathbf{z}} p_\eta(\mathbf{z} \mid \mathbf{x}) p_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{z}),$$

and we update on both sets of parameters (θ, η) using gradients depending on how retrieval affects the final generation. So the retriever learns to retrieve docs that make the generator's answers more likely / accurate.

However, this is hard and costly, since the summation over docs is huge, and retrieval itself involves discrete choices which means gradient cannot flow directly.

11.1 Advanced RAG Techniques

Query Rewriting Simple RAG implementation would likely struggle with the search step if a question is too verbose, or to refer to context in previous messages in the conversation. In this case, it is a good idea to use an LLM to rewrite the query into one that aids the retrieval step in getting the right information.

Multi-query RAG We can also extend the query rewriting to be able to search multiple queries if more than one is needed to answer a specific question. A classic example would be the user asking the following question: "Compare the data of ... in 2024 vs. 2025", we can perhaps find such a comparison directly, but we can also make two search queries and present the top results of both queries to the model for grounded generation. In addition, we can also give the query rewriter the option to determine if no search is required and if it can directly generate a confident answer without searching.

[25] This is also known as grounded generation.

Multi-hop RAG A more advanced question may require a series of sequential queries. Example: User Question: "Who are the largest car manufacturers in 2023? Do they each make EVs or not?"

To answer this, the system must first search for:

Step 1, Query 1: "largest car manufacturers 2023"

Then after it gets this information (the result being Toyota, Volkswagen, and Hyundai), it should ask follow-up questions:

Step 2, Query 1: "Toyota Motor Corporation electric vehicles"

Step 2, Query 2: "Volkswagen AG electric vehicles"

Step 2, Query 3: "Hyundai Motor Company electric vehicles"

Query Routing Give the model the ability to search multiple data sources.

Agentic RAG Delegates more and more responsibility to the LLM to solve more and more complex problems. This relies on the LLM's capability to gauge the required information needs as well as its ability to utilize multiple data sources.

11.2 RAG Evaluation

Fluency, Perceived utility (Whether the generated answer is helpful and informative.), Citation recall / precision, LLM-as-a-judge.

12 Parallelism

There are different types of parallelism, such as

- **Data Parallelism (DP):** Each GPU has a full copy of the model, different GPUs process different data batches, need to synchronize gradients so all model copies stay in sync. Examples incldue raw DDP, ZeRO-1/2/3, which is memory efficient. This is always used, and is the foundation.

- **Tensor Parallelism (TP):** Split the tensor (maths) inside a layer across GPUs, such as a giant weigt matrix $\mathbf{W} \in \mathbb{R}^{d \times d}$ can be row-sharded and column-sharded. Each forward/backward involves collective ops (all-reduce, all-gather). An example would be splitting a large weight matrix so that each GPU only holds a part of it, and they work together to do the full matrix multiplication (Each GPU only handles a fraction of the FLOPs.). Activations are then synced (e.g. via all-reduce or all-gather) to assemble the layer's output. However, this comes at the price of extra communication overhead. This is more about **mathematical parallelism** within layers. Example implementations include Megatron-LM tensor parallelism and Nvidia's Megatron-Deepspeed hybrid. This is needed when individual layers are too big for a single GPU.

- **Fully Sharded Data Parallelism (FSDP):** Shards all parameters, gradients, and optimizer states across devices. It works in a way that during forward pass: just before we need a layer's weights, they're gathered from all shards into full form, and after the forward/backward, they're sharded back. So, at any instant, only one layer's full weights need to live on a GPU, which brings memory usage down, but this comes at a price of lots of all-gather and reduce-scatter around each layer.

- **Hybrid:** One may think of combining TP + FSDP. Refer to the next subsection for an example of TP, FSDP and TP + FSDP. On top of this, we can also add DP, where an example of dimension 2 would be to divide 8 GPUs into 4 groups of 2 GPUs each, each group does TP + FSDP, and across groups we do DP.

- **Pipeline Parallelism (PP):** Split the model layers across GPUs (like an assembly line), each GPU owns a block of layers, passes activations to the next stage. There are techniques exist such as microbatching that keeps the pipeline busy. Examples include GPipe, PipeDream and DeepSpeed pipeline parallelism. This is used when the model is too deep to fit on a single GPU's memory.
- **Sequence Parallelism:** Split the sequence length dimension across GPUs.Each GPU processes a slice of the tokens. Reduces memory by distributing activations along the time axis. Examples: Megatron-LM introduced sequence parallelism for attention/MLP layers. This is used when very long context windows are present.
- **Expert / Mixture-of-Experts (MoE) Parallelism:** Explained before.

12.1 An example of TP, FSDP and TP + FSDP:

We will use a simple linear layer $\mathbf{Y} = \mathbf{X}\mathbf{W}$, $\mathbf{X} \in \mathbb{R}^{B \times d_{\text{in}}}$, $\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and $\mathbf{Y} \in \mathbb{R}^{B \times d_{\text{out}}}$ to illustrate how TP, FSDP and TP + FSDP works, and consider the case of 2 GPUs.

FSDP only:

- We let GPU 0 holds $\mathbf{W}_{0:} = \mathbf{W}[0 : \frac{d_{\text{in}}}{2}, :] \in \mathbb{R}^{\frac{d_{\text{in}}}{2} \times d_{\text{out}}}$ and GPU 1 holds $\mathbf{W}_{1:} = \mathbf{W}[\frac{d_{\text{in}}}{2} : d_{\text{in}}, :] \in \mathbb{R}^{\frac{d_{\text{in}}}{2} \times d_{\text{out}}}$.
- When computing:
  1. We all-gather $\mathbf{W}$, which lets both GPUs have the full $\mathbf{W}$.
  2. Each GPU then compute $\mathbf{Y} = \mathbf{W}\mathbf{X}$ locally.
  3. Backward pass is similar, we compute local gradients, then reduce-scatter to shard the gradients back to $\mathbf{W}_{0:}$ and $\mathbf{W}_{1:}$.
- Clearly this is memory efficient since no GPU holds full $\mathbf{W}$ permanently, but each GPU still does the full FLOPs.

TP only:

- We let GPU 0 holds $\mathbf{W}_{:0} = \mathbf{W}[:, 0 : \frac{d_{\text{out}}}{2}] \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}$ and GPU 1 holds $\mathbf{W}_{:1} = \mathbf{W}[:, \frac{d_{\text{out}}}{2} : d_{\text{out}}] \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}$.
- During forward:
  1. Rank 0 computes $\mathbf{Y}_0 = \mathbf{X}\mathbf{W}_{:0} \in \mathbb{R}^{B \times \frac{d_{\text{out}}}{2}}$.
  2. Rank 1 computes $\mathbf{Y}_1 = \mathbf{X}\mathbf{W}_{:1} \in \mathbb{R}^{B \times \frac{d_{\text{out}}}{2}}$.
  3. We all-gather activations to concatenate $\mathbf{Y} = [\mathbf{Y}_0, \mathbf{Y}_1] \in \mathbb{R}^{B \times d_{\text{out}}}$.
- During backward:
  1. The upstream gradient $\nabla\mathbf{Y} \in \mathbb{R}^{B \times d_{\text{out}}}$ is split consistently as $\nabla\mathbf{Y} = [\nabla\mathbf{Y}_{:0}, \nabla\mathbf{Y}_{:1}]$, where each rank receives only its local slice.

2. Each rank computes its local weight gradient:

$$\nabla \mathbf{W}_{:j} = \mathbf{X}^\top \nabla \mathbf{Y}_{:j} \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}, \quad j \in \{0,1\}.$$

   (No communication required.)

3. Each rank computes its partial input gradient:

$$\nabla \mathbf{X}_j = \nabla \mathbf{Y}_{:j} \mathbf{W}_{:j}^\top \in \mathbb{R}^{B \times d_{\text{in}}}.$$

4. We all-reduce (sum) these partials across ranks to obtain the full input gradient:

$$\nabla \mathbf{X} = \nabla \mathbf{X}_0 + \nabla \mathbf{X}_1.$$

   This is due to the formula based on chain rule:

$$\nabla \mathbf{X} = \nabla \mathbf{Y} \mathbf{W}^\top = \nabla \mathbf{Y}_{:0} \mathbf{W}_{:0}^\top + \nabla \mathbf{Y}_{:1} \mathbf{W}_{:1}^\top.$$

The formulas are easy to derive if we write out the sums explicitly[26].

- FLOPs are halved per GPU, Each GPU stores only its slice of $\mathbf{W}$, and we need synchronization for every layer's forward and backward.

TP + FSDP: Assume now we are in a little bit more complicated setting where we have 4 GPUs, and we want to do TP + FSDP with 2 GPUs for each.

- We will use index $i \in \{0, 1\}$ for FSDP shard and $j \in \{0, 1\}$ for TP shard.
- We first do TP splitting, where we have 2 GPUs in each TP group:

$$\mathbf{W} = [\mathbf{W}_{:0}, \mathbf{W}_{:1}], \quad \mathbf{W}_{:0} \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}, \mathbf{W}_{:1} \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}.$$

  Then we do FSDP splitting within each TP shard:

$$\mathbf{W}_{:j} = [\mathbf{W}_{0j}, \mathbf{W}_{1j}], \quad \mathbf{W}_{0j} \in \mathbb{R}^{\frac{d_{\text{in}}}{2} \times \frac{d_{\text{out}}}{2}}, \mathbf{W}_{1j} \in \mathbb{R}^{\frac{d_{\text{in}}}{2} \times \frac{d_{\text{out}}}{2}}.$$

- In the forward path:
  1. For the j-th TP block, we all-gather rows on the two GPU we have for FSDP, on each GPU, we reconstruct $\mathbf{W}_{:j} \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}$.
  2. We do local matmul and obtain

$$\mathbf{Y}_{:j} = \mathbf{X} \mathbf{W}_{:j} \in \mathbb{R}^{B \times \frac{d_{\text{out}}}{2}}$$

  3. Then we do TP all-gather activations, which allows us to obtain

$$\mathbf{Y} = [\mathbf{Y}_{:0}, \mathbf{Y}_{:1}] \in \mathbb{R}^{B \times d_{\text{out}}}$$

  4. Then we do FSDP reshard, each GPU only keeps $\mathbf{W}_{ij}$, free the temporary all-gathered parameter.
- In the backward path:

[26] A simple way of derivation would be to focus directly on the dimensions.

1. We are given the upstream gradient

$$\nabla \mathbf{Y} = [\nabla \mathbf{Y}_{:0}, \nabla \mathbf{Y}_{:1}], \qquad \nabla \mathbf{Y}_{:j} \in \mathbb{R}^{B \times \frac{d_{\text{out}}}{2}}.$$

2. FSDP all gather rows of each $\mathbf{W}_{:j}$ again.
3. Local weight gradient, then reduce scatter back into shards,

$$\nabla \mathbf{W}_{:j} = \mathbf{X}^\top \nabla \mathbf{Y}_{:j} \in \mathbb{R}^{d_{\text{in}} \times \frac{d_{\text{out}}}{2}}.$$

   and each rank (i, j) keeps $\nabla \mathbf{W}_{ij} \in \mathbb{R}^{\frac{d_{\text{in}}}{2} \times \frac{d_{\text{out}}}{2}}$.
4. Input gradient,

$$\nabla \mathbf{X}_j = \nabla \mathbf{Y}_{:j} \mathbf{W}_{:j}^\top \in \mathbb{R}^{B \times d_{\text{in}}}$$

   and then all reduce to

$$\nabla \mathbf{X} = \nabla \mathbf{X}_0 + \nabla \mathbf{X}_1.$$

- The memory cost now becomes $\frac{d_{\text{in}} d_{\text{out}}}{4}$ parameters per rank, this is easy to understand, because parameters, optimizer states and gradients all correspond to the $\frac{d_{\text{in}} d_{\text{out}}}{4}$ matrix.
- The computational baseline ($\mathbf{XW}$) costs $2Bd_{\text{in}}d_{\text{out}}$ FLOPs (1 addition plus 1 multiplication). While if we do TP only, then each rank needs $2Bd_{\text{in}}\frac{d_{\text{out}}}{2} = Bd_{\text{in}}d_{\text{out}}$ FLOPs which is half of baseline, thats why TP reduced computation per rank. In the case of TP + FSDP, within each TP group, each rank still need $2Bd_{\text{in}}\frac{d_{\text{out}}}{2} = Bd_{\text{in}}d_{\text{out}}$ because of internal FSDP, so the computation per rank is still halved. Notice that in this case, the total FLOPs is doubled, compared to the baseline, which is $4Bd_{\text{in}}d_{\text{out}}$, per rank is halved is only possible because we have 4 GPUs.

12.2 Common Distributed Training Frameworks

PyTorch DDP: Currently, the baseline of finetuning models in parallel is to use raw PyTorch DDP (distributed data parallel), where we have 1 GPU for each process.

Gradients are averaged via "torch.distributed.all_reduce".

Pros: • Native PyTorch, minimal dependencies, and we have full control over everything.

Cons: • We must handle everything, checkpointing, gradient accumulation, optimizer sharding and memory scaling, which makes it hard to handle for very large models (10B +)

Hugging face accelerate: This is pretty common nowadays, basically, we just use the PyTorch training loop and add decorators, now it would run on one or serveral GPUs, with optional DeepSeed/FSDP integration.

Under the hood, it is still DDP, but we

- Handles GPU assignment, logging only on rank 0, checkpoint saving.
- Can automatically switch to DeepSpeed ZeRO or FSDP with a config file.
- Hugely reduces boilerplate.

Common use cases include: SFT of 7B - 13B models across 4-8 A100 GPUs.

DeepSpeed ZeRO This enables massive models by sharding optimizer, gradients and parameters across GPUs. Its key idea is ZeRO which is zero redundancy optimizer, it includes stages:

- Stage 1: shard optimizer states.
- Stage 2 (ZeRO2): shard optimizer states + gradients. (Splits gradients and optimizer states accross GPUs, each GPU still holds a full copy of the model parameter.)
- Stage 3 (ZeRO3 with offload): shard everything. (including parameters) (Splits parameters, gradients and optimizer states across GPUs, no single GPU holds the full model, which makes train very large models even with limited GPUs possible.)
- **Offload**: Store some of the states in CPU RAM (or NVMe SSD) and bring them in when needed.

and extras such as CPU/NVMe offloading, activation checkpointing, communication optimizations.

Use cases: Training 13B-70B models across 64-512 GPUs, needed when VRAM is limiting, even with 80GB A100s. Often combined with LoRA/QLoRA for finetuning.

PyTorch FSDP This is basically a native PyTorch way to do what DeepSpeed ZeRO does. It works in a way that it (i) shards parameters and gradients across ranks, (ii) loads only parts of the model per GPU at a time. This also supports activation checkpointing.

HuggingFace Trainer Where we do not write loop, just pass datasets + model + arguments.

Nvidia's Megatron-LM Megatron-LM is a stand-alone training framework for large Transformer models (GPT, BERT, T5, etc.). Its main innovation was model parallelism — splitting one model within layers across multiple GPUs (tensor parallelism). In the very beginning, it looks like the following:

```
Megatron-LM
|-- Model code (Transformer layers, GPT, BERT)
|-- Parallelism engine (tensor + pipeline)
\-- Training loop (PyTorch)
```

It uses PyTorch Distributed (torch.distributed) directly — no DeepSpeed at all.

DeepSpeed (from Microsoft) was developed around the same time but with different goals:

- memory efficiency (ZeRO optimizer sharding)
- easy configuration (JSON configs)
- pipeline parallelism and offloading
- mixed precision training, etc.

DeepSpeed did not define models, but it could run any PyTorch model more efficiently.

Megatron-Deepspeed: Eventually, they are combined, resulting in the Megatron-DeepSpeed. This happens when people start to train very large models, they realize each framework solved part of the scaling problem: As a result, these two were merged side-by-side.

Problem	Who solved it best
Model too big for one GPU	Megatron's tensor parallelism
Optimizer too big for memory	DeepSpeed's ZeRO sharding
Need pipeline across nodes	Both, combined
Need fused high-speed kernels	Megatron
Need easy config & launch	DeepSpeed

Table 6: Comparison of Megatron and DeepSpeed capabilities.

```
[ PyTorch ]
     ↓
[ DeepSpeed Engine ] ← handles ZeRO, optimizer sharding, pipeline, etc.
     ↓
[ Megatron Model ] ← implements tensor parallel layers, fused ops
     ↓
[ CUDA / NCCL ]
```

This framework works, but was quite messy, and it is basically two frameworks glued together.

Megatron-Core: Since 2023, NVIDIA has refactored everything into a clean, modular library called Megatron-Core (inside the new Megatron-LM repo).

- Exposes official APIs for tensor, pipeline, and sequence parallelism.
- Integrates with NVIDIA Transformer Engine, FP8 training, FSDP, etc.
- Used internally by NVIDIA and by open-source projects like NeMo.

Summary As a summary

- Small models ($\leq$ 13B), we do SFTTrainer + LoRA + Accelerate (4-8) GPUs.
- Medium models (13B - 30B), we do Accelerate + DeepSpeed ZeRO-2/FSDP.
- Large models ($\geq$ 65B), DeepSpeed ZeRO-3 or FSDP across hundreds of GPUs. HuggingFace Trainer or custom PyTorch loop orchestrated by Accelerate.. Always with mixed precision (bf16/FP8)

12.3 Examples

Let us consider the following cases:

Case 1: How raw DDP works.

- **No ZeRO (normal DDP):** In this case, each GPU keeps a full copy of all N parameters, during training, each GPU does forward backward on its minibatch and gradients are averaged (all-reduce). The memory cost should be: each GPU stores N parameters + all optimizer states. The problem this that this does not scale to very large models, because each GPU must hold the entire model.

Case 2: How ZeRO2 works.

- **ZeRO2:** In this case, each GPU still has a full copy of the N parameters, but optimizer states, such as Adam's momentum m and second momentum v are sharded across GPUs. The memory cost would be the N parameters + only a slice of the optimizer states and grads, we save memory, but the full model still sits on every GPU.

Assuming there are 2 GPUs with N parameters and each GPU is proceeding with a microbatch of size 1 with gradient accumulation step 4.

1. Forward + backward on GPU 0, 1 with micro-batch=1, computing all gradients, then we do reduce scatter[27] (instead of all reduce in plain DDP), and each GPU holds a shard of gradients of size $\frac{N}{2}$, the shard is accumulated and later used to update the (second order) momentum.
2. We repeat the process so that the gradients shards gets accumulated.
3. At each optimizer step, every GPU only updates its shard of parameters using its shards of grads.
4. After the update, the parameters are all gathered, so all GPUs gets the fully freshed model for the next forward.

Basically, we are asking each GPU to be responsible for less parameters because some of them are just simply redundant, the final result is mathematically equivalent:

$$\theta_{\text{DDP}}^{t+1} = \theta_{\text{ZeRO-2}}^{t+1}. \tag{23}$$

Case 3: How ZeRO3 works.

- Now we also shard the parameters themselves, with Adam, we only need about $1/3$ of the memory on 1 GPU compared to that of DDP, instead of $2/3$ in ZeRO-2.

Assume we are in the same setting as before.

1. Forward passes: since now each GPU does not hold all params for a certain layer L (tensor parallelism) anymore, we need to call all gather for each GPU and they will temporarily assemble the full weight matrix for that layer, and the forward for layer L is computed locally on each GPU. After this layer is done, those gathered weights can be released (kept only in CPU/NVMe if offloading).
2. Backward passes: Before computing the grads for layer L, weights are all gathered and gradients are computed after that.
3. Reduce-Scatter: Instead of all reduce, gradients are reduce-scattered, so each GPU only keeps its shard.
4. Optimizer Step: Each GPU only updates its shard of optimizer state and parameters, no one ever touches the other shard.

Baically ZeRO-3 saves GPU memory but does more communication.

[27] Sharded after all reduce (averaging).

Case 4: How ZeRO-3 + CPU/NVMe offload works. The same sharding applies, but now we decide where the shards live when they're not in active use.

Assuming we are in the same setting as before.

1. **Parameter Offload:** Sharded params that are not immediately needed for the current layer can be moved to CPU RAM (or NVMe if RAM is still too small).[28] Before a layer executes, ZeRO all-gathers the shards back onto GPU(s). After use, they can be evicted back to CPU.

2. **Optimizer State Offload:** Adam's momentum states are big (2x param size), offloading lets each GPU keep only its shard in CPU RAM/NVMe, when optimizer step happens, that shard is pulled onto GPU, updated, and can be offloaded again. (This saves a huge amount of persistent GPU memory.)

3. **Gradient Offload:** Typically not offloaded explicitly but can be done, since grads are transient. They live in GPU memory just long enough to accumulate (over gradient accumulation steps).

13 Agent

13.1 AI agent based on LLMs

An AI agent based on large language models (LLMs) is an intelligent, modular system designed to extend a model's text generation ability into goal-directed reasoning and action. • At its core lies the LLM, which interprets user instructions, maintains contextual memory and decides the next action through natural-language reasoning.

• Surrounding this model is a runtime layer, often called an agent framework, that manages tool invocation, state tracking, and communication with the external environment.

• When the model determines that a task requires external information or computation, it emits a structured tool call (e.g., a JSON-formatted request) which the runtime routes to a corresponding tool handler implemented as a function, API, or MCP server. The tool executes the requested operation, such as running Python code, retrieving documents, or calling a database, and returns the result to the agent as structured data. The model then processes this observation, updates its reasoning trace, and may choose to plan further actions or present a final answer.

• This loop of **reason** → **act** → **observe** → **reflect** turns the LLM into an autonomous cognitive system, where the model provides the "mind" (language, reasoning, planning), and the MCP-connected tools form the "body" (sensing, acting, retrieving). Together, they enable a continuous cycle of grounded decision making, allowing the agent to interact with digital systems, perform iterative workflows, and accomplish complex multi-step objectives with minimal human intervention.

13.2 Model Context Protocol (MCP)

The Model Context Protocol (MCP) is an open communication standard designed to enable structured, secure, and interoperable interaction between large language models (LLMs) and external systems. Introduced to replace ad-hoc plugin and API formats, MCP defines a universal layer that allows an LLM to discover, describe, and call external tools, data sources, and services in a standardized way. An MCP ecosystem typically consists of three components:

- **MCP Client (Agent Runtime):** embedded alongside the LLM, this component exposes a structured list of available tools and mediates communication between the model and the external world.

[28] Originally, each GPU has to carry it otherwise it is gone. Now we store them in CPU and only load back the layer used. At each time, GPU memory only holds the weights for the current layer but not the whole shards.

- **MCP Server:** a backend service hosting one or more tools or data interfaces (e.g., file systems, code execution, API calls). Each server declares its capabilities using JSON schemas so that any compliant model can understand how to invoke them.
- **Protocol Layer:** defines message types such as list_tools, call_tool, and tool_result, enabling asynchronous, bi-directional exchange between the model and the environment via JSON-RPC over stdin / stdout or WebSocket.

An example would be: When a model operating through an MCP client decides to perform an action, such as summarizing a document or querying a database, it issues a tool invocation request that conforms to MCP's structured schema. The server executes the corresponding function and returns a structured result, which the model integrates into its reasoning process. This architecture allows any compliant model (OpenAI, Anthropic, local LLaMA, etc.) to interact seamlessly with tools built by independent developers, forming a modular ecosystem of interoperable agents and toolchains.

13.3 Training an AI Agent for Tool Use and Reasoning

Training an LLM-based AI agent involves teaching the model not only to generate text, but also to decide when and how to act through structured interactions. This is typically done in three progressive stages:

- **Supervised Instruction Tuning (SFT)**: The base LLM is fine-tuned on datasets of instruction-response pairs that include examples of **tool-augmented reasoning**. Each example demonstrates when to call a tool, how to format the call (e.g., JSON), and how to integrate tool results back into language output. Example training pair:

```
User: What's the weather in Paris?
Model: <tool_call name="get_weather" args={"city": "Paris"} />
```

  After executing the tool, the model learns to continue the dialogue:

```
Result: {"temp": 12, "condition": "Cloudy"}
Model: The temperature in Paris is 12°C with cloudy skies.
```

  This is when model learns how to make a proper response.

- **Reinforcement Learning with Feedback (RLHF, RLAIF):** Once the model can perform basic tool calls, it is refined using reinforcement signals, human or automated, to prefer efficient, accurate, and contextually appropriate actions. The reward model scores not only linguistic quality but also tool-use efficiency and task success, encouraging the LLM to call tools only when beneficial. This is when the model learns which responses are better.
  - **RLAIF**: The name alone already indicates that it is Reinforcement Learning from AI Feedback. The motivation is simply RLHF is powerful but expensive and slow, because it needs: (1) Humans to rate or rank model outputs, (2) Thousands or millions of examples to train a reward model. As a result, **RLAIF** replaces human rater with an AI rater, which typically is a stronger, safer or more instructed model (like GPT-4-Turbo, Claude Opus, etc.). Example: GPT-4 is asked: "Between response A and response B, which is more factual, helpful, and safe?" → The output "A is better" becomes the supervision signal. This produces a dataset of AI judgments, which is then distilled into a reward model.

  - **Why it works well**: (1) High-quality models can give consistent, cheap, and fast feedback. (2) Scales much better than hiring human raters. (3) Reduces bias/noise from inconsistent human annotation. (4) Enables bootstrapping: we can train Model B with feedback from Model A.
  - This is how Anthropic trained Claude 3 series, they call it **Constitutional AI**, which is essentially RLAIF guided by an ethical constitution rather than human labelers.

- **Agentic Meta-Training (Planning & Reflection Loops):** Advanced agents undergo iterative training in closed environments (e.g., code execution sandboxes or retrieval tasks) where they practice multi-step reasoning and self-correction. **During this phase, the model learns higher-level patterns: planning multiple actions, reflecting on previous outcomes, and maintaining stateful dialogues across steps.** Some frameworks (like OpenAI's "ReAct" and "Reflexion") explicitly simulate these loops, teaching the model to alternate between reasoning ("thoughts") and actions ("tool calls").

13.4 Agentic Meta-Training (Planning & Reflection Loops):

Traditional LLM training (pretraining + SFT + RLHF/RLAIF) teaches a model to produce good single responses, but:

- It doesn't train **multi-step planning**.
- It doesn't give **memory** across actions.
- It doesn't teach **self-correction** after failure.

Agentic meta-training addresses this by explicitly training or simulating interactive reasoning loops where the model must decide, act, observe, and adapt, just like a human or reinforcement-learning agent [29].

It has a shared foundation with RLHF or RLAIF, built upon the same optimization machinery: (i) staring from a pretrained language mode, (ii) collect interaction data (prompt - model output - feedback), (iii) and then optimize via SFT, RL (PPO/DPO/GRPO), or distillation.

They differ not in how gradients are computed, but in what kind of behaviors the data is designed to teach.

- In **RLHF** or **RLAIF**, we are trying to do alignment with respect to answers. Our goal is to make the model's final responses match human or AI preferences.

  Training data example:

```
    Prompt: "Explain black holes to a 10-year-old."
    Output_A: "Black holes are cosmic vacuum cleaners..."
    Output_B: "Black holes have infinite gravity..."
    Preference: A > B

```

  Reward model learns the preference and policy learns to output more A like answers.

- In **Agentic meta-training**, we are trying to do alignment with respect to process. Our goal is to make the model learn a multi-step reasoning pattern.

  Training data example:

[29] A reinforcement learning (RL) agent is an autonomous decision-making system that learns to achieve a goal by interacting with an environment and receiving feedback (rewards or penalties) for its actions.

```
    Question: "Who won the 2020 Olympics 100m race?"
    Thought: "I should search for the result."
    Action: search["2020 Olympics 100m winner"]
    Observation: "It was Marcell Jacobs."
    Thought: "Now I can answer."
    FinalAnswer: "Marcell Jacobs."
    Reflection: "Efficient. Next time, avoid redundant searches."

```

This teaches the model how to plan (decompose tasks), when to act (call tools, run code, query retrieval), and how to interpret results. In addition, we teach the model how to reflect on past performance. Mechanically, it is still SFT or RL on text, but the text now encodes reasoning trajectories.

We often combine the two: (i) Meta-train on reasoning trajectories (teach the process). (ii) Apply RLHF/RLAIF on the final answers (align the outcomes).

Examples: OpenAI's tool-use and reflection models, Anthropic's Constitutional AI with self-critique loops, Google DeepMind's "RFT" (Reasoning Fine-Tuning).

13.5 LLM Reasoning

"Chain of thought" (**CoT**) refers to the internal reasoning steps of the language model, the text it generates silently (or explicitly, during training) to reason before answering. That reasoning happens inside the LLM (for example, Mistral, GPT-4, or Llama-3)

ReAct. This is a framework (a prompting and interaction framework) proposed by Google to make large language models operate as interactive agents that can both think and do. The idea is to let the model generate two alternating kinds of outputs:

1. **Reasoning steps:** ("Thought"): the model explains what it's thinking or planning to do.
2. **Actions:** ("Act"): the model issues structured commands to tools or environments.

Then the system gives back an **Observation** (the result of that action), and the model continues reasoning based on it.

Its key ingradients include:

- **A prompt schema** enforcing the structure:

```
    Thought: ...
    Action: ...
    Observation: ...

```

- **A loop controller** in Python that:
  1. Feeds the accumulated context (Thought/Action/Observation history) to the model.
  2. Detects when the model emits an Action: line.
  3. Executes that action via a tool or API.
  4. Injects the Observation: result back into the next prompt.
- **A stopping condition**, usually when the model outputs Final Answer:.

Some times ReAct agents are combined with reflection (self-critique) and memory (persistent context across episodes).

14 Multi Modality

14.1 How to make LLMs understand images

As we know, LLMs understand sequences of embeddings. So for images, the whole game is to build a front-end that converts an image into a sequence of vectors that is compatible with the language model. This is what modern multimodel systems do: a **vision encoder** turns the image into visual features and then a **small bridge / connector** makes those features usable by the LLM.

From pixels to visual tokens: The standard starting point today is a vision transformer (ViT) style encoder: ViT explicitly treats an image as a sequence of patches, split the image into small blocks such as 16×16, flatten/project each patch, add positional information, and run a transformer over the resulting patch sequence. In this sense, patches play the role of tokens. Conceptually,

$$\mathbf{I} \in \mathbb{R}^{H \times W \times C} \to \{\mathbf{p}_1, \ldots, \mathbf{p}_N\} \to \{\mathbf{v}_1, \ldots, \mathbf{v}_N\}, \quad \mathbf{v}_i \in \mathbb{R}^{d_v},$$

where $\mathbf{p}_i$ are image patches, and $\mathbf{v}_i$ are learned visual embeddings. The image becomes a sequence.

In many practical systems, the vision encoder is not trained from scratch together with the LLM. Instead, people often start from a strong pretrained vision model such as CLIP or a ViT-like backbone already aligned with text. CLIP learns image representations from large-scale image-text pairing by predicting which caption matches which image, and that gives a very useful semantic visual space before touching the LLM at all.

What is CLIP: It is a dual-encoder model that learns a shared embedding space for images and text. One encoder maps an image to a vector (its image encoder is often a ViT backbone), another encoder maps a caption or text prompt to a vector, and training pushes matched image-text pairs closer while pushing mismatched pairs apart. In the original paper, OpenAI trained it on 400 million image-text pairs and showed strong zero-shot transfer across many vision tasks. Then CLIP normalizes these embeddings and uses their similarity, typically cosine similarity up to a learned temperature, to score whether image x_i matches text t_j. Training is basically a symmetric contrastive objective over a minibatch: each image should match its paired text and not the others, and vice versa. This is why CLIP is often described as "language-supervised representation learning" rather than generative modeling.

ViT as visual encoder: A ViT (Vision Transformer) is basically the image analogue of a token-based Transformer: instead of taking text tokens as input, it takes an image, chops it into patches, turns each patch into a vector, adds position information, and runs standard Transformer layers over the resulting sequence. Each patch is flattened and linearly projected into a hidden dimension d_{hid}, giving a sequence of vectors in $\mathbb{R}^{d_{\text{hid}}}$, which is the direct visual analogue of token embeddings in NLP.

In the original classification setup, ViT prepends a learned [CLS] token whose final hidden state is used for classification, although in multimodal systems people often care more about the full sequence of patch features than just a single pooled representation.

Concretely,

```
image -> patch embedding -> positional encoding -> stack of Transformer
 ↪ blocks (ViT) -> visual features.
```

Patch embedding in $\mathbb{R}^{d_{\text{model}}}$ is often results of projection of the patch vector in the space of $\mathbb{R}^{CP^2}$, where P is the patch size. This is often equivalent to a Conv2D with kernel size P and stride P. The choices of positional encodings are often: (1) Learned absolute positional embeddings. The classic ViT choice, and also very common in CLIP-style ViT backbones. Add a learned embedding to each patch token position. (2)

Fixed 2D sine-cosine embeddings: also common, especially in self-supervised ViT variants such as MAE. These are not learned in the same way and preserve a cleaner 2D geometric structure. (3) Relative position bias: common in hierarchical/windowed vision transformers such as Swin.

Raw visual tokens are not enough: Even after ViT or CLIP, we do not yet have something the LLM can naturally consume. This is due to that (1) LLM's token embeddings live in a language-centric space, while the image encoder outputs vectors in a different representation space. Also, (2) an image may produce hundreds or thousands of patch embeddings, which is expensive and often redundant.

So we need a **modality bridge**. This is the central design problem in multimodal LLM input pipelines.

- **Family A: simple projector into the LLM embedding space.** (Simplest and most common: Industrial default: We have a pretrained LLM, and a pretrained vision encoder, and we only need a connector that maps features into the language model's input space.) The pipeline in this case is

```
image -> vision encoder -> visual features -> linear / MLP projector
↪ -> LLM token space.
```

  To be specific, suppose that one image is $x \in \mathbb{R}^{H\times W\times C}$, a ViT-style vision encoder splits it into patches and outputs a sequence of visual feature vectors. Suppose there are N patches and the vision backbone hidden size is d_v, then the encoder output is typically $F \in \mathbb{R}^{N\times d_v}$. This is what we termed as visual features. The projector maps the last dimension from the vision space into LLM embedding space. As usual, we will denote the hidden size of LLM as d_{model}, then the simplest projector is just a linear map $\mathbf{W} \in \mathbb{R}^{d_v\times d_{\text{model}}}$. We can also use an MLP. In both cases the token count stays the same and the result becomes $\mathbb{R}^{N\times d_{\text{model}}}$. Some architectures do add a resampler / pooling / query bottleneck, which can reduce the number of visual tokens.

  The system usually does the two pipelines in parallel (text side / image side). Then they are assembled into one transformer input sequence.

  This is essentially the spirit of **LLaVA**: connect a pretrained vision encoder and an LLM with a trainable projection layer, then instruction-tune (multimodal SFT) the whole multimodal system on image-text instruction data. This design is attractive because it is simple, cheap, and compatible with standard autoregressive decoding. The downside is that a plain projector can be a relatively weak bottleneck for fine-grained spatial reasoning, OCR, and high-resolution detail. That is why later work keeps modifying the connector or increasing multi-scale visual input.

- **Family B: learned querying / resampling modules:** Instead of passing all patch tokens into the LLM, another strategy is to use a learned bottleneck that extracts only the most useful visual information. Two influential examples are:

  1. **BLIP-2: Q-Former**: BLIP-2[30] freezes the image encoder and the LLM, and inserts a lightweight Querying Transformer (Q-Former) between them (This is exactly what is trained.). A small number of learned query tokens attend to the image features and compress them into a compact set of visual representations that the LLM can use. This is a cleaner alignment mechanism than a bare linear projector.

```
image -> frozen image encoder
-> Q-Former -> linear projection -> frozen LLM
```

[30]Note that it is not the same as BLIP.

Querying bottleneck: the Q-former is a trainable transformer module placed between the frozen image encoder and the frozen LLM. Its job is to extract a fixed number of visual features from the image encoder output, independent of image resolution. Architecturally, the paper says Q-Former has two transformer submodules that share the same self-attention layers: an image transformer that interacts with the frozen image encoder, and a text transformer that can operate as a text encoder or decoder.

Conceptually: BLIP-2: use learned latent queries to read from visual tokens first, then pass only the compressed result onward.

The inputs to the Q-Former are (1) sequence of visual features output by frozen image encoder $F \in \mathbb{R}^{M \times d_v}$. (2) Self defined small set of learnable query tokens (32 queries each dimension 768) These query embeddings are model parameters, just like a learned prompt $Q^{(0)} \in \mathbb{R}^{K \times d_q}$. The image features are not treated as ordinary input tokens to be mixed directly into one long sequence at the beginning. Instead, the query tokens attend to the frozen image features through cross-attention. Specifically: suppose that at some layer the current hidden states of the learned query tokens are $H \in \mathbb{R}^{K \times d}$, and the frozen image features are the previous F.

$$Q_{\text{attn}} = HW_Q$$
$$K_{\text{attn}} = FW_K, \quad V_{\text{attn}} = FW_V.$$
$$\text{Attn}(H, F) = \text{Softmax}\left(\frac{Q_{\text{attn}} K_{\text{attn}}^\top}{\sqrt{d_k}}\right) V_{\text{attn}}$$

Training: 2-steps: (1) vision-language representation learning: Here, the Q-Former is attached to the frozen image encoder, but not yet really used as a generative interface to the LLM. The goal is to make the learned queries extract visual information that is strongly aligned with text. The paper describes this as teaching the Q-Former to learn visual representation most relevant to the text. (2) vision-to-language generative learning: Now the Q-Former output is linearly projected into the frozen LLM's embedding space and used as soft prompts. The goal here is different: the representation is no longer only "text-aligned", it must become interpretable by the LLM as a conditioning signal for generation.

Stage 1 teaches what visual information to extract; Stage 2 teaches how to present it to a frozen LLM.

More details on Stage 1: BLIP-2 uses image-text pairs (x, y), no frozen LLM, image x goes through visual encoder and becomes embedding F, which later interacts with query embeddings $Q^{(0)}$ and output summary $Z \in \mathbb{R}^{32 \times 768}$. Text y goes through text tokenizer and Q-former text transformer and becomes hidden vector, where $t \in \mathbb{R}^{768}$ is defined as the output embedding of [CLS] token (text representation). It is a single vector summarizing the whole text sequence.

(1) ITC: Image-Text Contrastive Learning: BLIP-2 wants the image representation and text representation to be close for matched pairs and far for mismatched pairs. In image-text contrastive learning, BLIP-2 aligns Z with t, specifically, it computes similarity between each query output z_i with t then takes the highest one as the image-text similarity. Then over the minibatch, BLIP-2 contrasts the score of the true pair (x, y) against mismatched pairs using in-batch negatives. (Contrastive Loss)

(2) ITG: Image-grounded Text Generation: Generate text conditioned on image queries. The attention mask is causual. So this is literally just language modeling conditioned on the query outputs. (Language Modelling loss)

(3) ITM: Image-Text Matching: Instead of computing similarity separately, the model lets the queries interact with the text tokens. Then do a binary classification to predict whether they are a pair. (Binary classfication loss.)

Each loss teaches the query vectors a different property.

2. **Flamingo: Perceiver Resampler + cross-attention:** Flamingo also bridges a pretrained vision model and a frozen language model, but uses a Perceiver Resampler to convert visual features into a fixed-size set of latent visual tokens. Then it inserts newly trained cross-attention layers inside the language model so the text stream can attend to visual memory while generating text. Flamingo was explicitly designed to handle interleaved text and visual inputs, including images and videos.

   Details: Imagine that the original prompt is a mixture of images and texts

```
(image_1, "This is a dog.", image_2, "What animal is this?")

```

   Flamingo converts the text side into something like

```
<image> This is a dog. <image> What animal is this?

```

   and then tokenizes that text. Meanwhile, each image is processed separately by the visual stack. A subtle but very important point: Flamingo later uses a masking rule so that each text token directly cross-attends only to the visual tokens of the most recent preceding image. Each image $\mathbf{I}^m \in \mathbb{R}^{H\times W\times C}$ is handled separately by the pretrained frozen vision encoder and results in the spatial visual features $\mathbf{X}_f^m \in \mathbb{R}^{S_m\times d_v}$, before it is sent to the perceiver resampler. [31] **In Flamingo, the vision encoder is a frozen feature extractor.**

   Details on Perceiver Resampler: An image or video may produce too many raw visual features, but the language model wants a manageable number of visual tokens. So the Perceiver Resampler compresses a variable-size set of visual features into a fixed-size set of learned visual tokens. This is one of Flamingo's key ideas. For video, the input can be regarded as $\mathbf{X}_f \in \mathbb{R}^{T\times S\times d_v}$, for an image we just take $T = 1$ which means that the number of frames is 1. S stands for the number of spatial locations per frame. Then module also uses (1) learned time pos embeddings $e \in \mathbb{R}^{T\times 1\times d}$ for frames, (2) R learned latent queries, with shape $\mathbb{R}^{R\times d_v}$.

```
def perceiver_resampler(
  x_f,  # The [T, S, d] visual features (T=time, S=space)
  time_embeddings,  # The [T, 1, d] time pos embeddings.
  x,  # R learned latents of shape [R, d]
  num_layers,  # Number of layers
):

  """The Perceiver Resampler model."""
  # Add the time position embeddings and flatten.
    x_f = x_f + time_embeddings
    x_f = flatten(x_f)  # [T, S, d] -> [T * S, d]
  # Apply the Perceiver Resampler layers.
  for i in range(num_layers):
    # Attention.
    x = x + attention_i(q=x, kv=concat([x_f, x]))
    # Feed forward.
    x = x + ffw_i(x)
  return x

```

[31] For video, frames are sampled at 1 FPS and encoded independently before being concatenated downstream.

(Flamingo concatenates the learned latents into the keys and values together with the visual features, and that this performed slightly better in their experiments.)

The output is $\mathbf{Z} \in \mathbb{R}^{R \times d}$, which is a fixed number of visual tokens, regardless of image resolution or video length. In the paper, the motivation is exactly to reduce the cost of vision-text cross-attention; in practice they use a small fixed number, stated as 64 outputs. Those $\mathbf{Z}$ are stored as visual memory.

Details on GATED XATTN-DENSE block: This is the component that lets the language model look at the visual tokens. The paper inserts these blocks between frozen LM layers. Each such block: (1) lets language states cross-attend to visual tokens, (2) passes the result through a feed-forward layer, (3) gates the new branch so that the original pretrained LM is not disturbed at initialization.

```
1   def gated_xattn_dense(
2     y,  # input language features (L, d)
3     x,  # input visual features (R, d)
4     alpha_xattn, # xattn gating parameter - init at 0.
5     alpha_dense, # ffw gating parameter - init at 0.
6   ):
7   """Applies a GATED XATTN-DENSE layer."""
8   # 1. Gated Cross Attention
9     y = y + tanh(alpha_xattn) * attention(q=y, kv=x)
10  # 2. Gated Feed Forward (dense) Layer
11    y = y + tanh(alpha_dense) * ffw(y)
12  # Regular self-attention + FFW on language
13    y = y + frozen_attention(q=y, kv=y)
14    y = y + frozen_ffw(y)
15  return y  # output visually informed language features
16
```

Line 9: In Flamingo, when multiple images are present, this is the stage where the image-aware masking is applied so each text position directly reads only the relevant preceding image's visual tokens. The residual connection is there because the block is not replacing the language state. It is adding a visually informed correction to it. This keeps the language stream stable and compatible with the pretrained LM. That is exactly the spirit of Flamingo: preserve the frozen LM behavior and inject visual information additively. Third, the frozen LM layers let the model continue standard language modeling while preserving the pretrained linguistic competence of the base LM. Refer to Figure 3 of the original paper for more details.

Why not a separate projector: The output of the Perceiver Resampler is used directly as the visual input to the GATED XATTN-DENSE block. There is no need for projection because the cross-attention layer already contains the needed projection mechanism. In Flamingo, the Perceiver Resampler is not just a compression gadget. It is one of the two learnable components that bridge the frozen vision encoder and the frozen language model.

How to obtain the latents: The latent queries are learned jointly with the Resampler parameters during multimodal training.

Training Recipe:

1. **Text LLM pretraining:** Pretrain the language model on text.

2. **Vision encoder pretraining:** Start from a strong vision encoder.
3. **Bridge:** Connect the visual stream to the LLM. This connector can be a simple projector, a resampler, cross-attention, or some other bridge.
4. **Aligment:** Do multimodal pretraining / alignment with a text objective (Train on image-text data, not raw text alone.). (Qwen2-VL: both text and image tokens appear in training, but supervision is only provided for the text tokens.) –> The image is just conditioning context.
5. **Multimodal SFT:** After coarse alignment, run supervised fine-tuning (SFT) on instruction-style data: image + question/instruction + answer.
6. Optionally do preference tuning / RL-style post-training.

14.2 How to make LLMs understand videos

The simplest and most common mental model is:

1. sample frames from the video.
2. run an image encoder on each frame.
3. attach some notion of temporal position / timestamp.
4. pool / resample / compress the resulting tokens.
5. feed them to the LLM through a **projector** or **cross-attention**.

Modern video LLMs differ mainly in whether time is modeled before the LLM, at the bridge, or inside the multimodal interaction layers.

1. **Family A (frame-as-image + temporal aggregation):** most current and most common public recipe.
2. Many recent open video LMMs still look like "strong image VLM + video sampling/compression tricks + projector + LLM," rather than a completely new native video architecture. LLaVA-Video builds on LLaVA-OneVision with a vision encoder plus LLM and explicit video representation choices, and Video-XL explicitly says it inherits the minimal LLaVA design of visual encoder + projector + LLM. Qwen2.5-VL also adds video support through dynamic FPS and absolute time encoding rather than abandoning the VLM paradigm.
3. **Family B: native spatiotemporal video encoder: (treats video as a first class modality)** ViViT and TimeSformer are still important and conceptually clean, but they were introduced as video-understanding backbones that model space and time directly, not as the dominant full-stack recipe for modern video assistants. In practice, this family is more "specialized and principled".
4. **Family C: compressed visual memory + cross-attention into the LLM:** Flamingo remains a foundational design because it explicitly supports images and videos via learned visual bridging and cross-attention.

We will now review family A since its most commonly used. This family actually starts from the very pragmatic observation that we already have very strong image encoders and image-VLM training pipelines. So instead of designing a native video model first, we sample the video into frames or short clips, encode them with an image backbone, inject temporal information, compress the result, and let the LLM reason over

that compressed sequence. So after per-frame encoding we have something like a tensor in $\mathbb{R}^{T\times N\times d_v}$ with T sampled frames, N visual tokens per frame, and hidden size d_v. Then we add temporal structure: timestamps, frame indices, slow/fast grouping, temporal pooling, or a temporal projector. And finally map to the LLM dimension d_{model}. Qwen2.5-VL explicitly highlights dynamic FPS training and absolute time encoding for video; LLaVA-Video describes a SlowFast-style representation to balance frame count and token budget.

The main weakness is that time is not native at the front end. If you sample poorly, you lose the event. If the motion is subtle, frame-wise image features may miss it. Apollo's design study is useful here: it reports that FPS-based sampling beats uniform frame sampling, and that Perceiver Resampler beats average pooling, which is a strong hint that in these systems the bottleneck is often not "language reasoning" but "how you preserve temporal evidence under a token budget".

Dynamic FPS training: The problem it tries to solve is simple: a video can be sampled at very different rates. The same 10 seconds event can be represented by 10 frams at 1 FPS or 20 frames at 20 FPS. If a model is always trained at one fixed sampling rate it tends to become brittle when inference uses another one. Qwen2.5-VL explicitly addresses this by introducing dynamic FPS sampling / training, described as extending dynamic resolution into the temporal dimension so the model can handle videos with varied sampling rates. The paper also says they dynamically sampled FPS during training to make the training distribution over FPS more even.

This is to make the model aware that "video understanding" should be robust to whether frames were taken every 1 second, every 0.5 seconds, or at some other interval. That improves generalization to different video lengths, different motion speeds, and different inference-time token budgets. [32]

But dynamic FPS training alone is not enough. It tells the model to survive different sampling rates, but it does not by itself tell the model the real elapsed time between sampled frames. That is exactly why absolute time encoding is introduced.

Absolute time encoding Older frame-based schemes often use temporal positions like $1, 2, \ldots, T$ which tells the model about the order, but not real time. This does not account for content speed changes or the absolute timing of events. Qwen2.5 VL's fix is to align the temporal component of MRoPE to absolute time. Instead of saying only "this is the next frame", the model gets a temporal encoding whose intervals reflect real timestamps. The paper states that by aligning MRoPE IDs directly with timestamps, the model can understand tempo through the intervals between temporal IDs, without adding extra grounding heads. This is crucial for questions like: What happens around 1 minute 35 seconds? The benefits are two fold: first: it helps with tempo: fast-changing and slow-changing clips are no longer temporally ambiguous. Second, it helps with temporal localization: the model can better map language about time to actual moments in the video.

MRoPE This refers to Multimodal extension of the Rotary Position Embedding. In Qwen2-VL, this is the multimodal extension of 1D RoPE: instead of having only one positional axis, it splits the positional encoding for visual tokens into three components: temporal, height and width.

1. For **text** tokens, the three position IDs are set identically, so it behaves like standard 1D RoPE.
2. For **images**, the temporal ID is fixed while height/width vary by patch location
3. For **videos**, height / width vary within each frame and the temporal ID changes across frames.

[32] Right now, most video LLMs samples only at 1-8 FPS, while human visual system behaves like a 60-100 FPS sensor. This is far below human temporal resolution, which means that they probably relies more on semantic redundancy rather than true motion understanding.

SlowFast-style representation This is about how we spend our token budget. The original SlowFast idea from video recognition is: use two pathways. A Slow pathway runs at low frame rate and focuses on spatial semantics; a Fast pathway runs at high frame rate and focuses on motion, while being lightweight. For video LLMs, the same idea gets adapted into a token-budget strategy.

Usually, we cannot afford: (1) many frames (2) each with many spatial tokens (3) all at once.

So we compromise: (1) for a small subset of important frames, keep more spatial detail, (2) for many other frames, keep less spatial detail but more temporal coverage.

LLaVA-Video uses the same general idea in a more implementation-oriented form. It says it follows the classic SlowFast idea to balance the number of frames against the number of visual tokens under limited LLM context and GPU memory. It divides frames into slow and fast groups, and applies different pooling rates to the two groups. In other words, some frames are represented richly, others more cheaply.

14.3 How LLM understand audio

For the previous mentioned family A: a general patter is that

```
modality encoder + alignment / connector + LLM.
```

In that sense, for audio you usually replace the vision encoder with an audio encoder, then use a projector / pooler / Q-Former-like connector to map audio features into something the LLM can condition on. This is exactly the basic architecture used by models such as Qwen-Audio / Qwen2-Audio and SALMONN. Qwen2-Audio is described as an audio encoder + large language model.

There are two importants caveats.

1. Audio is often more **temporally dense** than images or short videos. Audio models often need stronger temporal compression than image models. In Qwen2-Audio, raw waveform is resampled to 16 kHz, converted to a 128-channel mel spectrogram, and even after pooling, each encoder output frame still corresponds to only about 40 ms of original audio. So long audio produces very long feature sequences very quickly.

2. Second, "audio" is not one single semantic type. It may contain speech content, speaker identity, prosody/emotion, music structure, and non-speech events all at once. Because of that, some systems use **multiple audio encoders** rather than a single one. SALMONN is a clean example: it combines a Whisper speech encoder with a BEATs audio encoder, then fuses them through a window-level Q-Former before passing them to the LLM.

Pooling Sometimes we see pooling a lot in relevant papers: this term simply means that we take many nearby features/tokens and compress them into fewer features/tokens by aggregation. The usual purpose is simple, to reduce sequence length while keeping most of the useful information. In multimodal LLMs, pooling is often a compute-saving compression step before the modality features are handed to the LLM.

14.4 Generateing Images

A modern multimodal assistant that can "draw an image" often works like this

```
[user prompt] -> [LLM] -> [structured image prompt / tool args] -> [image
    ↪ model] -> [image]
```

The LLM may do things like: (1) infer missing details, (2) resolve ambiguity, (3) maintain conversation context, (4) apply safety rules, (5) decide aspect ratio / style / number of images, (6) send a refined prompt to the image model.

So in that sense, the LLM is the controller or brain. But the actual image synthesis is usually done by a different model, for example a diffusion model.

How can a model output images There are two types of models:

- **Autoregressive image generation:** The idea is that we first convert an image into a sequence of discrete visual tokens, and then we train a transformer to predict those tokens one by one. This is very similar to LLM, because it is still next-token prediction, just over visual tokens instead of words.

  Typical Recipe: (1) Train an image tokenizer, often with something like VQ-VAE / VQGAN. (2) Represent each image as a grid/sequence of codebook indices. (3) Train a transformer conditioned on text to autoregressively predict these indices. (4) Decode the indices back to pixels.

  Typically, we have long token sequences for images, and the sampling is slow. Historically, it results in weaker image quality than strong diffusion systems.

- **Diffusion / flow-style generation:** This is the dominant idea for high-quality image generation[33]. The idea is that instead of predicting image tokens one by one, we ask the model to learns to iteratively denoise a noisy image latent.

Top level facts: (1) Diffusion learns a reverse stochastic process from noise to data. (2) Flow matching / rectified flow learns a deterministic ODE flow from noise to data.

Let p_{data} be the target data distribution on $\mathbb{R}^d$, for images $d = HWC$ in pixel space, or much more commonly in practice, d is the dimension of a learned latent code[34].

Diffusion Models The core idea is

1. Define a forward process that gradually destroys structure by adding noise.
2. Learn a reverse process that gradually removes the noise.
3. To generate a sample, start from pure noise and run the learned reverse process.

So baically, the diffusion models turn the hard problem of sample $x_0 \sim p_{\text{data}}$ into many easier local denoising problems.

The forward diffusion process The forward process is fixed, at each step, we add a small amount of Gaussian noise to the data sample (we will use x_0 to denote the initial clean data sample).

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t \mathbf{I}_d),$$

We are taking the mean as a scaled version of the previous state, $\alpha_t = 1 - \beta_t$, this gradually destroys information over time. β_t controls the amount of noise added. This is equivalent to the sampling equation

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}_d).$$

[33] More precisely, today it is usually diffusion or a very close relative such as flow matching / rectified-flow-style models.

[34] Latent diffusion models explicitly move the diffusion process into latent space for efficiency, and use cross-attention to inject text conditioning.

After many steps, the sample becomes almost pure Gaussion noise. A very important fact is that we can write x_t as a noisy version of x_0,

$$q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)\mathbf{I}_d).$$

where $\bar{\alpha}_t$ is defined as

$$\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s.$$

This matters because during it means that during training, we do not need to simulate the whole chain step by step. We can sample a random t and construct x_t directly from x_0.

The reverse process: the real generative model The reverse process is what we learn. Let $x_{0:T}$ be the whole collection of $\{x_0, x_1, \ldots, x_T\}$. The diffusion model defines a joint distribution over the whole reverse trajectory:

$$p_\theta(x_{0:T}) = p(x_T) \cdot \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t). \tag{24}$$

Here θ is the trainable parameter, x_T is the starting point of the reverse process, which is standard Gaussian noise:

$$p(x_T) = \mathcal{N}(x_T; 0, \mathbf{I}_d).$$

This means that before any denoising happens, the state x_T is sampled from a standard normal distribution. From (24), we know that once x_T is sampled, we repeatedly sample x_{t-1} from $p_\theta(x_{t-1} \mid x_t)$, as we are gradually denoising. This formula is built upon the underlying Markov assumption.

Note that the true object we care about is the x_0, but not the whole chain. So after defining the joint data distribution $p_\theta(x_{0:T})$, we obtain the model's distribution on clean samples by marginalizing out the intermediate states:

$$p_\theta(x_0) = \int p_\theta(x_{0:T}) dx_1 \ldots dx_T.$$

This basically means that in order to compute the probability of a clean sample x_0, we must account for all possible noisy trajectories that could have led to it.

How do we generate a sample in practice?

1. Step 1: sample pure noise: $x_T \sim \mathcal{N}(0, \mathbf{I})$, which gives a completely random Gaussian vector.
2. Step 2: denoise one step, and repeat: $x_{T-1} \sim p_\theta(x_{T-1} \mid x_T)$, and repeat.
3. Step 3: Output the final clean sample.

One might ask, how do we actually parametrize $p_\theta(x_{t-1} \mid x_t)$? A conditional probability distribution can be very complicated. The standard choice is to assume $p_\theta(x_{t-1} \mid x_t)$ is Gaussian.

Unfinished.

15 Miscellaneous

15.1 Chinchilla Optimality

Chinchilla optimalilty refers to the balance between model size (number of parameters) and dataset size (number of training tokens) for a fixed budget of compute.

- Before Chinchilla, the common practice (e.g. GPT-3) was to train very large models on relatively small datasets. This made the model undertrained, which means that they had more parameters than the dataset could properly fit.
- The Chinchilla paper by Deepmind in 2022 systematically showed that for a fixed budget of compute, the optimal stragety is not to make the model as large as possible, but instead to use a smaller model and train it on much more data.
- Roughly speaking, they found that the optimal scaling law would be letting the number of training tokens scale linearly with the number of parameters
- The empirical results show that the optimal data-to-model ratio to be around 20 tokens per paramete ($1\times$ case)[35]. If we train with fewer tokens per parameter such as $4 \sim 5$ tokens per parameter, the model would be under-trained, and if we train with more tokens per parameter such as $50 \sim 100$ tokens per parameter, the model would be over-trained.

In practice, GPT-3 had 175B parameters but was trained on 300B tokens. Chinchilla showed that 70B parameters trained on 1.4T tokens (with the same compute budget) achieves much better performance.

15.2 Metrics (nats, bits, bpb)

The standard language modeling loss is the average of the negative log likelihood of the next token.

$$\mathcal{L}_{\text{LM}} = -\frac{1}{L}\sum_{t=1}^{L} \log \mathbb{P}_\theta\left(\mathbf{x}_t \mid \mathbf{x}_{<t}\right),$$

where $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_L)$ is a sequence of tokens. The loss it self can be reported in nats (natural log base e) L_{nats} or bits (log base 2) L_{bits}. If we exponentiate the average loss, we get perplexity.

There is another related measure, namely bits per bytes (bpb), coming from viewing the model as a compression system, which connects to Shannon's source coding theorem. That is, the average number of bits needed to encode a symbol is equal to its entropy (expected negative log likelihood). Under this view, the bits L_{bits} indicates how many bits are needed to encode each token. This already gives us bits per token. However, tokens are not uniform units, because they vary in length, so bits per token is not directly comparable across tokenization schemes. This is where bpb comes in. Instead of normalizing by token count, normalize by raw UTF-8 byte count of the text. If we denote the dataset text length as T-tokens and B-bytes, then

$$\text{bpb} = \frac{T}{B} \cdot L_{\text{bits}}.$$

This makes the result more stable and comparable across models with different tokenizer.

[35] We often see $2\times$ or $4\times$ cases where we increase the number of tokens 2 or 4 times.

15.3 Model Distillation

Model distillation (knowledge distillation) is the process where a large "teacher" model trains a smaller "student" model to imitate its behavior. The student learns from:

- the teacher's soft logits
- intermediate features
- responses / reasoning traces
- tool usage / actions
- synthetic data generated by the teacher

The concept of distillation is now new, but recently it becomes popular because of LLMs. It can be used to perform cost reduction since we can distill a 70B or 405B model into a 3B - 8B model that retains 90% performance. Also the technique can be applied for fast inferencing.

We can also add safety + controllability, where we distill "desired behaviors" and filter out bad outputs. This is frequently done after RLHF or safety tuning. Llama 3.1-8B, Qwen2.5-7B, Mistral-7B are all extremely good because they were trained with teacher signals from larger "pre-RLHF" and "post-RLHF" teachers.

Logit distillation: This is the classical way, where the student tries to match the teacher logits on tokens. It is a strong signal, but requires access to teacher logits, but teacher is often closed models and will not provide such information.

If we think about it, it is nothing more than forcing the student to imitate the teacher's probability distribution over the vocabulary at every training step.

Let us imagine we have an input batch of token ids $\mathbf{x} \in \mathbb{R}^{B\times L}$, we can first run the forward pass of the teacher and obtain teacher hidden state $\mathbf{H}^T \in \mathbb{R}^{B\times L\times d}$, and the teacher logits $\mathbf{Z}^T \in \mathbb{R}^{B\times L\times V}$.

```
with torch.no_grad():
  teacher_logits = teacher(x).logits       # shape [B, L, V]
```

For temperature $T > 0$, we have

$$\mathbf{P}^T = \text{softmax}(\frac{\mathbf{Z}^T}{T}).$$

```
teacher_probs = softmax(teacher_logits / temperature, dim=-1) # shape [B, L,
  ↪ V]
```

For student model, we can also run forward and obtain hidden state $\mathbf{H}^S \in \mathbb{R}^{B\times L\times d'}$ ($d' \leq d$), and $\mathbf{Z}^S \in \mathbb{R}^{B\times L\times V}$.

```
student_logits = student(x).logits   # shape [B, L, V]
student_probs  = softmax(student_logits / temperature, dim=-1)
```

Distillation loss via KL divergence. Then we define the distillation loss:

$$\mathcal{L}_{\text{distill}} = T^2 \cdot \mathrm{D}_{\mathrm{KL}}\left(\mathbf{P}^T \parallel \mathbf{P}^S\right).$$

```
log_student = log_softmax(student_logits / temperature, dim=-1)
loss_distill = (temperature**2) * torch.nn.functional.kl_div(
    log_student, teacher_probs, reduction="batchmean"
)
```

Note that

$$\mathrm{D}_{\mathrm{KL}}\left(P \parallel Q\right) = \sum_i P_i \left(\log P_i - \log Q_i\right).$$

Optional Cross-Entropy loss. If the label $\mathbf{y} \in \mathbb{R}^{B\times L}$ exists, then

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{BL}\sum_{b,l} \log \mathbf{P}^S(y_{b,l})$$

```
loss_ce = cross_entropy(
    student_logits.view(B*L, V),
    y.view(B*L),
  )
```

Then we can take the combination of the two losses.

Response-style distillation (instruction tuning). This is the modern workhorse and there is no teacher's logits required. We only use the response generated by the teacher. In this case, each sample is consisted of the instruction $\mathbf{x} \in \mathbb{R}^{L_x}$, and the response generated by the teacher $\mathbf{y}^t \in \mathbb{R}^{L_y}$, the student then perform standard supervised learning on the input batched input consisted of $\mathbf{x}$ and $\mathbf{y}^t$, and we will only compute the CE loss on the teacher generated part.

```
logits = student(input_ids)                    # [B, Lx+Ly, V]
loss = cross_entropy(
    logits[:, shift: , :].reshape(-1, V),
    y_teach_tokens.reshape(-1),
)
```

This is done in models such as LLaMA-3 Instruct, Qwen2.5-Instruct and etc. It is easy to see why this works, since teacher's generated text encodes reasoning trajectory, style, task formatting, chain-of-thought (if included), tool-use patterns and how to structure answers. The student simply learns the distribution of responses. This is cheaper but weaker than logit distillation

- **Pros:** (i) works with closed models; (ii) much easier and cheaper; (iii) easy dataset scaling; (iv) captures high-level style and reasoning patterns.
- **Cons:** (i) student never sees teacher's full token distribution; (ii) sensitive to sampling biases from teacher and cannot fully preserve teacher's performance ceiling.

Reasoning Distillation (CoT / ReAct / Reflection). Student learns step-by-step reasoning patterns generated by a stronger teacher. These methods do not aim to imitate final answers, instead they teach the internal structure of reasoning.

Synthetic data distillation. The core idea is that instead of using a real human dataset, we ask a strong teacher model to invent the dataset itself. The student then trains on this synthetic dataset. The technique has become one of the most powerful techniques for frontier models.

Take math for example:

- We first decide topic / difficulty / format (GSM8K-style, MATH-style, AMC/AIME-style), which defines the task distribution we want the teacher to generate.
- Then, we prepare a meta-prompt for the teacher:

```
Generate a new math problem
Then generate a correct solution
Then produce the final numeric or symbolic answer
Follow a consistent format
Avoid trivial rephrasings of known datasets
Ensure the problem is solvable
```

  This fully controls the structure of the synthetic sample.
- We then ask the teacher to generate a new math problem, then immediately follow a solution, which is then combined into a training example.
- Verification pass is needed, typically, a separate or the same teacher instance or verifier model is asked to check if the reasoning is correct and if the final answer matches the reasoning. Based on whether it is correct or not, we decide to keep or regenerate the response. This ensures the quality of the dataset.
- We may also want to do deduplication and filtering, which filters out near-duplicate problems, structurally similar templates, formatting problem or ill-posed problem and invalid numeric parameters. This ensures dataset diversity and correctness.
- Curriculum tuning: we often organize the synthetic data in a way that they become increasingly difficult.

Training & Evaluations:

- Math:
  - **Training:** GSM8K-synthetic expansions, MATH, MathQA.
  - **Evaluations:** GSM8K, MATH, MATH500 (Minerva subset), AMC 10/12, AIME, OlympiadBench.
- Code:
  - **Training:** The Stack v1/v2, GitHub repositories (filtered), CodeSearchNet.
  - **Evaluations:** HumanEval, MBPP, APPS (Intro / Inter / Hard)
- Logic and Generic Reasoning:
  - **Training:** Synthetic CoT datasets (massive), Self-generated reasoning traces (DeepSeek-R1 style), GSM8K-style CoT explanations.

  - **Evaluations:** BBH (Big Bench Hard), StrategyQA, CommonsenseQA.

- Other aspects include: Multi-hop & knowledge QA, instruction following (general SFT), Tool Use & Function Calling, Safety and alignment.

- There are also certain meta benchmarks such as MMLU (57 subjects; "IQ-test" for LLMs), MMLU-Pro (harder reasoning + domain knowledge), HELM (holistic evaluation).

Feature Distillation Student imitates internal representations of the teacher, not just logits or final outputs. This happens when we want a smaller model to behave like a larger one and we want better generalization than response distillation alone. It is commonly used in tiny transformers and quantized models.

15.4 Model Compression

Modern LLMs (7B to 405B) are expensive to run. Inference cost is dominated by memory bandwidth (moving weights around), activation sizes, KV cache growth in long-context inference. In general there are 4 types of model compression, which we have seen most of them.

- **Quantization:** Convert high precision weights/activations into low precision. This includes
  - **Weight-only quantization:** W4A16, W3A16 and W2A16. This is practical for most LLMs, and preserve accuracy superisingly well.
  - **Activation quantization:** Needed for fully low-precision inference.
  - **KV-cache quantization:** Large gains for long-context inference.
  - **Group quantization:** reduces precision but protects outliers. (weight vector splitted further into small-sized groups.)

- **Pruning:** Pruning removes unnecessary weights or even the entire neurons.
  - **Unstructured pruning:** Remove individual weights based on magnitude. Hard to accelerate on GPU.
  - **Structured pruning:** Remove entire attention heads, MLP channels, blocks, leads to real speedups.

  This will be discussed in detail later.

- **Distillaton:** As discussed previously, teacher and student framework.

- **Low-rank factorization:**
  - **Training time compression:** LoRA reduces trainable parameters.
  - **Inference time compression:** An example would be we have a big matrix $\mathbf{W}$, we can compute an SVD or similar factorization offline decomposing it into $\mathbf{W} \simeq \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r = \mathbf{AB}$, and we store only $\mathbf{A}, \mathbf{B}$ or compute $\mathbf{A}(\mathbf{B}x)$ when inferencing, which reduces memory footprint or compute.

15.5 Pruning

The general idea would be to remove weights, neurons or even the entire attention heads / MLP blocks while trying to preserve accuracy. This helps reduce memory footprint, inference latency, compute cost. Typically, there are two types of pruning.

Unstructured Pruning (weight pruning): Removes individual weights based on magnitude or other criteria.

- **Pros:** We could achieve high compression rate, and it is flexible.
- **Cons:** This creates irregular sparsity, which is not easy to use by hardwares, real speed ups require specialized sparse-kernel implementations.

This is still used in some places, but not an ideal technique for real development.

Structured Pruning: This is where most practical LLM pruning happens. We remove groups of weights: such as the entire neurons in MLP, the entire attention head, the entire FFN channels. Sometimes an rare case is that we remove the entire layer.

- **Pros:** Gives actual speedup on standard hardware, easy to implement, and plays nicely with quantization and distilation.

Modern structured pruning methods include: Wanda, SparseGPT, LLaMA-Pro block pruning, outlier-aware pruning and GPTQ-aware channel pruning.

Activation / dynamic pruning: These methods disable computation for tokens or layers depending on runtime signals. This is more like runtime optimization rather than classical pruning.

Specific Pruning Algorithms: Here, we review some of the pruning methods.

- **Simple Magnitude Pruning:** It could be (i) Compute the magnitude of weights $|\mathbf{W}_{ij}|$ and drop the smallest. (ii) For structured pruning, sum over groups $g_k = \|\mathbf{W}_k\|_2$ and remove groups with the smallest norms.
- **SparseGPT:** (Mostly niche now.) post-training, weight-pruning method designed for LLM, it prunes billions of weights without retraining, while keeping perplexity degradation small.

Pruning is done layer-by-layer, block-by-block by solving a local quadratic minimization problem. The **core motivation** here would be that we know pruning must change the weights, but we must not let the change of weights change the layer's output distribution. Otherwise the error will propagates through dozen's of layers and the whole model collapses.

To keep the layer output almost unchanged, we need a way to predict how zeroing some weights will change the output. Let us take an linear layer for example $\mathbf{y} = \mathbf{W}\mathbf{x}$. Assume that we apply pruning to $\Delta\mathbf{W}$ to $\mathbf{W} \in \mathbb{R}^{m\times d}$, i.e.,

$$\mathbf{W}' = \mathbf{W} + \Delta\mathbf{W}.$$

The output will now become

$$\mathcal{L}(\Delta\mathbf{W}) = \mathbf{y}' = (\mathbf{W} + \Delta\mathbf{W})\,\mathbf{x} = \mathbf{y} + \Delta\mathbf{W}\mathbf{x}.$$

We would like to have $\mathbf{y}' \simeq \mathbf{y}$, so we minimize

$$\min_{\Delta\mathbf{W}} = \mathbb{E}_{\mathbf{x}}\left[\|\Delta\mathbf{W}x\|^2\right].$$

Suppose that we have a calibration batch of N activation vectors $\mathbf{X} \in \mathbb{R}^{d \times N}$, each column corresponds to each sample. We approximate the expectation via empirical mean:

$$\begin{aligned}\mathcal{L}(\Delta\mathbf{W}) \simeq \frac{1}{N}\sum_{i=1}^{N}\|\Delta\mathbf{W}\mathbf{x}_i\| &= \frac{1}{N}\|\Delta\mathbf{W}\mathbf{X}\|_F^2 \\ &= \frac{1}{N}\mathrm{tr}\left(\mathbf{X}^\top(\Delta\mathbf{W})^\top\Delta\mathbf{W}\mathbf{X}\right) \\ &= \mathrm{tr}\left(\Delta\mathbf{W}\left(\frac{1}{N}\mathbf{X}\mathbf{X}^\top\right)(\Delta\mathbf{W})^\top\right).\end{aligned}$$

Define the empirical input variance $\mathbf{C} = \frac{1}{N}\mathbf{X}\mathbf{X}^\top \in \mathbb{R}^{d\times d}$, we can write $\mathcal{L}_{\Delta\mathbf{W}} = \mathrm{tr}\left(\Delta\mathbf{W}\mathbf{C}(\Delta\mathbf{W})^\top\right)$. Now, if we write $\delta = \mathrm{vec}\,(\Delta\mathbf{W})$, we can rewrite the loss function using the trace vec identity as

$$\mathcal{L}(\delta) = \delta^\top\left(\mathbf{C}\otimes\mathbf{I}_m\right)\delta.$$

Denote $\omega = \mathrm{vec}\,(\mathbf{W})$ and $\omega' = \mathrm{vec}\,(\mathbf{W}')$, we have $\omega' = \omega + \delta$, and since we are doing pruning, let $\mathcal{S}$ be the set of indices of pruned coordinates, we have

$$\delta_i = \begin{cases} -\omega_i, & \text{if } i \in \mathcal{S}, \\ 0, & \text{otherwise.}\end{cases}$$

We can actually reorder the coordinates of δ, so that

$$\delta = \left[\delta_{\mathcal{S}}, \delta_{\bar{\mathcal{S}}}\right]^\top.$$

Based on this new order, we can reorder $\mathbf{C}\otimes\mathbf{I}_m$ in the same way, obtaining

$$\mathbf{H} = \begin{bmatrix}\mathbf{H}_{\mathcal{S}\mathcal{S}} & \mathbf{H}_{\mathcal{S}\bar{\mathcal{S}}} \\ \mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}} & \mathbf{H}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\end{bmatrix}$$

where $\mathbf{H}_{\mathcal{S}\mathcal{S}}$ means that rows in $\mathcal{S}$ and columns in $\mathcal{S}$, and others have similar meanings. We will often refer to this as the Hessian surrogate or the curvature matrix. Plug this into the original loss inequality (notice that change of order does not change the loss), we obtain

$$\mathcal{L}(\delta) = \delta^\top\mathbf{H}\delta = \begin{bmatrix}\delta_{\mathcal{S}} \\ \delta_{\bar{\mathcal{S}}}\end{bmatrix}^\top \begin{bmatrix}\mathbf{H}_{\mathcal{S}\mathcal{S}} & \mathbf{H}_{\mathcal{S}\bar{\mathcal{S}}} \\ \mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}} & \mathbf{H}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\end{bmatrix}\begin{bmatrix}\delta_{\mathcal{S}} \\ \delta_{\bar{\mathcal{S}}}\end{bmatrix}.$$

Expanding it, we get

$$\mathcal{L}(\delta) = \delta_{\mathcal{S}}^\top\mathbf{H}_{\mathcal{S}\mathcal{S}}\,\delta_{\mathcal{S}} \;+\; 2\,\delta_{\bar{\mathcal{S}}}^\top\mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}}\,\delta_{\mathcal{S}} \;+\; \delta_{\bar{\mathcal{S}}}^\top\mathbf{H}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\,\delta_{\bar{\mathcal{S}}}.$$

Now, notice that we can treat $\delta_{\mathcal{S}}$ as constant (assume that we have already selected the coordinates in some way, which we will dicuss later, and because the values of those coordinates are fixed (as $-\omega_i$ to make the original parameter zero.), they are constants), we can take the gradient w.r.t. to $\delta_{\bar{\mathcal{S}}}$. [36] As a result, we can write the minimization problem as

$$\min_{\delta_{\bar{\mathcal{S}}}} \mathcal{L}(\delta_{\bar{\mathcal{S}}}; \delta_{\mathcal{S}}).$$

[36] We can think of this as a simplification on purpose: originally, we have to select which of the coordinates to prune and by how much on purpose, but now we choose to do it in a sequential manner, i.e., we first select the group of parameters to prune according to some standard, which allows us to solve the subsequent optimization problem via standard tools.

Notice that for this specific problem, we can actually use the optimality condition

$$2\mathbf{H}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\delta_{\bar{\mathcal{S}}} + 2\mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}}\delta_{\mathcal{S}} = 0,$$

Solving gives

$$\delta^{\star}_{\bar{\mathcal{S}}} = -\mathbf{H}^{-1}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}}\delta_{\mathcal{S}} = \mathbf{H}^{-1}_{\bar{\mathcal{S}}\bar{\mathcal{S}}}\mathbf{H}_{\bar{\mathcal{S}}\mathcal{S}}\omega_{\mathcal{S}}.$$

Note that the the matrix $\mathbf{H}_{\bar{\mathcal{S}}_c\bar{\mathcal{S}}}$, while PSD, is not necessarily invertible. In practice what we do is that we add a $\lambda \cdot \mathbf{I}$, which is actually equivalent to adding a regularizer $\lambda \cdot \|\delta_{\bar{\mathcal{S}}}\|^2$ to the objective. This is often refers to as **damping**. The final weights are now given by

$$\omega'_{\mathcal{S}} = 0, \qquad \omega'_{\bar{\mathcal{S}}} = \omega_{\bar{\mathcal{S}}} + \delta^{\star}_{\bar{\mathcal{S}}}.$$

Now, we answer the remaining question of which of the parameters we wants to prune. This is done with approximations using a cheap importance score.

Importance of pruning a single weight: If we prune only coordinate i, i.e.,

$$\delta_i = -\omega_i, \quad \delta_j = 0 (j \neq i)$$

then we have

$$\mathcal{L}(\delta) = \omega_i^2 \mathbf{H}_{ii},$$

We will treat this as the importance score, and the larger it is, the more sensitive this direction is (notice that it is always PSD), and we should be less willing to prune this.

In large LLMs, $\mathbf{H}$ is huge typically, so we actually divide it into small blocks (e.g., 128×128) and compute the importance scores for weights in that block, and prune the lowest-scoring weights to hit the sparsity target.

A final remark on sparseGPT is that this type of pruning actually introduces unstructured sparsity, which is not useful for hardware acceleration. However, it is not designed for this purpose in the begining.

• **GPTQ:** The idea is very similar to sparseGPT. Imagine for a linear layer $\mathbf{W}$, we want to quantized it to replace $\mathbf{W}$ by a quantized matrix $\mathbf{Q}_{\mathbf{W}}$. Define the perturbation,

$$\Delta\mathbf{W} = \mathbf{Q}_{\mathbf{W}} - \mathbf{W}, \qquad \delta = \text{vec}\left(\Delta\mathbf{W}\right).$$

We want to minimize the output distortion

$$\mathcal{L}(\delta) = \mathbb{E}\left[\|\Delta\mathbf{W}x\|^2\right] \simeq \delta^{\top}\mathbf{H}\delta,$$

with certain constraints (instead of zero, we put on some grid) on $\mathbf{Q}_{\mathbf{W}}$, we can therefore use a similar way to approximate this optimization process.

• **SparseGPT-variants**: With nearly the same mathematics, SparseGPT can be extended to multiple other sparsity pattern.

1. **Block sparsity:** Fixed size blocks (e.g. 8×8, 16×16 are either kept or removed), sometimes the block sizes can be asymmetric, and typically, $50 - 75\%$ sparsity. Mainly in GPU-oriented research and less common then $N : M$ sparsity.

2. **N:M sparsity:** one of the most important parttern today (as of 2025), which is semi-structured. An example would be 2:4 sparsity, which means that in every 4 weights, exactly 2 are zero.

   One may wonder why such a pattern leads to acceleration. In fact, the hardware (Ampere, Hopper) knows that in every contiguous group of 4 weights along a specific dimension, exactly 2 must be zero in advance. The key is that the groups are contiguous in memory and the hardware encodes is aware of a compacted mask, so it could reduce the number of FMA (Fused-Multiply-Add) operations.

   The Nvidia Tensor Cores have dedicated spares MMA instructions which half the multiplications. This results in $1.5 \sim 2$ speedups.

- **Wanda:** Wanda is a first order, activation-aware pruning method that esitmates weights importance using only activation statistics and weight magnitudes, without any second-order (Hessian) compensation.

SparseGPT shows that second order pruning works extremely well, but it is computationally heavy because of damping and inverse. A natural question based on that is that:

```
Can we get most of the benefit of SparseGPT without computing or
inverting any Hessian?
```

Let us consider the same linear layer with $\mathbf{W}$, sparseGPT uses $\omega_i \mathbf{H}_{ii}$ as the importance score, note that

$$\mathbf{H}_{ii} = \mathbb{E}\left[x_j^2\right],$$

for the corresponding input channel. This is justifed, since that for each coordinate i in ω, there must be a corresponding (r, j) in $\mathbf{W}$ where r is the index in the out dimension and j is the index in the in dimension, we have $i = (j-1)m + r$. Thus,

$$\mathbf{H}_{ii} = \mathbf{C}_{jj} = \mathbb{E}\left[x_j^2\right],$$

where $x \in \mathbb{R}^d$ is the input. **Wanda** makes the following approximation: $\mathbf{H}_{ii} \simeq \mathbb{E}\left[|x_j|\right]$, i.e.

$$\mathbf{H}_{ii} \simeq \|\mathbf{X}_{j,:}\|_1 \text{ or } \mathbf{H}_{ii} \simeq \|\mathbf{X}_{j,:}\|_2,$$

i.e, we take all i-th coordinates over the calibration batch and use 1 or 2 norm. And we replace the original pruning score as.

$$\text{score}_i = |\omega_i| \, \|\mathbf{X}_{j,:}\|.$$

After we determine which set of the parameters to prune, we do the pruning, and thats it, no compensation. Roughly speaking, Wanda is effectively replacing the sparseGPT score $\omega_i^2 \mathbf{H}_i i$ with $|\omega_i| \cdot \sqrt{\mathbf{H}_i i}$, which is just a first order proxy of the second order sparseGPT criterion. Now that we do not adjust the weights of the unpruned. This works empirically, and the intuition is roughly that this is just first order approximations.

15.6 Low-Precision Inference Techniques for LLMs

Beyond model architecture and algorithmic improvements, precision reduction has emerged as a key strategy to reduce memory footprint, improve throughput, and lower inference cost.

AWQ: Activation-Aware Weight Quanization (AWQ), is a widely adopted post-training quantization method and is supported by several production inference frameworks, including vLLM and TensorRT-LLM. It is commonly used for INT4-weight only inference on GPUs [37].

Motivation:

[37] This refers to that model weights are stored in 4 bits integers and activations and most computations remain in higher precision (FP16 / BF16 / sometimes INT 8), this setup is executed efficiently on GPUs.

15.7 Softmax Tricks

The usual softmax of a vector $\mathbf{x}$, is given by the following formula:

$$\mathrm{softmax}(\mathbf{x})_i = \frac{e^{\mathbf{x}_i}}{\sum_{j=1}^{d} e^{\mathbf{x}_j}}.$$

The problem is, if certain x_i are too large, e^{x_i} will overflow and become "inf", which causes trouble in training stability.

Safe softmax trick: This refers to the following practice,

$$\mathrm{softmax}(\mathbf{x})_i = \frac{e^{\mathbf{x}_i - m}}{\sum_{j=1}^{d} e^{\mathbf{x}_j - m}}, \quad \text{where } m := \max_i \mathbf{x}_i.$$

The two are equivalent in mathmatics, because we only divide each term by e^m, every $e^{\mathbf{x}_i - m}$ is between $[0, 1]$, and the overflow problem is much more controllable.

The log-sum-exp trick: Now, if we imagine that we have obtained a logit vector $\mathbf{x}$, we first compute the softmax to transform it into a probability vector, then take the log and finally we compute the negative log likelihood as the cross entropy, basically, we have

$$\mathrm{NLL}(\mathbf{x}) = -\log \frac{e^{\mathbf{x}_y}}{\sum_j e^{\mathbf{x}_j}},$$

where we actually faces the same problem as before. But notice we can simplify the expression a bit, by

$$\begin{aligned}
-\log \frac{e^{\mathbf{x}_y}}{\sum_j e^{\mathbf{x}_j}} &= -\left(\mathbf{x}_y + \log \frac{1}{\sum_j e^{\mathbf{x}_j}}\right) \\
&= -\left(\mathbf{x}_y + \log \frac{1}{e^m \sum_j e^{\mathbf{x}_j - m}}\right) \\
&= -\left(\mathbf{x}_y - m + \log \frac{1}{\sum_j e^{\mathbf{x}_j - m}}\right) \\
&= -\mathbf{x}_y + m + \log \sum_j e^{\mathbf{x}_j - m}.
\end{aligned}$$

15.8 KL divergence and Cross Entropy

Let us first give the definition, let $\mathbf{q}, \mathbf{p}$ be a discrete probability distribution over the vocabulary $\mathcal{V}$, then we have the term entropy defined as

$$H(\mathbf{q}) = -\sum_{v \in \mathcal{V}} \mathbf{q}(v) \cdot \log \mathbf{q}(v). \tag{Entropy}$$

and cross entropy defined as

$$H(\mathbf{q}, \mathbf{p}) = -\sum_{v \in \mathcal{V}} \mathbf{q}(v) \log \mathbf{p}(v). \tag{Cross Entropy}$$

The KL divergence is defined as

$$\mathrm{KL}(\mathbf{q}||\mathbf{p}) = \sum_{v \in \mathcal{V}} \mathbf{q}(v) \log \frac{\mathbf{q}(v)}{\mathbf{p}(v)}. \tag{KL Divergence}$$

It is easy to check that the three definitions satisfies

$$\mathrm{KL}(\mathbf{q}||\mathbf{p}) = -H(\mathbf{q}) + H(\mathbf{q}, \mathbf{p}). \tag{25}$$

One may notice that in the cross entropy loss, we use $\mathbf{q}$ as the goal (data-generating) distribution, and $\mathbf{p}$ as the model / parametric distribution, hence, minimizing the cross entropy loss is just equivalent to minimizing the KL divergence via (25). To this end, one can interpret the minimization as a type of projection based on KL divergence.

15.9 Contiguous

Many PyTorch operations (especially '.view') assume the tensor is laid out contiguously in memory. If it isn't, the operation either fails or silently does extra work (copies), which affects correctness assumptions and performance.

Definition: A tensor is contiguous if its elements are stored in memory in the same order as its logical indexing. Example:

```
x = tensor( [[a, b, c],
            [d, e, f]] )
```

It has a memory lay out of

```
[a, b, c, d, e, f],
```

which is contiguous.

How tensors become non-contiguous: Slicing, transposing, permuting does not move data, it only changes strides.

```
x = torch.randn(4, 5, 6)
y = x[:, :-1, :]      # slicing
z = x.transpose(1, 2)
```

Both y and z are usually non-contiguous. They are views pointing to the same memory, but with different strides.

.view() .view() is a pure reinterpretation of memory. It requires: new shape is compatible with existing strides.

```
x = torch.randn(2, 3, 4)
y = x[:, :-1, :]      # non-contiguous
y.view(-1, 4)         # RuntimeError
```

Because the memory layout is no longer "linear" in the way .view() expects.

```
y = y.contiguous()
y.view(-1, 4)
```

works because .contiguous() physically copies the data into a contiguous block.

.reshape() It is kind of smarter in the sense that in the case that the tensor is contiguous, it returns a view (zero cost), but in the other cases, it will silently allocate a contiguous copy. However, if we use this function, we lose control of when memory is allocated, how much memory is allocated and performance predictability. That's why we often write

```
y = y.contiguous().view(...)
```

to make the cost explicit.

All in one sentence: If we slice / transpose / permute a tensor and then call .view(), we should almost always call .contiguous() first.

Stride: Strides are the rule that tells PyTorch how to walk through memory when we index a tensor. For a tensor $\mathbf{x}$ with shape $(d_0, d1, \ldots, d_n)$, stride at dimension k is how many memory elements we must skip to move the index by $+1$ along that dimension. If $\mathbf{x}$ has base pointer ptr and strides $s = (s_0, s_1, \ldots, s_n)$ then the address of $\mathbf{x}\,[i_0, i_1, \ldots, i_n]$ is given by

$$\text{ptr} + \sum_{k=0}^{n} i_k \cdot s_k.$$

15.10 Binary Classification, AUC and ROC

In binary classification tasks, we often construct the confusion matrix, which has the following four categories:

- TP: Ground Truth Positive and Prediction Positive
- FP: Ground Truth Negative and Prediction Positive
- TN: Ground Truth Positive and Prediction Negative
- FN: Ground Truth Negative and Prediction Negative

We define two key rates

- Recall: (True Positive Rate)

  $$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

  This stands for the percentage of TP in the total correct examples.

- Sensitivity: (False Positive Rate)

  $$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

  This stands for the percentage of FP in the total wrong examples.

Then the ROC (Receiver Operating Characteristic) curve is the curve that

- x axis: False Positive Rate (FPR)
- y axis: True Positive Rate (TPR)

As the threshold go from 0 to 1 (assume the score output by the model is in $[0, 1]$), FPR, TPR changes thats why we get a curve, for threshold 0, we predict all positive, so y should be 1 (because FN $= 0$), and x should be 1 (because TN $= 0$). When we set threshold to be 1 (all negative predictions), we get $(0, 0)$

AUC (Area under the ROC curve) measures how well a model ranks positive examples above negative ones, i.e., AUC is the probability that a randomly chosen positive sample gets a higher score than a randomly chosen negative sample. Mathematically:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, \mathbf{d}\, \text{FPR}.$$

Now notice that each in FPR corresponds to letting one more negative pass the threshold (FP increases, and FPR increases), the corresponding TPR tells you how many positives are already above that negative.

$$\text{TPR}(t) = \frac{|\{x^+ : s(x^+) \geq t\}|}{\text{P}}$$
$$\text{FPR}(t) = \frac{|\{x^+ : s(x^+) < t\}|}{N}$$

Fix a ground-truth negative sample x^- with score s^-, at the threshold $t = s^-$, this negative just becomes predicted positive so FPR increases $1/\text{N}$, and if we plug in $t = s^-$, we realize TPR becomes the fraction of postive samples that has a higher score than this negative example. Since each negative contributes:

$$\Delta\text{FPR} = \frac{1}{\text{N}},$$

Assume no ties, the integral becomes a summation of

$$\text{AUC} = \frac{1}{\text{N}} \sum_{x^-} \frac{|\{x^+ : s(x^+) > s(x^-)\}|}{\text{P}} = \frac{1}{\text{PN}} \sum_{x^-} |\left\{x^+ : s(x^+) > s(x^-)\right\}|$$

The numerator counts all positive negative pairs where $s(x^+) > s(x^-)$ and the denominator is the total number of pairs. Thus

$$\text{AUC} = \Pr\left\{s(x^+) > s(x^-)\right\},$$

because we first computed for each x^- the probability of a x^+ that has a larger $s(x^+) \geq s(x^-)$, then average over x^-.

The case where ties happens: It could happen that $s(x^+) = s(x^-)$, in this case, we define

$$\phi(x^+, x^-) = \begin{cases} 1, & s(x^+) > s(x^-) \\ \frac{1}{2}, & s(x^+) = s(x^-) \\ 0, & s(x^+) < s(x^-). \end{cases}$$

Then AUC becomes

$$\text{AUC} = \frac{1}{\text{PN}} \sum_{x^+} \sum_{x^-} \phi(x^+, x^-) = \Pr\left\{s(x^+) > s(x^-)\right\} + \frac{1}{2}\Pr\left\{s(x^+) = s(x^-)\right\}.$$

Practical Hint. Assume we have the ground truth labels $y_i \in \{0, 1\}$ and the model score $s_i \in \mathbb{R}$.

- Sort all samples decreasingly in terms of their s_i.
- Move threshold from $+\infty$ to each point s_i and compute $(\text{FPR}_k, \text{TPR}_k)$.
- Compute the area, given ROC points we have generated, we have

$$\text{AUC} = \sum_k (\text{FPR}_{k+1} - \text{FPR}_k) * \frac{\text{TPR}_k + \text{TPR}_{k+1}}{2}.$$

We may use the Mann-Whiteney formula (uses the pairwise ranking view)

- Rank all samples by score and record their ranking r_i, with average ranks are used for ties.
- $$\text{AUC} = \frac{\text{R}_+ - \frac{\text{P}(\text{P}+1)}{2}}{\text{PN}}$$

  where R_+ is sum of ranks of positive samples, which is mathematically identical to ROC integration and very efficient.

```
from sklearn.metrics import roc_auc_score
auc = roc_auc_score(labels.cpu(), scores.cpu())
```

Remark: Random guessing gives AUC around 0.5 numerically.

15.11 Random Interview Questions

Write the GRPO loss. Instead of learning a value function, GPRO uses relative ranking inside the group, that is, for an prompt $\mathbf{x}$, we prepare a group of rollouts $\mathbf{y}_i$, and we denote

$$A_i = r(\mathbf{y}_i, \mathbf{x}) - \frac{1}{G} \sum_{i=1}^{G} r(\mathbf{y}_i, \mathbf{x}),$$

where G is the size of the group. We optionally normalized them with

$$A_i = \frac{A_i}{\sqrt{\text{Var}(A) + \varepsilon}}.$$

The GRPO loss can be written as

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{G} \sum_{i=1}^{G} \frac{\pi_\theta(\mathbf{y}_i \mid \mathbf{x})}{\pi_{\theta_{\text{old}}}(\mathbf{y}_i \mid \mathbf{x})} A_i$$

In practice sometimes we clipped the noise

$$\mathcal{L}_{\text{GRPO-CLIP}} = -\frac{1}{G} \sum_{i=1}^{G} \min \left\{ \rho_i A_i, \text{Clip}(\rho_i, 1-\varepsilon, 1+\varepsilon) A_i \right\},$$

where $\rho_i = \frac{\pi_\theta(\mathbf{y}_i|\mathbf{x})}{\pi_{\theta_{\text{old}}}(\mathbf{y}_i|\mathbf{x})}$. Very commonly, we apply KL divergence to the loss

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{GRPO}} + \beta \cdot \text{D}_{\text{KL}} \left(\pi_\theta(\cdot \mid \mathbf{x}) \parallel \pi_{\theta_{\text{ref}}}(\cdot \mid \mathbf{x}) \right).$$

- Sample multiple candidate answers to the same interview question.
- Rank them relative to each other.
- Push probability up for answers better than the group average.
- Push probability down for worse ones.
- No value network, no reward baseline model.

GRPO (like PPO) performs multiple gradient update steps on rollouts generated by θ_{old}, but only within a trust region.

Write the PPO loss. Given prompt $\mathbf{x}_t$ sample response $\mathbf{y}_t$ and advantage A_t, let ρ_t be

$$\rho_t = \frac{\pi_\theta(\mathbf{y}_t \mid \mathbf{x}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{y}_t \mid \mathbf{x}_t)}.$$

We can write $\mathcal{L}_{\text{PPO}}$ as

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_t \left[\min \left\{\rho_t A_t, \text{Clip}\left(\rho_t, 1-\varepsilon, 1+\varepsilon\right) A_t\right\}\right]$$

We also add the KL to reference

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PPO}} + \beta \cdot \mathbb{E}_t \left[\mathrm{D}_{\text{KL}}\left(\pi_\theta(\cdot \mid \mathbf{x}) \parallel \pi_{\theta_{\text{ref}}}(\cdot \mid \mathbf{x})\right)\right].$$

A_t is the $r(\mathbf{x}_t, \mathbf{y}_t) - V_\phi(\mathbf{x}_t)$ (how good above average), we use the value head to estimates the expected absolute reward for a prompt. And we add the value (critic) loss

$$\mathcal{L}_{\text{value}} = \mathbb{E}_t \left[\left(V_\phi(\mathbf{x}_t) - G_t\right)^2\right],$$

where G_t is the observed return from rollout with

$$\mathbb{E}\left[G_t \mid \mathbf{x}_t\right] = V^\pi(x_t) = \mathbb{E}_{\mathbf{y}\sim\pi}\left[r(x, y)\right].$$

We use this loss to update the value head instead of the policy. In classical RL PPO, we add entropy bonus, but in LLM RL, this is replaced by KL divergence control. In GRPO, we are basically replacing the estimated V_ϕ with the value head with the group average reard. PPO learns what "average" means, GRPO computes it directly from the samples.

GRPO loss oscillates back and forth If GRPO loss oscillates back and forth, it usually means the updates are too noisy or too aggressive, or your advantage signal is unstable.

- **Updates are too aggressive (most common)**
- Symptoms:
  1. Loss spikes every few steps/epochs.
  2. KL to the reference (or to θ_old) jumps.
  3. Policy starts changing too fast; group advantages flip sign frequently.
- Fixes:

1. Lower learning rate (first thing to try)
2. Fewer PPO-style update epochs per rollout (e.g., from 4-8 down to 1-2) (Note that this is similar to local training.)
3. Tighter clipping: reduce ε (e.g., $0.2 \to 0.1 \to 0.05$).
4. Add / increase KL penalty to a reference model; use adaptive KL if you have it.
5. Early stop on KL: stop updates when KL exceeds a target.

- **Group-relative advantage is too noisy**: GRPO baseline is the group mean. If the group is small or low-diversity, the mean is a noisy estimator, so advantages bounce.

- Reasons

  1. Group size G too small (e.g., 2-4).
  2. Sampling is too random (high temperature/top-p), making rewards noisy.
  3. Reward model is noisy or inconsistent for your task.

- Fixes

  1. Increase group size G.
  2. Normalize advantages within each group.
  3. Clip advantages to avoid extreme updates.
  4. Stabilize sampling, slightly lower temperature, reduce top-p, or use a fixed decoding policy during training rollouts

- **Reward scale / reward hacking / reward distribution shift**

- Symptoms:

  1. Rewards have heavy tails (rare huge scores)
  2. Rewards drift upward/downward rapidly across steps
  3. Loss oscillation correlates with reward outliers

- Solutions:

  1. Reward normalization (per batch or per prompt-group)
  2. Track and limit response length; uncontrolled length often destabilizes reward signals
  3. Winsorize rewards (cap extreme values)

15.12 About Fast Inferencing

There are several techniques exist for fast inferencing in practice.

- (1) We have already talked about one of them, which is the V cache optimization. When we perform autoregressive inferencing, the main cost of computing attention comes from reading the KV cache, with MQA / GQA, we can reduce the size of the KV cache by asking some of the heads to share Key vector and Value vector.

- (2) Another technique is the Paged KV Cache, which is used in vLLM. This is a memory-management technique introduced in the vLLM inference system. The key idea is to manage the KV cache like an operating system manages virtual memory pages rather than storing it as one long contiguous block.

Traditionally, most implementations allocate contiguous memory for each request. Example memory layout:

```
Request A
[token1 token2 token3 token4 ... token120]

Request B
[token1 token2 ... token15]

Request C
[token1 ... token450]
```

Each sequence occupies one continuous block. However, in real inference servers: <1> requests arrive and finish at different times <2> sequences have different lengths.

Initial memory:

```
A: 120 tokens
B: 15 tokens
C: 450 tokens
D: 8 tokens
```

Memory after B finishes:

```
A: 120 tokens
[free space of 15 tokens]
C: 450 tokens
D: 8 tokens
```

The system cannot use the 15-token hole, so it must allocate memory elsewhere. This is memory fragmentation, which results in GPU memory wasted, less requests catered and throughput decreases. Instead of storing tokens contiguously, vLLM splits the KV cache into fixed-size pages. Each request stores a list of pages, not one continuous block. The physical pages can be anywhere. This is exactly the same idea as virtual memory paging in operating systems.

This also enables continous batching (new requests can be inserted between decoding steps).

In naive batching: we do

1. collect a batch of requests
2. run decoding for that batch
3. wait until some finish
4. build a new batch later

So the batch is relatively static.

Example

```
Batch at start: A, B, C
decode step 1
decode step 2
decode step 3
...
```

If a new request D arrives while A, B, C are already being decoded, the system may not insert D immediately. It may wait for some convenient rebatching point.

In continous batching, the server does this instead:

1. after each decode step, it can
2. remove finished requests
3. add newly arrived requests
4. keep unfinished requests

This means the GPU is kept busy with a continuously refreshed set of sequences. With the previous contiguous design, we need to make sure there are enough space because we do not have the concept of pages. This causes problem for continuous batching.

About vLLM: vLLM is an inference and serving engine for LLMs. In plain English: it is a system we use after training, when we want to run a model efficiently for generation or expose it as an API. Its official docs position it around two main workflows: **offline batched inference** and **online serving** through an OpenAI-compatible server [38]. It is known for high throughput and memory-efficient serving, with techniques like paged attention / paged KV cache.

Example usage:

```python
from vllm import LLM, SamplingParams

llm = LLM(model="facebook/opt-125m")
sampling_params = SamplingParams(temperature=0.8, top_p=0.95, max_tokens=64)

prompts = [
  "Explain what a transformer is in one paragraph.",
  "What is the difference between SGD and Adam?",
]

outputs = llm.generate(prompts, sampling_params)

for output in outputs:
  print(output.prompt)
  print(output.outputs[0].text)
```

- (3) Continous Batching: itself is a fast inferencing technique.
- (4) Flash Attention: It does not change the mathematical definition of attention, it only changes how attention is scheduled on GPU memory.

Let us consider a classical setting when we compute attention for one head: we have

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{L \times d}$$

and the attention (scaled dot product attention) is computed as

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}, \quad \mathbf{P} = \text{softmax}(\mathbf{S}), \quad \mathbf{O} = \mathbf{P}\mathbf{V}.$$

The problem is that if we look at $\mathbf{S}$ and $\mathbf{P}$, their sizes are $L \times L$, which means that they are huge, especially when the context lengths are long.

GPU has HBM[39] / global memory, which is large but slow, and it also has SRAM[40] / shared memory / registers which are much smaller but much faster.

In the naive implementation, we typically do things like:

[38] lets your app call a local/self-hosted model almost like it calls the OpenAI API.
[39] High Bandwidth Memory: very wide memory bus, many parallel channels
[40] Static Random Access Memory: This is on-chip memory, built directly on the GPU die.

1. read $\mathbf{Q}$ and $\mathbf{K}$
2. compute all scores $\mathbf{S}$
3. write $\mathbf{S}$ out to HBM
4. read $\mathbf{S}$ back
5. compute softmax
6. maybe write/read again
7. multiply by $\mathbf{V}$

So even if the arithmetic is fine, the algorithm moves a lot of data back and forth between slow and fast memory. FlashAttention was explicitly introduced as an IO-aware exact attention algorithm to reduce these HBM reads/writes.

FlashAttention avoids materializing the attention matrix by computing the needed pieces block by block and immediately folds them into the final output. The core idea is tiling.

Tiling. Instead of treating $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ as one large tensor, we partition them along the sequence dimension

$$\mathbf{Q} = \begin{bmatrix} \mathbf{Q}^{(1)} \\ \mathbf{Q}^{(2)} \\ \vdots \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}^{(1)} \\ \mathbf{K}^{(2)} \\ \vdots \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} \mathbf{V}^{(1)} \\ \mathbf{V}^{(2)} \\ \vdots \end{bmatrix}.$$

Each tile contains a contiguous chunk of tokens.

For a fixed query block $\mathbf{Q}^{(i)}$, we iterate over the key/value tiles $\mathbf{K}^{(j)}$ and $\mathbf{V}^{(j)}$ and compute the score tile

$$\mathbf{S}^{(i,j)} = \frac{\mathbf{Q}^{(i)}(\mathbf{K}^{(j)})^\top}{\sqrt{d}}.$$

Instead of forming the full attention matrix

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}},$$

the computation proceeds tile by tile so that the working data fits in on-chip memory.

Softmax. Attention probabilities are obtained by applying a row-wise softmax

$$\mathbf{P} = \mathrm{softmax}(\mathbf{S}).$$

For a single query row r the attention weights are

$$p_{rj} = \frac{e^{S_{rj}}}{\sum_{t=1}^{T} e^{S_{rt}}}, \qquad j = 1, \ldots, T.$$

The denominator involves all keys, so it cannot be computed independently within each tile.

Online Softmax. To avoid materializing the entire score row, we process the keys sequentially and maintain two running statistics

$$m_t = \max_{1\le j\le t} S_{rj},$$
$$l_t = \sum_{j=1}^{t} e^{S_{rj}-m_t}, \qquad t = 1,\dots,T.$$

Assume that after processing keys $1,\dots,t-1$ we have

$$m_{t-1} = \max_{1\le j<t} S_{rj},$$
$$l_{t-1} = \sum_{j=1}^{t-1} e^{S_{rj}-m_{t-1}}.$$

When a new score S_{rt} is observed, the running maximum becomes

$$m_t = \max(m_{t-1}, S_{rt}).$$

The normalization term satisfies

$$\begin{aligned} l_t &= \sum_{j=1}^{t} e^{S_{rj}-m_t} \\ &= e^{m_{t-1}-m_t} l_{t-1} + e^{S_{rt}-m_t}. \end{aligned}$$

Attention Output. The attention output for query r is

$$\mathbf{o}_r = \frac{\sum_{j=1}^{T} e^{S_{rj}} \mathbf{v}_j}{\sum_{j=1}^{T} e^{S_{rj}}}.$$

Note that this is basically $\mathrm{softmax}(\mathbf{S})\mathbf{V}$. Define the running numerator (numerically stable)

$$\mathbf{o}_t = \sum_{j=1}^{t} e^{S_{rj}-m_t} \mathbf{v}_j.$$

Using the same rescaling argument,

$$\mathbf{o}_t = e^{m_{t-1}-m_t} \mathbf{o}_{t-1} + e^{S_{rt}-m_t} \mathbf{v}_t.$$

After all keys have been processed, the final attention output is

$$\mathbf{o}_r = \frac{\mathbf{o}_T}{l_T}.$$

In practice the algorithm applies this update simultaneously to every row of the current query tile while iterating over key/value tiles. The full attention matrix is never materialized; only the current tile and the running statistics are maintained.

It is worth noting that the memory consumption changes from $\mathcal{O}(T^2)$ to $\mathcal{O}(T)$, this is because in FlashAttention, we store:

1. $\mathbf{Q}, \mathbf{K}, \mathbf{V}$: already part of the model computation

2. output $\mathbf{O} \in \mathbb{R}^{T \times d}$
3. per-row running statistics like max and normalization
4. one or a few temporary tiles in SRAM

So the extra memory for the attention computation scales linearly in T not quadratically. The papers are precise that the savings come from reducing HBM traffic and not materializing the full attention matrix. However, the computation is still roughly **quadratic**. What changes is memory footprint and IO cost, which is often the real bottleneck on GPU. So FlashAttention is better thought of as: same exact attention math, much better memory movement pattern.

Why recomputation appears? Because FlashAttention refuses to store large intermediates, in backward pass it may recompute some local quantities instead of saving everything from forward. This is a classic time-vs-memory tradeoff, but on GPUs it often wins because memory traffic is so expensive. The original paper emphasizes that reducing memory reads/writes can outperform approaches that try to save every intermediate.

Three generations of FlashAttention

1. **FlashAttention:** The original 2022 work introduced the tiled, IO-aware exact attention algorithm that avoids materializing the full attention matrix and substantially reduces HBM traffic.
2. **FlashAttention-2:** keeps the same basic philosophy but reorganize the kernel so the GPU stays busy much more of the time. What FlashAttention-2 changes is how the work is partitioned and scheduled on the GPU, so it runs much faster in practice.
   - **Fewer non-matmul operations:** more time is spent in tensor-core-friendly matrix multiplications and less time is wasted on bookkeeping, rescaling, and other operations that GPUs execute less efficiently.
   - **Better parallelism across the sequence dimension:** FlashAttention-1 can underutilize the GPU when batch size or number of heads is small. FlashAttention-2 parallelizes attention computation more aggressively, including for a single head, across multiple thread blocks, so GPU occupancy is higher.
   - **Better work partitioning within a thread block:** Inside each block, FlashAttention-2 distributes work across warps[41] more carefully, which reduces synchronization and shared-memory traffic. That means less internal overhead and better effective speed.

   The FlashAttention-2 paper reports roughly 2x speedup over FlashAttention-1 in many settings, and says it reaches about 50-73% of theoretical peak FLOPs/s on A100, which is much closer to GEMM-like efficiency.
3. **FlashAttention-3:** FlashAttention-3 targets Hopper GPUs such as H100 and uses Hopper-specific features. (1) exploiting asynchrony and TMA[42] to overlap compute and data movement, (2) interleaving blockwise matmul and softmax operations more aggressively, and (3) using low-precision support such as FP8-related techniques.

[41] A warp is a group of 32 threads that execute the same instruction at the same time on an NVIDIA GPU. When you launch a CUDA kernel with many threads, the GPU hardware internally schedules them in blocks of 32-thread bundles, called warps. The GPU scheduler operates at the warp level, not the thread level.

[42] Tensor Memory Accelerator

Warp: A warp is a group of 32 threads that execute the same instruction at the same time on an NVIDIA GPU. When you launch a CUDA kernel with many threads, the GPU hardware internally schedules them in blocks of 32-thread bundles, called warps. The GPU scheduler operates at the warp level, not the thread level.

Suppose a warp runs this kernel:

```
  y[i] = a * x[i] + b
```

Warp execution looks conceptually like

```
Thread 0   : y[0]  = a*x[0]  + b
Thread 1   : y[1]  = a*x[1]  + b
...
Thread 31  : y[31] = a*x[31] + b
```

But the GPU issues one instruction for the whole warp. Because warps share the same instruction stream, branches can hurt performance. Example:

```
if (x[i] > 0)
  y[i] = x[i]
else
  y[i] = 0
```

If inside a warp, some threads take the if, others take the else, then GPU executes both paths sequentially, masking threads. This is called warp divergence. When people say: FlashAttention-2 distributes work across warps more carefully, they mean the kernel assigns computation like query blocks, key blocks, reduction work to different warps, instead of having one warp doing sequential steps.

The original FlashAttention had patterns like:

```
warp 0:
  load K tile
  compute partial attention
  update running softmax
  write results
```

FlashAttention-2 reorganizes so that

```
warp 0 → compute QK^T
warp 1 → compute softmax stats
warp 2 → multiply V
warp 3 → reductions
```

This improves warp occupancy, memory throughput and SM utilization.

The execution hierarchy is

```
GPU
 - SM (streaming multiprocessor)
     - thread blocks
         - warps (32 threads)
             - threads
```

FlashAttention kernels are tuned around: warp-level matrix multiply, warp-level reductions and warp-level memory loads. FlashAttention-2 improved performance largely by designing the kernel around warp-level parallelism instead of thread-level loops.

- (5) Speculative Decoding: The core idea is

GPU Level	CPU Analogy
thread	scalar instruction
warp	vector instruction
block	CPU core workload
SM	CPU core

Table 7: A useful mental model relating CUDA execution hierarchy to CPU concepts.

1. a small fast model (the draft model) proposes several next tokens at once
2. the large target model then checks those proposals in parallel
3. every proposal that passes verification is accepted
4. generation then continues from the first rejection point or from the end of the accepted chunk.

So instead of the big model doing one token per forward pass, speculative decoding tries to let the small model "guess ahead" a few steps, and then uses one target-model pass to verify multiple guesses at once. That is why the technique can **xreduce inter-token latency**, especially in memory-bound inference settings.

The speed up depends on (1) draft cost: The draft model must be much cheaper than the target model; (2) Acceptance rate: The draft model's guesses must often be accepted.

- (6) Quantization: **Reduce memory bandwidth**.
- (7) Kernel Fusion: Kernel fusion means combining several GPU operations that would normally run as separate kernels into one kernel. TensorRT's glossary defines fusion exactly in this spirit: combining multiple layers into a single operation to reduce**memory bandwidth** use and **kernel launch overhead**.
- (8) Prefill / Decode Separation: LLM inference is not one uniform workload. It has two very different phases:

1. **Prefill**: consume the whole prompt and build the KV cache.
2. **Decode**: generate new tokens one by one using that KV cache.

These two phases stress the hardware in very different ways, so good systems often treat them differently in scheduling, batching, kernels, and even cluster layout. vLLM explicitly supports this idea; its docs describe disaggregated prefilling as running prefill and decode in different vLLM instances, so we can optimize TTFT(Time To First Token)[43] and inter-token latency separately. vLLM also documents chunked prefill and different thresholds for prefill-heavy versus decode-only batches.

- (9) Tensor Parallel Inference: Megatron / TensorRT-LLM. Multiple GPUs assigned for tensor parallelism.
- (10) Activation Recomputation Removal.
- (11) Prefix Cache: If multiple requests share the same prefix (e.g. system prompt), we can cache KV cache. Often seen in ChatGPT, vLLM.

LLM inference is often memory-bound: Read model weights; Read KV cache. Training is often compute bound because of large batches.

15.13 About PPO

In RLHF, we usually have three stages:

[43] Users care about at least two latency metrics: TTFT: time to first token, ITL: inter-token latency

1. **Supervised fine-tuning (SFT):** Train the language model on high-quality demonstrations.
2. **Reward model (RM):** Train a model to score outputs according to human preferences.
3. **RL optimization with PPO:** Further update the policy model so it generates responses with higher reward, while not drifting too far from the original SFT model.

• The policy is the language model itself: $\pi_\theta(\mathbf{y} \mid \mathbf{x})$, where $\mathbf{x}$ is the prompt and $\mathbf{y}$ is the generated response.

• **Vanilla policy gradient** can be unstable: one update can change the model too much and destroy language quality. **PPO** is a stabilized policy-gradient method that prevents overly large policy updates.

• RLHF objective: Denote the reward model as $r_\phi(\mathbf{x}, \mathbf{y})$ and reference model π_{ref}, a common scalar reward is

$$R(\mathbf{x},\mathbf{y}) = r_\phi(\mathbf{x},\mathbf{y}) - \beta\,\mathrm{KL}(\pi_\theta(\cdot \mid \mathbf{x}) \,\|\, \pi_{\text{ref}}(\cdot \mid \mathbf{x}))\,.$$

or in practice often a token-level KL penalty approximation such as

$$R(\mathbf{x},\mathbf{y}) \approx r_\phi(\mathbf{x},\mathbf{y}) - \beta \sum_{t=1}^{T} [\log \pi_\theta(y_t \mid \mathbf{x},\mathbf{y}_{<t}) - \log \pi_{\text{ref}}(y_t \mid \mathbf{x},\mathbf{y}_{<t})]\,.$$

The KL penalty stops the model from deviate too far from the reference. We will not use the reward directly, having a reward (or return) is not enough for stable policy gradient updates, and the advantage A reduces variance and tells the optimizer whether an action was better or worse than expected. So we use reward to estimate A.

• PPO surrogate objective: Suppose data is collected using the old policy π_{old}: Define the importance ratio

$$\rho_t(\theta) = \frac{\pi_\theta(y_t \mid \mathbf{x},\mathbf{y}_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid \mathbf{x},\mathbf{y}_{<t})}.$$

Let $\hat{A}_t$ be the estimated advantage at token t. PPO maximizes

$$L_{\text{PPO}}(\theta) = \mathbb{E}\left[\min\left(\rho_t(\theta)\hat{A}_t,\, \mathrm{clip}\left(\rho_t(\theta),1-\epsilon,1+\epsilon\right)\hat{A}_t\right)\right].$$

This clipping is the defining trick. If $\rho_t(\theta)$ becomes too large or too small, PPO truncates the incentive to push it further. So even if the gradient wants to make a huge update, PPO resists that.

• Where does the advantage come from? In RLHF for LLMs, after sampling a completion, one must convert the sequence-level reward into learning signals for the policy. There are three common ways to construct the advantage.

Sequence-level handling. A simple approximation treats the entire completion as a single action. After sampling

$$y = (y_1, \ldots, y_T), \qquad y \sim \pi_\theta(\cdot \mid x),$$

the reward model produces a scalar score

$$r_\phi(x,y).$$

A KL penalty against a frozen reference policy π_{ref} is added

$$\mathrm{KL}(x,y) = \sum_{t=1}^{T} \big(\log \pi_\theta(y_t \mid x,y_{<t}) - \log \pi_{\text{ref}}(y_t \mid x,y_{<t})\big).$$

The sequence-level reward becomes

$$R(x,y) = r_\phi(x,y) - \beta\,\mathrm{KL}(x,y).$$

A baseline or value estimate $V_\psi(x)$ predicts the expected reward for prompt x, giving the sequence-level advantage

$$\hat{A}(x,y) = R(x,y) - V_\psi(x).$$

In practice this scalar advantage is often broadcast to all tokens in the sampled completion when updating the policy.

Token-level returns without GAE. More commonly, the sequence reward is converted into token-level rewards. Define the token-level state

$$s_t = (x, y_{<t}),$$

and compute the per-token KL

$$\mathrm{KL}_t = \log \pi_\theta(y_t \mid s_t) - \log \pi_{\mathrm{ref}}(y_t \mid s_t).$$

The token rewards are constructed as

$$r_t = \begin{cases} -\beta\,\mathrm{KL}_t, & t < T, \\ r_\phi(x,y) - \beta\,\mathrm{KL}_T, & t = T. \end{cases}$$

From these we compute the return

$$G_t = \sum_{k=t}^{T} r_k.$$

A value model $V_\psi(s_t)$ predicts the expected return from state s_t, and the advantage used by PPO is

$$\hat{A}_t = G_t - V_\psi(s_t).$$

Token-level advantages with GAE. To reduce variance, many implementations use generalized advantage estimation (GAE). First define the temporal-difference residual

$$\delta_t = r_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t),$$

and then compute the advantage as

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}.$$

The resulting token-level advantages $\hat{A}_t$ are then used in the PPO loss to update the policy.

Intuitively, the reward model provides a sequence-level signal, which is then converted into token-level advantages indicating whether each sampled action y_t led to a better or worse outcome than expected. GAE typically produces lower-variance estimates and therefore more stable training.

So PPO in RLHF is usually actor-critic:

1. actor: policy LM,
2. critic: value head/model,
3. reward model: separate learned preference scorer.

- **Typical loop:** A typical loop is:

1. Sample prompts $\mathbf{x}$.
2. Generate responses $\mathbf{y} \sim \pi_{\theta_{\text{old}}}(\cdot \mid \mathbf{x})$.
3. Compute:
   - log-probs from current/old policy,
   - log-probs from reference model,
   - reward model score,
   - KL penalty,
   - total reward.
4. Estimate advantages $\hat{A}_t$ and returns.
5. Update:
   - policy with PPO clipped objective,
   - value model with regression loss,
   - often also include entropy regularization.
6. Repeat.

- **Drawbacks of PPO-based RLHF**.

PPO-based RLHF works, but it has real drawbacks:

1. **Expensive.**

   You must generate rollouts online and run:

   - policy model,
   - reference model,
   - reward model,
   - value model.

   That is a lot of compute.

2. **Sensitive / unstable.**

   Training can be delicate:

   - KL coefficient β,
   - clipping threshold ϵ,
   - reward scaling,
   - value loss balance,
   - rollout length.

   These all matter.

3. **Reward hacking.**

   The policy may exploit imperfections in the reward model.

4. **Credit assignment is crude.**

   A single sequence-level preference score is often spread across many tokens, which is not always ideal.

5. **Not always the best modern choice.**

   Many newer alignment methods avoid full PPO because it is heavy and complicated.

16 Megatron

- Megatron-Core: the modular library of distributed-training components.
- Megatron-LM: the reference training framework built on top of Megatron Core with ready-made scripts and recipes.

Megatron exists to train Transformer models that are too large, too deep, or too sequence-heavy for a single GPU or even a naïve multi-GPU setup. Megatron Core provides composable building blocks for Transformer training together with multiple parallelism strategies that can be combined across many GPUs. The most important ones to understand first are data parallelism (DP), tensor parallelism (TP), pipeline parallelism (PP), context parallelism (CP), and expert parallelism (EP).

Typical usage:

- Forward TP: All-Gather
- Backward TP: Reduce-Scatter
- Data Parallel Gradients: All-Reduce

Two styles of pipeline parallelism

1. **GPipe style:** Do all forward microbatches first, then all backward.

2. **1F1B (one-forward-one-backward):** After warmup, alternate forward and backward more tightly.

Megatron commonly uses 1F1B-style scheduling, because it is more memory-efficient and practical at scale.

Context Parallelism With context parallelism, we split the sequence length across GPUs. So instead of each rank storing the full sequence, each rank stores a chunk of the context.

Motivation: Long context is expensive because activations and attention memory grow strongly with sequence length. Even when the model itself fits, the activation memory for long sequences may not fit.

CP is mainly a solution of long sequence training, activation memory pressure from sequence length and scaling to larger context windows.

The hidden states are partitioned along sequence dimension. However, this introduces a problem for attention computation. In self-attention, each token can attend to many other tokens. So CP requires communication to make attention work across partitioned sequence chunks. This is due to that even though tokens are sharded, attention dependency is global over the context window. So CP introduces communication around K/V or attention-related states so that each shard can attend correctly.

TP splits **features**. CP splits **tokens**.

Combined Parallelism One of the main strengths of Megatron is these strategies are not alternatives in a menu. They are meant to be composed.

A typical world size decomposition looks like:

```
world size = TP x PP x DP.
```

The role of "parallel_state": The official API docs describe it as the place that manages ranks and groups for tensor, data, context, pipeline, and expert model parallelism.

Before training can begin, Megatron must answer questions like:

1. Which ranks are in my tensor-parallel group?
2. Which ranks are in my pipeline stage group?
3. Which ranks are data-parallel replicas of each other?
4. If I am an MoE rank, what is my expert group?
5. If context parallelism is enabled, who shares my sequence partition?

Activation Checkpointing: During a normal forward pass, the model computes many intermediate tensors, called activations, and keeps them in memory because the backward pass will need them to compute gradients.

With activation checkpointing, we do not save all of those intermediate activations. Instead, we save only a small subset of them, called checkpoints. Then, during backward, when some missing activation is needed, we recompute it by rerunning part of the forward pass. So the tradeoff is between less GPU memory and more computation time. This is mainly to reduce the amount of memory needed by activations, which can be huge if the model is deep, the sequence length is long or the batch size is large. A simple example would be

$$x \mapsto f_1 \mapsto f_2 \mapsto f_3 \mapsto f_4 \mapsto y,$$

Without checkpointing, we need to store $f_1(x), f_2, f_3, f_4$, so backward can use them directly. With gradient checkpointing, we may only store x, f_2, y, then during backward, if we need the output of f_3, we recompute it from the nearest saved point. Basically we are replacing memory usage with recomputation.

In practice, once models become moderately large, people very often combine data / tensor / pipeline parallelism, mixed precision and activation checkpointing.

In practice, activation checkpointing and gradient checkpointing usually means the same thing.

Prelinminaries:

1.
```
    parallel_state.initialize_model_parallel():

```

After initializing PyTorch distributed, Megatron Core is initialized with model-parallel settings, including tensor-model-parallel and pipeline-model-parallel sizes. It read the desired decomposition of the world, builds the process groups for TP, PP, DP, and possibly others and stores rank/group metadata so later modules know how to communicate.

2.
```
    TransformerConfig():

```

This is the model-level object we instantiate for a GPT-style stack. In Megatron, the "model" is not just a plain PyTorch stack. Its layers and execution are aware of the distributed topology. So when we instantiate a Megatron model, we are not merely allocating tensors; we are instantiating a distributed computation structure.

3. Forward Step functions: Megatron Core uses schedules.py to run the model. Define a forward step function that takes the data iterator and the model as input and produces the output tensor and a loss function.

```
    def forward_step_func(data_iterator, model):
      # pass
      return output_tensor, partial(loss_func, loss_mask)

```

The partial creates something behaving like:

```
    def reduced_loss_func(output_tensor):
      return loss_func(loss_mask, output_tensor)

```

4. Distributed Checkpoint: Megatron Core uses distributed checkpoints for loading and saving models. This allows you to convert the model from one parallel setting to another when you load it. For example, a model trained with tensor parallel size 2, can be loaded again as a tensor model with parallel size 4.

```
    def save_distributed_checkpoint(checkpoint_path, gpt_model):
      sharded_state_dict = gpt_model.sharded_state_dict(prefix='')
      dist_checkpointing.save(sharded_state_dict=sharded_state_dict,
      checkpoint_dir=checkpoint_path)

  def load_distributed_checkpoint(checkpoint_path, gpt_model):
      sharded_state_dict=gpt_model.sharded_state_dict(prefix='')
      checkpoint = dist_checkpointing.load(sharded_state_dict=
      sharded_state_dict, checkpoint_dir=checkpoint_path)
      gpt_model.load_state_dict(checkpoint)
      return gpt_model

```

Context parallel package Context Parallelism ("CP") is a parallelization scheme on the dimension of sequence length. Unlike prior SP (sequence parallelism) which only splits the sequence of Dropout and LayerNorm activations CP partitions the network inputs and all activations along sequence dimension.

With CP, all modules except attention (e.g., Linear, LayerNorm, etc.) can work as usual without any changes, because they do not have inter-token operations.

As for attention:

1. The $\mathbf{Q}$ of each token needs to compute with the KV (key and value) of all tokens in the same sequence. Hence, CP requires additional all-gather across GPUs to collect the full sequence of $\mathbf{K}, \mathbf{V}$. Correspondingly, reduce-scatter should be applied to the activation gradients of $\mathbf{K}, \mathbf{V}$ in backward propagation. (Recall that in forward, every rank temporarily used the full $\mathbf{K}, \mathbf{V}$. In backward, gradients with respect to $\mathbf{K}, \mathbf{V}$ are produced from all those attention computations. However, each original $\mathbf{K}, \mathbf{V}$, chunk belongs to one owning rank, so the gradients must be accumulated and sent back to the correct owner.)

A small example would be when computing attention forward, imagine we have 2 GPUs, we initially have

```
GPU0: Q0, K0, V0
GPU1: Q1, K1, V1
```

To compute the attention, we need to call all_gather, after which the situation becomes

```
GPU0: Q0, [K0, K1], [V0, V1] --> computing O0
GPU1: Q1, [K0, K1], [V0, V1] --> computing O1
```

When the backward happens, the loss gradient arriving from above is

```
GPU0: d L / d O0
GPU1: d L / d O1
```

Right now, on GPU0, contributions to the gradient of $\mathbf{Q}_0$, $\mathbf{K}_0$, $\mathbf{V}_0$, $\mathbf{K}_1$, $\mathbf{V}_1$ will be generated, we write the chain rule formula

$$\frac{\partial L}{\partial \mathbf{Q}_0} = \frac{\partial L}{\partial \mathbf{O}_0} \cdot \frac{\partial \mathbf{O}_0}{\partial \mathbf{Q}_0}$$
$$\frac{\partial L}{\partial \mathbf{K}_0} = \frac{\partial L}{\partial \mathbf{O}_0}\frac{\partial \mathbf{O}_0}{\partial \mathbf{K}_0} + \frac{\partial L}{\partial \mathbf{O}_1}\frac{\partial \mathbf{O}_1}{\partial \mathbf{K}_0}.$$

The contribution of $\mathbf{Q}_0$ in gradient can be easily computed, however, for $\mathbf{K}_0$, it participates also the computation of $\mathbf{O}_1$ on GPU1, which requires us to do first reduce (sum), and then scatter [44].

2. An important memory optimization: during forward, each rank stores only its local $\mathbf{K}, \mathbf{V}$ activations rather than permanently storing the full-sequence $\mathbf{K}, \mathbf{V}$. In backward, the needed $\mathbf{K}, \mathbf{V}$ are gathered again, which preserves the memory benefits of CP.

3. CP is enabled by simply setting context_parallel_size=<CP_SIZE> in command line.

4. All-gather and reduce-scatter are implemented under the hood as point-to-point communication in a ring topology. This implementation is optimized. Logically, the model still needs the effect of all-gather and reduce-scatter. Physically, that communication can be carried out as staged ring exchanges for better efficiency.

5. The docs explicitly point out that CP benefits from MQA and GQA because those attention variants reduce the number of key/value heads, which reduces the size of communicated $\mathbf{K}$,$\mathbf{V}$ tensors.

6. **Why CP is useful for long context?** Long-context training often becomes memory-bound because activation memory grows with sequence length. Full activation recomputation can add around 30% overhead, while simply increasing tensor parallelism can also become inefficient because communication becomes harder to hide. CP is presented as a better fit when the real bottleneck is sequence length.

[44] These matrix products are really backprop shorthand, not strict elementary scalar chain rule notation.

The underlying idea is simple: TP helps when layers are too large, CP helps when sequences are too long. With CP size C, each GPU owns about $1/C$ of the sequence, so activation memory per GPU drops accordingly, while the communication is focused mainly around attention rather than every part of the model.

A somewhat more complicated example is the following: 2 TP + 2 CP, this is presented in the Nvidia's Megatron docs.

TP groups: (GPU0, GPU1) for seq chunk 0, and (GPU2, GPU3) for seq chunk 1. This means that GPU0 and GPU1 both work on the same local sequence chunk, but with different TP shards. GPU2 and GPU3 do the same for the other sequence chunk. GPU0 exchanges context information with GPU2, GPU1 exchanges context information with GPU3.

Suppose the input hidden states are $\mathbf{X} \in \mathbb{R}^{B\times S\times H}$, with CP$=2$, the sequence is split into two chunks:

$$\mathbf{X}^0 \in \mathbb{R}^{B\times \frac{S}{2}\times H}, \quad \mathbf{X}^1 \in \mathbb{R}^{B\times \frac{S}{2}\times H}.$$

The first yellow AG / RS: TP-side activation layout bookkeeping These yellow boxes are best understood as the TP-side communication/layout transitions that appear when Megatron combines TP with sequence-parallel-style activation handling. To be specific, inside each TP group, Megatron is not storing the full $\mathbf{X}^0$ on both TP ranks all the time. To save activation memory, it keeps the activations sequence-sharded across the TP ranks, so for the top group, $\mathbf{X}^0$ is split again along the sequence dimension:

$$\mathbf{X}^0 = \left[\mathbf{X}^{00}, \mathbf{X}^{01}\right], \text{ where } \mathbf{X}^{00}, \mathbf{X}^{01} \in \mathbb{R}^{B\times \frac{S}{4}\times H}.$$

We do all gather here so each GPU now has an full input for the corresponding chunk.

Now consider the QKV part, which we assume is a column parallel linear layer, the weight is splitted along the output dimension:

$$\mathbf{W}_A = \left[\mathbf{W}_A^0, \mathbf{W}_A^1\right], \quad A \in Q, K, V$$

Here, we compute both $\mathbf{X}^0\mathbf{W}_A^0, \mathbf{X}^0\mathbf{W}_A^1$, which are both $\mathbb{R}^{B\times \frac{S}{2}\times \frac{H}{2}}$, which we will denote as $\mathbf{Q}^{00}, \mathbf{Q}^{01}$ respectively.

Now comes the blue AG, this is because each query token need key and value from all tokens in the same sequence. After the blue AG, on GPU0, we have

$$\mathbf{Q}^{00} \in \mathbb{R}^{B\times \frac{S}{2}\times \frac{H}{2}}, \quad [\mathbf{K}^{00}, \mathbf{K}^{10}] \in \mathbb{R}^{B\times S\times \frac{H}{2}}, \quad [\mathbf{V}^{00}, \mathbf{V}^{10}],$$

similarly on GPU1, we have

$$\mathbf{Q}^{01} \in \mathbb{R}^{B\times \frac{S}{2}\times \frac{H}{2}}, \quad [\mathbf{K}^{01}, \mathbf{K}^{11}] \in \mathbb{R}^{B\times S\times \frac{H}{2}}, \quad [\mathbf{V}^{01}, \mathbf{V}^{11}],$$

Notice that the tensor parallelism is kept, and the attnetion is computed in the following Attn block, resulting in $\mathbf{O}^{00}, \mathbf{O}^{01} \in \mathbb{R}^{B\times \frac{S}{2}\times \frac{H}{2}}$ on GPU0, GPU1 respectively.

Then in the Attn out layer, we do projection. This is the part that is row-parallel. RowParallelLinear means the weight is split along the input dimension, and the input is assumed to already be split across TP ranks. The reason for doing so is obvious, if we notice that on each GPU, we keep the input matrix in the shape of $B \times \frac{S}{2} \times \frac{H}{2}$, we naturally need $\mathbf{W}_O^{00}, \mathbf{W}_O^{01} \in \mathbb{R}^{\frac{H}{2}\times H}$ So on GPU0, we have

$$\mathbf{P}^{00} = \mathbf{O}^{00} \times \mathbf{W}_O^{00} \in \mathbb{R}^{B\times \frac{S}{2}\times H},$$

and similar on GPU1

$$\mathbf{P}^{01} = \mathbf{O}^{01} \times \mathbf{W}_O^{01} \in \mathbb{R}^{B \times \frac{S}{2} \times H}$$

These are all partial sums, and the result is given by

$$\mathbf{Y}^0 = \mathbf{P}^{00} + \mathbf{P}^{01} \in \mathbb{R}^{B \times \frac{S}{2} \times H}.$$

This is why another reduce-scatter is called here.

During the next reduce-scatter, we reduce to get $\mathbf{Y}^0$ and $\mathbf{Y}^1$ and reshard them along the sequence dimension.

Notice that the rest does not involve CP anymore because there is no inter token actions.

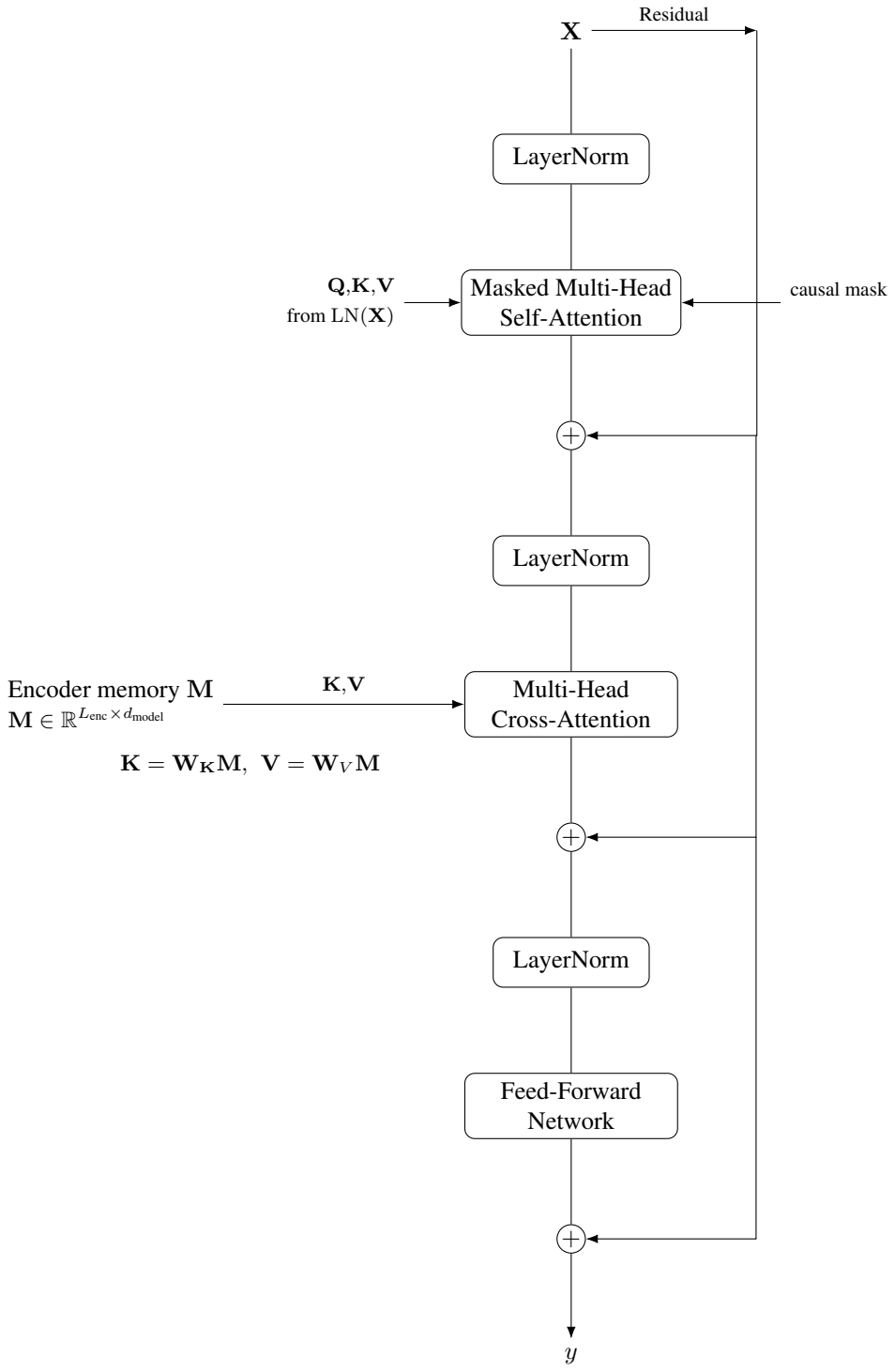


Figure 1: Pre-LN Transformer **Decoder** block. Each sublayer consumes a LayerNormed input. The masked self-attention applies a *causal mask* (no future tokens). Cross-attention queries come from the decoder stream, while keys/values are derived from the encoder memory $\mathbf{M}$ (encoder hidden states: $\mathbf{K} = \mathbf{W_K}\mathbf{M}$, $\mathbf{V} = \mathbf{W_V}\mathbf{M}$).

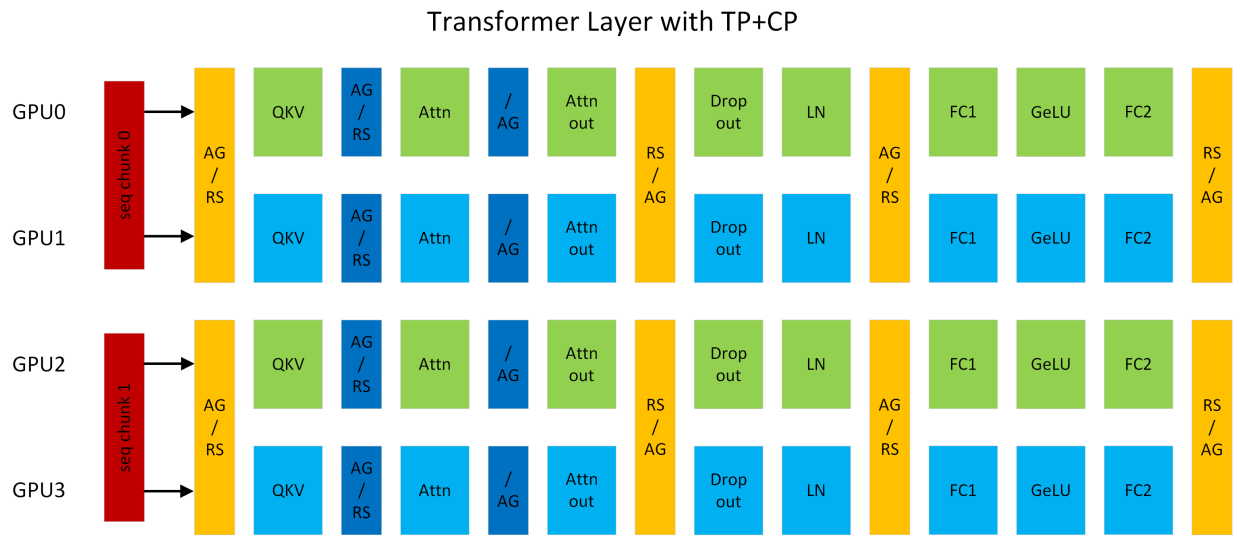


Figure 2: 2 TP + 2 CP